

HIGH-SPEED AND POWER EFFICIENT INTEGRATED ELECTRO-OPTIC
MODULATORS

by

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A thesis submitted in conformity with the requirements
for the degree of Doctor of Philosophy

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University of Toronto

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2025

Abstract

Communication networks worldwide are experiencing exponential growth in data traffic. The foundation of international communication is currently provided by fiber-optic communication networks, which manage more than half of the internet traffic. Increasing the capacity as well as reducing the energy consumption of the optical communication links is essential to meet the current demands. An electro-optic (EO) modulator, which translates the digital bit stream into an optical signal, is one of the essential components that determines the data rate of a fiber-optic link. The deployment of thousands of externally modulated laser links in a data center requires EO modulators with low energy consumption and small footprints, which enable high-density integration at a low cost. The standard silicon-on-insulator (SOI) platform and the more recently developed lithium niobate-on-insulator (LNOI) platform are the foundations of photonic integrated circuits (PIC), which have emerged as a promising technology to meet the needs of the optical communication industry. While effective optical modulators have been shown on both platforms, higher-speed and more compact devices are required to keep up with the increasing demand of internet traffic.

Therefore, to enhance the bandwidth and to reduce the power consumption I demonstrated two SOI platform-based device architectures in this thesis: one based on the Mach-Zehnder interferometer (MZI) silicon modulator to enhance the bandwidth and the other on the microring resonator (MRR) silicon modulator to support higher order modulation formats. In addition to that, I also present a modulator with new device concept based on the LNOI platform which lowers the voltage needed to drive the state-of-the-art thin film lithium niobate (TFLN) modulators.

I initially presented the silicon dioxide cladding capacitively coupled silicon (CC-Si) modulator in **Chapter 2**. I reported our observation of plasma dispersion effects for the first time in silicon strip waveguides fabricated in the multi-project wafer (MPW) run of a commercial foundry. In a proof-of-concept experiment, I generated sidebands up to 50 MHz around a carrier signal near telecom wavelengths by applying sinusoidal modulation signals at one arm of the MZI. Using a

self-heterodyne measurement system I estimated a half-wave-voltage-length product ($V_{\pi}L$) of ~ 9.4 V.cm at 20 MHz modulation signal. The EO measurement of the proposed modulator can generate the first order sidebands up to 68 GHz of radio-frequency (RF) signals and almost a flat frequency response was observed. The modulator has an estimated 3-dB bandwidth (f_{3dB}) of ~ 9.3 GHz and with further optimization the bandwidth can be enhanced beyond 40 GHz. Later, I demonstrated a silicon strip waveguide-based modulator in a different configuration which is also known as the series push-pull (SPP) configuration. This modulator also shows a flat electro-optics response up to 68 GHz. A $V_{\pi}L$ of ~ 87.8 V.cm is estimated from the power of generated sideband near a C-band optical carrier by a 15 GHz RF modulation signal. I also have demonstrated fabrication process of the CC-Si modulator with barium titanate cladding which was partially fabricated at a commercial foundry and the rest of the part was fabricated in several back-end-of-line (BEOL) post-fabrication processes at the university fabrication facility. I demonstrated an MZI based CC-Si modulator with SPP configuration. For a 2 mm long device I generated sidebands up to 67 GHz and observed a flat frequency response. I further demonstrated a $V_{\pi}L$ of ~ 2 V.m at 15 GHz at a C-band carrier wavelength.

In **Chapter 3**, I have demonstrated a SOI platform-based optical phase shifter by combining the through and drop ports of a silicon add-drop MRR modulator. The measured intensity spectra for different bias voltages show that, our proposed phase shifter combining the through and the drop ports show ~ 4.29 dB less maximum intensity variation compared to the maximum intensity variation observed at the through port for $\sim 2\pi$ phase shift. I also extracted the phase spectrum by placing such an optical phase shifter in one arm of an integrated MZI structure. I also achieved a full 2π phase shift of a continuous wave (CW) carrier optical signal with maximum ~ 4.38 dB of intensity variation by driving the silicon MRR modulator with a forward bias voltage which gives us a half-wave voltage (V_{π}) of 250 mV for a telecommunication CW optical carrier. Finally, I created a constellation map of 8-ary phase shift keying (8-PSK) based on the extracted phase and the intensity at different forward bias voltages.

In **Chapter 4**, I proposed a novel EO modulator based on the LNOI platform where a cladding material of very high dielectric constant i.e., barium titanate (BTO) is used around an etched lithium niobate (LN) waveguide. Lumerical Mode and Device simulations have been used to optimize a MZI based EO modulator that shows a $V_{\pi}L$ as low as ~ 1.36 V.cm maintaining the optical loss due to metal absorption of ~ 1 dB/cm. The 3 dB EO bandwidth of ~ 132 GHz has been estimated using

the data generated by the microwave simulations carried out in Computer Simulation Technology (CST) Studio Suite.

I would like to dedicate my PhD degree to the brave students and general people of my home country, Bangladesh who were martyred by the autocratic government of Sheikha Hasina in July-August 2024.

Acknowledgements

Alhamdulillah! All praise is due to Allah.

I would like to express my profound gratitude to my supervisor and esteemed professional colleague, Prof. J. Stewart Aitchison, for his steadfast support, guidance, and trust throughout my PhD journey. Prof. Aitchison played a pivotal role in my career by welcoming me into his research group at a critical juncture when I was contemplating leaving the University of Toronto. His mentorship has been instrumental in my development as both a researcher and an individual. His belief in my potential, coupled with the knowledge, skills, and confidence he imparted, has been invaluable to my success.

I also wish to extend my heartfelt thanks to the members of my PhD supervisory committee, Prof. Nazir P. Kherani and Prof. Amr S. Helmy, for their generous allocation of time and insightful guidance. Furthermore, I would like to acknowledge Prof. Trevor James Hall from the University of Ottawa and Prof. Ali Sheikholeslami from the University of Toronto for their meticulous review of my PhD thesis and for the constructive feedback they provided as the external and internal examiners, respectively.

Moreover, I am thankful to my former supervisors and colleagues at the Technical University of Denmark. My direct supervisor, Prof. Kresten Yvind, along with my co-supervisors, Prof. Michael Galili, Prof. Leif Katsuo Oxenløwe, and Dr. Yunhong Ding, provided exceptional support during my tenure in Denmark. Their assistance was particularly significant during a difficult time when my wife's Danish visa encountered multiple rejections due to certain unreasonable foreign policies of the Danish government, especially as I had recently entered into marriage.

I would also like to take this opportunity to express my deep appreciation to my previous supervisor, Prof. Joyce Poon, who initially welcomed me into her group at the University of Toronto. Although I had to transition to Prof. Aitchison's group due to family obligations, she played a crucial role in facilitating a substantial portion of my research. She graciously allowed me to utilize her lab, which enabled me to produce one conference paper and four journal articles.

Additionally, I wish to extend my thanks to our collaborators at the National Institute of Scientific Research (INRS) in Quebec, Canada, particularly Prof. José Azaña, Dr. Hao Sun, Robin Helsten, Md Mahadi Masnad, and Dr. Imtiaz Alamgir. Their collaboration played a crucial role in enabling me to publish two conference papers and one journal article.

I extend my gratitude to Prof. George V. Eleftheriades and his team member, Alex Mackay, for lending us the high-power RF signal generator and RF power meters, which facilitated the writing of three journal papers. I am also thankful to Prof. Li Qian and her team member, Alexander Greenwood, for providing us with the high-resolution optical spectrum analyzer, which contributed to the production of three journal papers.

Finally, I wish to express my heartfelt appreciation to my family—my wife, Fabiha Nawar, my

parents, my two sisters, and my in-laws—for their unwavering support throughout my PhD journey. I have been fortunate to welcome a daughter, Afeefah, and a son, Arham, during this period. Without the support of my family, particularly my wife, this journey would have been significantly more challenging and prolonged.

Ahmed Shariful Alam

February 2025

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Associated Publications

The research works presented in this thesis have resulted in 9 papers as a first author. Out of these 9 papers 5 of them are journal papers and 4 of them are conference papers. In these works, I conceived and verified the idea, designed and performed the experiment, and wrote the paper. The other co-authors contributed to discussing the idea, procurement of some devices, and editing the manuscripts. Additionally, through collaboration with members of the group of Prof. J. Stewart Aitchison at the University of Toronto and researchers from other research groups i.e., the group of Prof. José Azaña at National Institute of Scientific Research (INRS), the group of Prof. Joyce Poon at the University of Toronto, and the Quantum-Nano Fabrication and Characterization Facility (QNFCF) at the University of Waterloo.

Journal Articles Related to the Thesis

1. **A. S. Alam**, S. Nasif, J. K. S. Poon, J. Azaña, and J. S. Aitchison, “High-Bandwidth Silicon Strip Waveguide-Based Optical Modulator in Push-Pull Configuration,” (Submitted).
2. **A. S. Alam**, M. A. Islam, J. K. S. Poon, J. Azaña, and J. S. Aitchison, “Wideband Capacitively Coupled Silicon Modulator with Barium Titanate Cladding,” (Submitted).
3. **A. S. Alam**, H. Sun, M. M. Masnad, I. Alamgir, J. K. S. Poon, J. Azaña, and J. S. Aitchison, “Capacitively Coupled Silicon Modulator Fabricated on the Standard Silicon-on-Insulator Platform,” accepted in IEEE Photonics Journal.
DOI: [10.1109/JPHOT.2025.3539248](https://doi.org/10.1109/JPHOT.2025.3539248).
4. **A. S. Alam**, H. Sun, J. Azaña, J. K. S. Poon, and J. S. Aitchison, “Silicon Ring Resonator Based Optical Phase Shifter for Higher Order Modulation Formats,” in Optics Express 32, 46203-46212 (2024).
DOI: [10.1364/OE.536872](https://doi.org/10.1364/OE.536872).
5. **A. S. Alam** and J. S. Aitchison, “Optimization of a Broadband Lithium Niobate-Barium Titanate Hybrid Modulator With Low Half-Wave-Voltage-Length Product,” in IEEE Photonics Journal, vol. 15, no. 6, pp. 1-7, Dec. 2023, Art no. 6602407.
DOI: [10.1109/JPHOT.2023.3329781](https://doi.org/10.1109/JPHOT.2023.3329781).

Conference Articles Related to the Thesis

1. **A. S. Alam**, H. Sun, M. M. Masnad, I. Alamgir, J. Azaña, and J. S. Aitchison, “Bandwidth Estimation of a Silicon Strip Waveguide Based Optical Modulator,” in FiO LS 2024, Technical Digest Series (Optica Publishing Group, 2024).
DOI: [10.1364/FIO.2024.JW4A.50](https://doi.org/10.1364/FIO.2024.JW4A.50).
2. **A. S. Alam**, H. Sun, M. M. Masnad, I. Alamgir, J. Azaña, and J. S. Aitchison, “Carrier Dispersion Modulation in a Silicon Strip Waveguide,” in CLEO 2024, Technical Digest Series (Optica Publishing Group, 2024), paper JTh2A.195.
DOI: [10.1364/CLEO_AT.2024.JTh2A.195](https://doi.org/10.1364/CLEO_AT.2024.JTh2A.195).
3. **A. S. Alam**, J. C. Mikkelsen, J. K. S. Poon, and J. S. Aitchison, “Silicon Ring Modulator Based Optical Phase Shifter,” in CLEO 2023, Technical Digest Series (Optica Publishing Group, 2023), paper SF2K.7.
DOI: [10.1364/CLEO_SI.2023.SF2K.7](https://doi.org/10.1364/CLEO_SI.2023.SF2K.7).
4. **A. S. Alam** and J. S. Aitchison, “Low Half-Wave-Voltage Lithium Niobate Modulator Using High-K Dielectric Material Cladding,” in Optica Advanced Photonics Congress 2022, Technical Digest Series (Optica Publishing Group, 2022), paper JTt2A.9.
DOI: [10.1364/BGPPM.2022.JTt2A.9](https://doi.org/10.1364/BGPPM.2022.JTt2A.9).

Acronyms

16-APSK 16-level amplitude-phase shift keying (APSK)

16-QAM 16-level quadrature amplitude modulation (QAM)

3D Three dimensional

8-PSK 8-ary phase shift keying

AIM American Institute for Manufacturing Integrated

AMF Advanced Micro Foundry

AOM acousto-optic modulator

APSK amplitude-phase shift keying

BEOL back-end-of-line

BOX buried oxide

BTO barium titanate

CC-Si capacitively coupled silicon

CC-SOH capacitively coupled silicon-organic hybrid

CEA Leti Information Technology Electronics Laboratory

CPS coplanar striplines

CPU central-processing-units

CPW coplanar waveguide

CST Computer Simulation Technology

CW continuous wave

DC direct current

EO electro-optic

ESA electrical spectrum analyzer

FDTD Finite-Difference Time-Domain

FSR free spectral range

GS ground-signal

GSG ground-signal-ground

GSSG ground-signal-signal-ground

IC integrated circuit

IMEC Interuniversity Microelectronics Centre

InP indium phosphide

IoT internet-of-things

IQ in-phase and quadrature-phase

LN lithium niobate

LNOI lithium niobate-on-insulator

MBE molecular beam epitaxy

MMI multimode interferometers

MOCVD metalorganic chemical vapour deposition

MOS metal/oxide/semiconductor

MPW multi-project wafer

MRR microring resonator

MZI Mach-Zehnder interferometer

OSA optical spectrum analyzer

OVA optical vector analyze

PAM pulse amplitude modulation

PAM-2 2-level pulse amplitude modulation (PAM)

PAM-4 4-level PAM

PC polarization controller

PD photodiode

PDK process design kits

PIC photonic integrated circuits

PIN P-type/Intrinsic/N-type

PMMA polymethyl methacrylate

PN P-type/N-type

PSK phase shift keying

QAM quadrature amplitude modulation

QPSK quadrature phase shift keying

RC resistance-capacitance

RC-Si resistively coupled silicon

RC-SOH resistively coupled silicon-organic hybrid

RF radio-frequency

sccm standard cubic centimeter per minute

SEM scanning electron microscope

SG signal-ground

SiN silicon nitride

SOH silicon-organic hybrid

SOI silicon-on-insulator

SPP series push-pull

TE transverse electric

TFLN thin film lithium niobate

TiN titanium nitride

TLS tunable laser source

TMM transfer matrix model

TW-MZM traveling-wave Mach-Zehnder modulators

TWE Travelling wave electrodes

UV15 ultra-violet curable adhesive

VNA vector network analyzer

WDM wavelength division multiplexers

Chapter 1

Introduction

In the age of information and technology fast communication across the world is essential in almost every aspect of our lives. Due to the emerging internet-of-things (IoT), cloud-based services, and smart devices virtually generated information is shared instantaneously. This resulted in an unprecedented demand for the volume and speed of data transfers. According to a forecast by Cisco, the global internet traffic is rising by 26% in every year and this demand will continue to grow undoubtedly [1]. Long-haul optical communications such as transatlantic or transpacific submarine cables and short-reach optical communications like intra or inter-datacenter communications are the backbone of the global communication network. It has been observed that almost 75% of all data flows inside or among the datacenters [2]. To keep up with such explosive increase of data rates, the capacity of short-reach optical links in these datacenters needs to be increased. At the same time, central-processing-units (CPU) of the future zettascale supercomputers require high-speed electrical to optical and optical to electrical signal conversions (optical interconnects) to transfer data to other CPUs and to the respective memory units. These supercomputers are power hungry, and the majority portion of the power is consumed by the interconnects. As a matter of fact, transferring data to the nearest chip can consume 200 times more power than performing an operation on the same data [3]. Therefore, reducing the energy consumption of the interconnects is equally important to enhancing the bandwidth.

The electro-optic (EO) modulator is a key optical interconnect which translates the information signal from the electrical domain to the optical domain by controlling various physical properties of light i.e., intensity, phase, or polarization. For a long time, lithium niobate (LN or LiNbO_3) based traditional EO modulators have been the workhorse of long-haul optical communication systems [4]. Yet, these EO modulators are not convenient for short-reach optical because of their bulky size and power consumption. Since the beginning of 1990 photonic integrated circuit (PIC) [5] technologies have become more mature and been providing advanced photonic functionalities ensuring compact chips with extraordinary integration density [6]. The efficiency of EO modulators depends on 3 major parameters – the halfwave voltage (V_π) or the voltage which is required to achieve a 180° phase shift for an optical signal, the optical propagation loss which is usually quantified in dB/cm, and the 3-dB EO bandwidth ($f_{3\text{dB}}$). A variety of material platforms such as, silicon-on-insulator (SOI), indium phosphide (InP), lithium niobate-on-insulator (LNOI) etc. have been investigated and com-

mercialized in the field of PICs to optimize these performance parameters and efficient modulators have been demonstrated on each of these platforms. This thesis focuses primarily on the SOI and LNOI platform based EO modulators. Hence, this chapter includes a short overview of the most common modulators based on these two PIC platforms.

1.1 SOI Platform-Based Modulators

The foundation of silicon PICs has been mainly defined by the SOI technology for over the past two decades which leads this platform to commercial success in short-reach optical communication systems and high-speed optical transceiver products. SOI platform-based silicon waveguides are capable of having less dispersion and tight optical confinement compared to conventional silicon-built waveguides [7]. However, in terms of EO effects silicon is lagging the other materials such as LN, InP etc. Silicon has a very low Kerr coefficient in the order of $10^{-18} \text{ m}^2/\text{W}$ [8] and the Pockels effect is absent in bulk silicon at the telecommunication wavelengths of $1.3 \mu\text{m}$ and $1.55 \mu\text{m}$ [9]. Therefore, alternative methods are required to modulate light in silicon waveguides. One way is to perform thermal modulation since silicon has large thermo-optic effects. But the speed of thermo-optic modulators is limited to a few hundred kHz which is too slow for modern days' GHz speed optical communications [10].

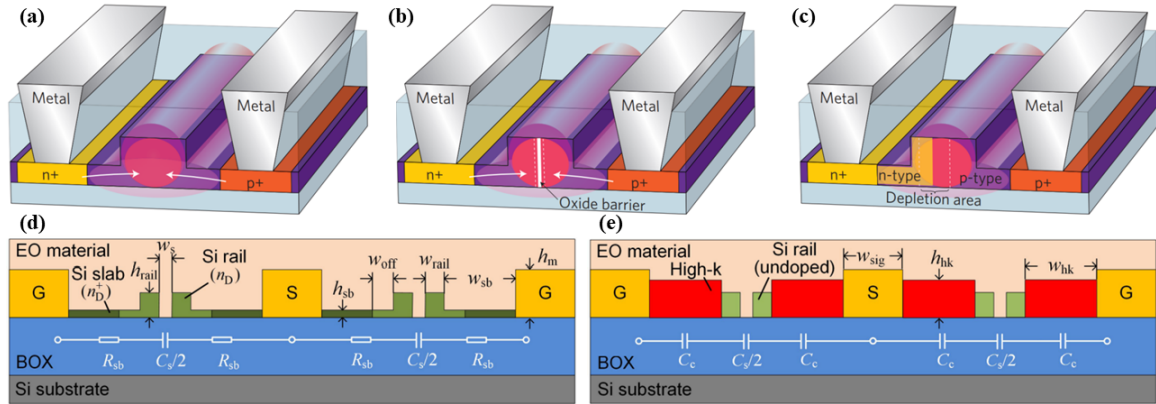


Figure 1.1: Three dimensional (3D) schematic view of (a) a carrier injection modulator, (b) a carrier accumulation modulator, and (c) a carrier depletion modulator. Cross-sectional view of (d) an resistively coupled silicon-organic hybrid (RC-SOH) modulator, and (e) a capacitively coupled silicon-organic hybrid (CC-SOH) modulator. Figures were reproduced from Ref. [11] and [12].

Conventional silicon modulators are mainly based on the plasma dispersion effect where the complex refractive index of silicon can be manipulated by varying the carrier concentrations [13]. Modulation of both the real and the imaginary part of the refractive index causes amplitude-phase coupling i.e., pure phase modulation without changing the light intensity is not possible. Also, the change of both the real and imaginary part of the refractive index is not completely linear which compromises the signal quality by creating undesirable chirps. But, with carefully design these drawbacks can be minimized. There are mainly three different types of silicon modulators based on the plasma dispersion effect – carrier injection modulator, carrier accumulation modulator and carrier depletion modulator. In carrier injection modulator, optical mode confined in a P-type/Intrinsic/N-type (PIN) structured rib waveguide is modulated by applying a forward bias voltage in order to modulate

the carrier concentration i.e., the complex effective refractive index exploiting the carrier injection phenomena. To avoid higher optical loss the optical mode is mostly confined within the intrinsic region of the waveguide as depicted in Fig. 1.1a. Although these modulators can achieve V_π below 1 V because of high forward bias current, f_{3dB} of such modulators are limited to about 100 MHz because of longer transit times of the carriers in the forward bias regime [14]. The EO bandwidth of silicon modulators were enhanced to ~ 42 GHz range by introducing a vertical dielectric layer (silicon dioxide) in the middle of a silicon rib waveguide by accumulating free carriers on p and n-doped sides of the waveguide (as shown in Fig. 1.1b) by applying a reverse bias voltage rather than injecting carriers by applying a forward bias voltage. These modulators are known as the carrier accumulation modulators or metal/oxide/semiconductor (MOS) type modulators. Even though this type of modulator has a half-wave-voltage-length product ($V_\pi L$) of around ~ 1.5 V.cm, the actual V_π of this device is very high since the length of this is still limited to a few hundred μm . This is done in order to reduce the microwave propagation loss and to keep the overall effect of resistance-capacitance (RC) time constant small since the dielectric layer in the middle of the waveguide acts as an extra capacitor [15]. Carrier depletion by applying a reverse bias voltage across a lightly doped P-type/N-type (PN) diode rib waveguide is another mechanism to bypass the speed limitation due to the minority carrier lifetime. In a carrier depletion modulator, the carrier concentrations are varied by varying the width of the depletion region as shown in Fig. 1.1c. Unlike the MOS type modulator, the PN type modulator does not have an extra capacitor. Therefore, the PN type modulators have lower RC time constant and are expected to be the fastest among all silicon modulators. The $V_\pi L$ of such modulators can be as low as ~ 1.4 V.cm and the f_{3dB} can be as high as 12 GHz [16]. Longer modulators are necessary to keep the effective V_π low and hence the bandwidth for such modulators is limited by the microwave propagation loss, and velocity mismatch between the optical and microwave modes. Both the driving voltage and the bandwidth of PN type modulators have been further enhanced by extensive engineering for different modulator structures such as the traveling-wave Mach-Zehnder modulators (TW-MZM) with $f_{3dB} = 41$ GHz and $V_\pi = 7$ V [17], the segmented TW-MZMs with $f_{3dB} = 67$ GHz and $V_\pi = 5$ V [18], and slow-light modulators with $f_{3dB} = 110$ GHz and $V_\pi = 10$ V [19].

A hybrid approach of EO organic polymers with silicon solves the problem of amplitude-phase coupling. This creates a new class of SOI platform-based modulators, the silicon-organic hybrid (SOH) modulators, where the optical mode is still guided by a silicon slot waveguide, but the EO operation is performed by an EO organic polymer which fills the slot of the waveguide (Fig. 1.1d). Because of the high index contrast the optical field intensity is very high in the EO organic material filled silicon slot. But at the same time due to the leaky slot modes and the silicon sidewall roughness the optical propagation loss of slot waveguides is almost 40 times higher compared to the silicon strip waveguides in dB scale [20]. The conventional SOH modulators, also known as the resistively coupled silicon-organic hybrid (RC-SOH) modulators, can achieve an f_{3dB} of 43 GHz which is still limited by the RC time constant contributed by the highly doped thin silicon stripload and the slot capacitance. Although an extremely low $V_\pi L$ of 0.032 V.cm can be achieved by these modulators, these devices can have optical propagation loss as high as 37.5 dB/cm [21]. The bandwidth of the SOH devices can be incredibly enhanced (by almost 3 times) using capacitive coupling between the electrical and optical signal [22, 23] as depicted in Fig. 1.1e. These capacitively coupled silicon-

organic hybrid (CC-SOH) modulators can achieve an $f_{3\text{dB}}$ of 78 GHz with a $V_{\pi}L$ of 0.13 V.cm [12, 24]. Despite having high bandwidth and low driving voltage SOH modulators suffer from high optical loss and instability of the EO polymers prone to the thermal and environmental exposure [25].

One of the objectives of this thesis is to employ the capacitive coupling approach to enhance the bandwidth of a conventional PN diode silicon modulator. Previously mentioned conventional silicon modulators suffer less from problems like high optical loss and thermal & environmental instability. Capacitive coupling in order to avoid the RC time limitation problem has not been tried for conventional silicon PN diode modulators. To the best of our knowledge, I have demonstrated the CC-Si modulators for the first time experimentally. I have demonstrated two different types of structures – 1) the CC-Si modulators with silicon dioxide cladding which are compatible with the fabrication process steps of the standard silicon foundries, and 2) the CC-Si modulators with a high dielectric constant material cladding like barium titanate (BTO or BaTiO_3) which are not compatible with the fabrication process steps of the standard silicon foundries and need back-end of line (BEOL) post-fabrication processes.

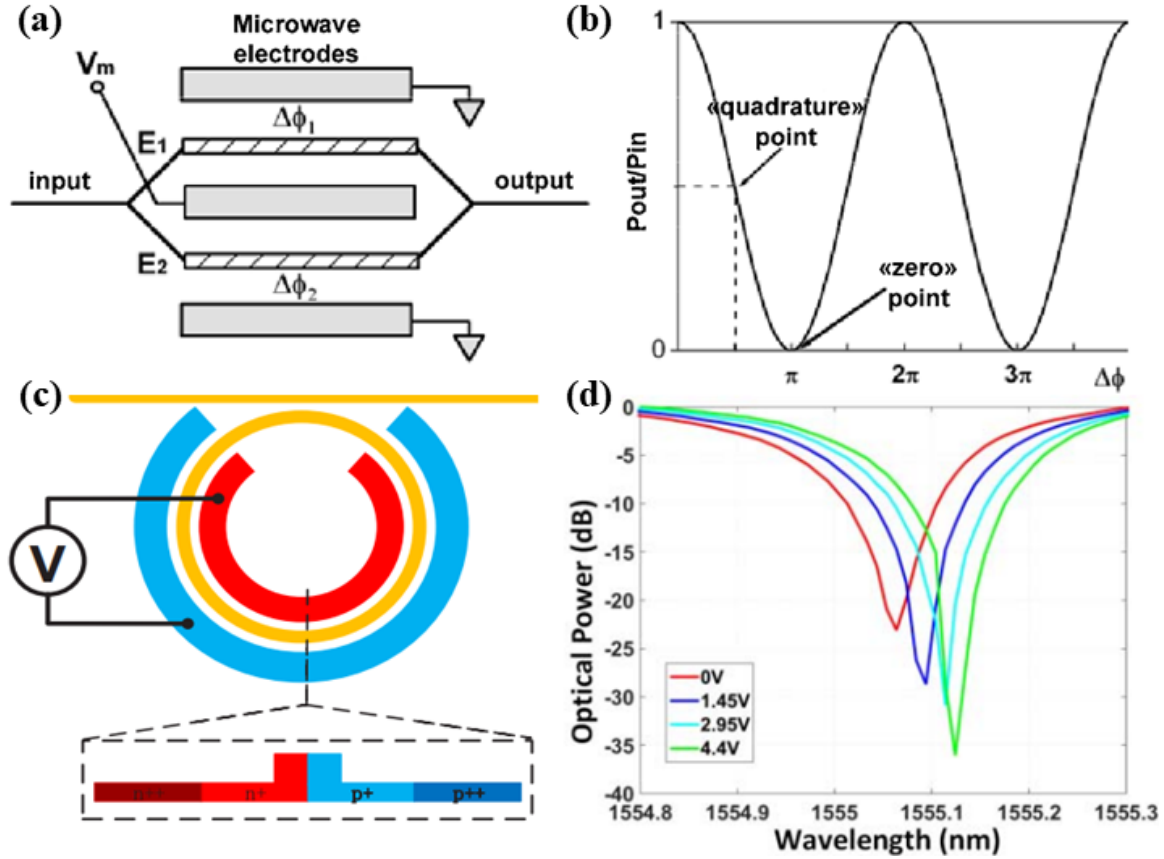


Figure 1.2: (a) Configuration and (b) transfer function of an MZI-based modulator. (c) Configuration and (d) intensity spectra of an MRR-based modulator. Figures were reproduced from Ref. [26] and [27].

The abovementioned silicon modulators are basically straight waveguide-based optical phase shifters which are mostly used in one or both arms of an Mach-Zehnder interferometer (MZI) structure to re-

alize an intensity modulator (Fig. 1.2a & 1.2b). Two MZI-based intensity modulators can be placed in two arms of another MZI structure, known as the in-phase and quadrature-phase (IQ) modulator, which can be used in higher order modulation formats. As mentioned earlier, these phase shifters are rather long (several millimeters) which introduces higher microwave loss and suffer from velocity mismatch between the optical and microwave signals. On the contrary, silicon microring resonator (MRR) based modulators are compact, hence can be considered as lumped elements in the microwave regime. Therefore, the MRR modulators have higher bandwidth since velocity matching between the optical and microwave modes is not very critical. But silicon MRR modulators can only be used as intensity modulators because of their resonant behavior and thus, are not compatible for higher order modulation formats (Fig. 1.2c & 1.2d). Several MRR-based phase shifters have been demonstrated previously using over-coupled rings which suffer from high intensity variation while shifting the phase [28, 29, 30].

Another objective of this thesis is to demonstrate a silicon MRR-based optical phase shifter for higher order modulation formats. In this thesis, I have demonstrated a silicon MRR modulator based optical phase shifter which can show $\sim 2\pi$ phase shift with very low intensity variation. I have also showed the prospect of using the phase shifter in a higher order modulation format, 8-level phase shift keying (8-PSK) signal.

1.2 LNOI Platform-Based Modulators

LN is a widely used material for its excellent optical properties such as large second order nonlinearity, decent EO effect and wide optical transparent window. With such unique properties LN EO modulators have been used in telecommunication for long time despite integration challenges associated with the low index contrast waveguides of ~ 0.02 fabricated by Titanium (Ti) ion diffusion [31]. This results in large optical mode with a mode field diameter of $\sim 10 \mu\text{m}$ and the metal electrodes are placed far away from the optical waveguides like Fig. 1.3a in X-cut LN, or the low index buffer layer must be thick enough like Fig. 1.3b & 1.3c Z-cut LN to avoid large metal absorption losses. Therefore, the electrical field strength of the RF mode is low which affects the $V_\pi L$ of the device. Bulk LN modulators show a $V_\pi L$ of roughly 12-32 V.cm [32]. Increasing modulator length may reduce the V_π of the device but limits the $f_{3\text{dB}}$ owing to higher microwave loss and velocity mismatch between the optical and the RF modes. Travelling wave electrodes (TWE) can achieve velocity matching between the optical and the RF modes for a ridge waveguide to enhance the modulator speed to $\sim 100 \text{ GHz}$ but because of the microwave loss the V_π remains still high [33].

In recent years, LNOI wafers have become commercially available where TFLN is bonded on silicon dioxide (also known as buried oxide (BOX)) on silicon after crystal ion slicing [37]. Since etching of LN is very difficult initially striploaded waveguides were used to realize the first generation of TFLN based modulators. An example of such a modulator is shown in Fig. 1.3d. In certain instances, a thin layer of silicon [38], silicon nitride [35], or alternative materials [39] is deposited, while in other cases, an LN thin film is bonded on pre-patterned silicon waveguides [40] to create strip-loaded waveguides. These waveguides are familiar for their simpler fabrication process. But the mode size of these devices is still large and hence, the electrodes are placed far away from the

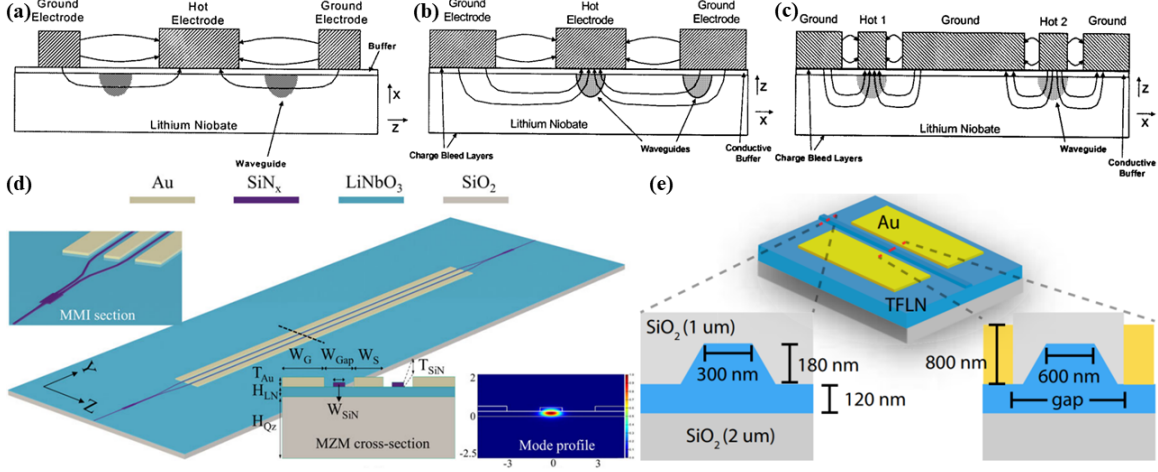


Figure 1.3: Cross-sectional view of (a) an X-cut, (b) a single drive Z-cut, and (c) a dual drive Z-cut bulk lithium niobate modulator. (d) A Three dimensional (3D) and a cross-sectional view of a striploaded TFLN modulator with silicon nitride stripload. (e) A Three dimensional (3D) and a cross-sectional view of a rib waveguide based etched TFLN modulator. Figures were reproduced from Ref. [34], [35], and [36].

waveguide region to avoid excess optical loss due to metal. The best $V_\pi L$ and optical loss achieved by a striploaded waveguide based TFLN modulator are 2.11 V.cm and ~ 2.25 dB/cm [35] respectively which is certainly a decent improvement (at least 5 times less) in terms of $V_\pi L$ compared to the bulk LN modulators which comes at a price of higher optical loss due to metal absorption. Additionally, efficient transfer of light from one layer to another layer is quite challenging for these devices despite some recent demonstrations of Three dimensional (3D) integration between TFLN and the striploaded waveguiding layers [41, 42]. Recent improvements of controlled physical dry etching techniques of LN have resulted in some excellent TFLN modulators based on high contrast rib waveguides (Fig. 1.3e) [43, 44, 45]. Having low scattering losses these TFLN waveguides made the fabrication of chip-scale LN modulators possible. The TFLN-based rib waveguides can show metal absorption loss as low as 0.03 dB/cm and the etched rib waveguide-based modulators can have a $V_\pi L$ of 2.2 V.cm [43]. Even in these modulators only the optical confinement is improved. Microwave mode confinement still remains as an issue for the rib waveguide based etched TFLN modulators because of the low dielectric constant silicon dioxide cladding. By using a segmented electrode instead of a uniform coplanar waveguide (CPW) electrode the bandwidth of such modulators can be improved, but the $V_\pi L$ of the device remains the same.

The final objective of this thesis is to propose a new modulator architecture where the low dielectric constant silicon dioxide cladding is replaced with a high dielectric constant BTO cladding to improve both optical and microwave confinement which is known as the LN-BTO hybrid modulators.

1.3 Thesis Organization

This thesis is organized in five chapters. **Chapter 1** provides a general background to the research topics and outlines the scope and the extent of this thesis.

Chapter 2 I briefly discussed the background physics of modulation of optical mode in silicon using an electrical signal. I further discussed the method of estimating the V_π and the f_{3dB} of the modulator. Additionally, the modulation mechanism of the conventional silicon modulator and the capacitively coupled silicon (CC-Si) modulator for reverse bias have been discussed. Finally, the experimental demonstrations of CC-Si modulators with different cladding materials have been shown.

Chapter 3 I will explain about the higher order modulation formats using optical phase shifters and the fundamentals of an add-drop MRR. This chapter also explains the idea of an optical phase shifter based on a silicon MRR. I further discussed the forward bias operation of the lateral PN diode-based silicon phase modulator and its use to perform intensity or phase modulation by tuning the effective refractive index in the ring. Finally, the experimental demonstration of the silicon add-drop MRR-based optical phase shifter was experimentally demonstrated in this chapter.

Chapter 4 I will discuss about the second order nonlinear effects or linear EO effect of LN. In addition, I will discuss about the working principle of the etched TFLN modulators which exploit this effect for EO modulation. Finally, this chapter presents the new concept of rib waveguide-based etched TFLN modulator – the LN-BTO hybrid modulator and its optimization procedure.

Chapter 5 concludes the thesis by summarizing the key points, achievements and the scope of future works.

Chapter 2

Capacitively Coupled Silicon Modulators

The previous chapter (Chapter 1) gives only a brief description of different types of modulators based on the SOI platform. But, this chapter (Chapter 2) discusses the fundamental physics of modulation or the plasma dispersion phenomena and the use of this phenomena in MZI structures. This chapter is organized as follows. In Section 2.1, the plasma dispersion effect in silicon, its use in a lateral PN junction based optical waveguide to realize a phase modulator, the idea of capacitively coupled PN junction based optical modulation approaches in silicon, and how this phase modulation concept is used in an MZI structure to realize an intensity modulator are discussed. Additionally, the method of half-wave voltage (V_π) and 3-dB bandwidth (f_{3dB}) estimation are discussed in this section. In Section 2.2, the first experimental demonstration the CC-Si modulator with silicon dioxide cladding is presented and various important modulator parameters like the V_π & the f_{3dB} are estimated. I also discussed the CC-Si modulator with BTO cladding in this Chapter. I measured the EO response of this device and estimated the V_π . Finally, this chapter is concluded in Section 2.3.

2.1 Background

The modulation of the phase and/or amplitude of optical signals in silicon waveguides can be achieved by changing the refractive index and optical absorption of silicon by exploiting the plasma dispersion effect. In this section, I discussed the physics behind this effect as well as the fundamentals of an MZI to perform intensity modulation. I further discussed the method of estimating the V_π from the sidebands generated by applying a low frequency microwave signal around an optical carrier signal in the telecommunication wavelengths. I also discussed the method of f_{3dB} estimation from small signal analysis and S-parameter measurement of the electrical transmission line of the modulator.

2.1.1 Plasma Dispersion Effect

In general, plasma means a charged gas. But, in semiconductor plasma refers to the electrons and holes which can freely move within the material. In 1987, the change of refractive index and optical absorption in silicon as a function of the change of free carriers was predicted by Soref and Bennett

[9]. This phenomenon is known as the plasma dispersion effect in silicon. Since the inversion symmetric crystal structure of silicon does not have any $\chi^{(2)}$ -nonlinearities, plasma dispersion effect is used to perform optical modulation using an electrical signal. A brief overview of the $\chi^{(2)}$ -nonlinearity is discussed in Chapter 4.

The motion of an electron in silicon structures is governed by Newton's Law,

$$\vec{F}_{\text{tot}} = m \frac{d\vec{v}}{dt} = \vec{F}_{\text{ext}} - \vec{F}_{\text{damp}} \quad (2.1)$$

Here, $\vec{F}_{\text{tot}}$ is the total applied force on the electron. $\vec{F}_{\text{ext}} = q(\vec{E} + \vec{v} \times \vec{B})$ is the external force on the electron. $\vec{F}_{\text{ext}}$ can cause because of the electrical field, $\vec{E}$ originated from the propagating light through the material, and because any external magnetic field, $\vec{B}$. Since the velocity of electron, $\vec{v}$ is much slower than the speed of light, c (i.e. the $\vec{E}$ field), the second term of $\vec{F}_{\text{ext}}$ can be neglected and the equation becomes $\vec{F}_{\text{ext}} \approx q\vec{E}$ [46]. $\vec{F}_{\text{damp}} = m\vec{v}/\tau$ is the damping force induced from the change of momentum when the electron is involved in a band transition. Here, τ is the relaxation time constant i.e., average time between momentum destroying collisions. Then the equation of the motion of the electron can be rearranged as [47],

$$q\vec{E} = m \frac{d\vec{v}}{dt} + m \frac{\vec{v}}{\tau} \quad (2.2)$$

If we consider a monochromatic light passing through silicon then we can consider the optical electric field by, $\vec{E} = \vec{E}_0 e^{-i\omega t}$, where ω represents the frequency of the light wave. Now the electron velocity can be expressed as,

$$\vec{v} = \frac{q\vec{E}}{-i\omega m + m/\tau} \quad (2.3)$$

Inserting the expression of $\vec{v}$ in Maxwell's equation, we get,

$$\begin{aligned} \nabla \times \vec{H} &= \frac{\partial \vec{D}}{\partial t} + \vec{J}_p \\ &= -i\omega\epsilon\vec{E} + Nq\vec{v} \\ &= -i\omega\epsilon_0 \left[\epsilon_r + i\epsilon_i - \frac{(\omega_p\tau)^2 (\omega\tau)^2 - i\omega\tau}{(\omega\tau)^2 + 1} \right] \vec{E} \\ &= -i\omega\epsilon\vec{E} \end{aligned} \quad (2.4)$$

and,

$$\epsilon \equiv (n + ik)^2 = \epsilon_r + i\epsilon_i - \frac{(\omega_p\tau)^2 (\omega\tau)^2 - i\omega\tau}{(\omega\tau)^2 + 1} \quad (2.5)$$

Here, $\omega_p = \sqrt{q^2 N / m\epsilon_0}$ is the plasma frequency of the electron, $\vec{J}_p$ is the polarization current density, m is the carrier effective mass, N is the free carrier density, ϵ_r & ϵ_i are the real & the imaginary parts of relative permittivity respectively in absence of polarization current, ϵ_0 & $\epsilon \equiv (n + ik)^2$ are the permittivity of free space & the effective permittivity in the medium, respectively. After separating the real and imaginary part of the relative permittivity, ϵ we obtain the relation between the real

part, n and the imaginary part, k of the refractive index of the medium.

$$n^2 - k^2 = \epsilon_r - \frac{(\omega_p \tau)^2}{(\omega \tau)^2 + 1} \quad (2.6)$$

$$2nk = \epsilon_i - \frac{(\omega_p \tau)^2}{\omega \tau} \cdot \frac{1}{(\omega \tau)^2 + 1} \quad (2.7)$$

If the total force is set to zero, the electron mobility $\mu = q\tau/m$ is found from Eq. 2.1 and 2.2. Substituting $\tau = \mu m/q$, $n = n_0 + \Delta n$ and $n = n_0 + \Delta n$ into Eq. 2.6 and 2.7, the change of real part of the refractive index and the change of the optical power absorption coefficient are obtained,

$$\Delta n = -\frac{e^2 \lambda^2}{8\pi^2 c^2 \epsilon_0 n_0} \cdot \frac{N}{m} \quad (2.8)$$

$$\Delta \alpha = \frac{4\pi \Delta k}{\lambda} = \frac{e^3 \lambda^2}{4\pi^2 c^3 \epsilon_0 n_0} \cdot \frac{N}{m^2 \mu} \quad (2.9)$$

Here, for the telecommunication wavelengths we can safely assume $\omega \gg 1/\tau$, $n \gg k$ and $n_0 \gg \Delta n$ [48]. n_0 and n_0 are the real and imaginary parts of the refractive index respectively without free carriers. When the contribution from both electrons and holes are taken into account, the total refractive index change and the total absorption coefficient can be expressed by,

$$\Delta n = -\frac{e^2 \lambda^2}{8\pi^2 c^2 \epsilon_0 n_0} \left(\frac{\Delta n_e}{m_e} + \frac{\Delta n_h}{m_h} \right) \quad (2.10)$$

$$\Delta \alpha = \frac{4\pi \Delta k}{\lambda} = \frac{e^3 \lambda^2}{4\pi^2 c^3 \epsilon_0 n_0} \left(\frac{\Delta n_e}{m_e^2 \mu_e} + \frac{\Delta n_h}{m_h^2 \mu_h} \right) \quad (2.11)$$

Here, the subscripts 'e' & 'h' refer to electrons & holes respectively and Δn_e & Δn_h are free electron & hole densities respectively. The changes in the refractive index, Δn was evaluated using the experimentally produced absorption curves by Soref and Bennett for a wide range of Δn_e and Δn_h , over a wide range of wavelengths. They quantified Δn and $\Delta \alpha$ with respect to Δn_e and Δn_h in silicon at a wavelength of 1550 nm which can be expressed by,

$$\Delta n = - (8.8 \times 10^{-22} \text{cm}^3) \Delta n_e - (8.5 \times 10^{-18} \text{cm}^{2.4}) (\Delta n_h)^{0.8} \quad (2.12)$$

$$\Delta \alpha = (8.5 \times 10^{-18} \text{cm}^2) \Delta n_e + (6.0 \times 10^{-18} \text{cm}^2) \Delta n_h \quad (2.13)$$

These equations (Eq. 2.12 & 2.13) from the empirical fit were used in the simulations in this thesis.

2.1.2 Silicon Waveguides

In silicon integrated devices light does not propagate as a plane wave, but as an optical mode confined in the core region of a waveguide. Therefore, instead of considering the change of refractive index, Δn we consider the change of effective refractive index, Δn_{eff} while designing the modulator. The two most common silicon waveguides used in the standard SOI platform are the strip waveguide where the silicon device layer is fully etched around the core regions and the rib waveguide where the silicon device layer is partially etched leaving a thin silicon slab layer around the core region. Fig.

2.1a and 2.1b show the fundamental transverse electric (TE) modes in a single mode 500 nm wide strip waveguide and a single mode 500 nm wide rib waveguide based on the standard SOI platform with 220 nm thick silicon device layer and 90 nm thick silicon slab layer. The Lumerical Mode Solver was used to calculate Δn_{eff} and generate the optical field distribution of the TE modes of both waveguides. Usually, such strip waveguides have an $\Delta n_{\text{eff}} \approx 2.445557$ and are good for optical routing, and such rib waveguides have an $\Delta n_{\text{eff}} \approx 2.567589$ and are used for active phase shifting using plasma dispersion effect considering the core material (silicon) refractive index, $n_{\text{Si}} = 3.48$, and the cladding material (silicon dioxide) refractive index, $\Delta n_{\text{SiO}_2} = 1.44$ near 1550 nm wavelength [49].

In the course of our experiment, rib waveguides were employed within the active region, specifically in the electro-optic (EO) phase shifter component of the modulator. In contrast, strip waveguides were utilized in all other instances, including the coupling of light into the chip from a fiber, the coupling of light out of the chip to a fiber, as well as during the processes of power splitting and power combining.

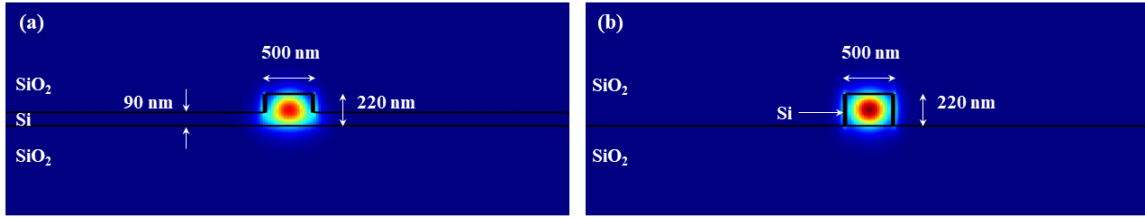


Figure 2.1: Intensity plots of the fundamental optical TE modes at 1550 nm in silicon (a) rib, and (b) strip waveguides.

2.1.3 Conventional/Resistively Coupled Silicon (RC-Si) Modulator

As we have mentioned in Chapter 1, for high-speed operation depletion type silicon modulators are most suitable. Fig. 2.2a shows the cross-sectional view the most common depletion type silicon modulator which has a silicon rib waveguide in the middle and two aluminium metal lines on two sides of the rib waveguide core [50, 51]. These aluminium metal lines are known as coplanar striplines (CPS) which are placed on the silicon slab of the rib waveguide. The waveguide core is divided into lightly P and N-doped regions which expand into slab regions towards the left and right sides of the waveguide core respectively as shown in Fig. 2.1a to maintain low optical loss. Usually, the PN junction is in the middle of the core region. The rest of the silicon slab regions are highly P-doped (P+) on the left-hand side and highly N-doped (N+) on the right-hand side of the core respectively. The aluminium CPS are mostly placed on these highly doped regions for the low loss ohmic contacts. For high-speed operation the PN diode needs to operate in the reverse bias regime since the dominant drift current during the reverse bias operation involves higher velocity carriers than the dominant diffusion current during the forward bias operation. Therefore, the right aluminium metal line is considered as the signal contact which is connected to the P-doped side of the phase modulator and the left aluminium metal line is considered as the ground contact which is connected to the N-doped side of the phase modulator. By applying an external reverse bias voltage, the width of the depletion region is changed. The change in depletion width leads to the change of carrier concentrations in the

core region which eventually leads to a change of n_{eff} (Δn_{eff}) of the optical mode shown in Fig 2.1a. The Lumerical Device simulations are shown Fig. 2.2b to realize how the width of the depletion region, and the electron and hole concentrations in that region change because of different applied reverse bias voltages. One thing should be kept in mind that, the optical loss, α_{eff} of the waveguide also changes with the applied reverse bias voltage which introduces amplitude-phase coupling.

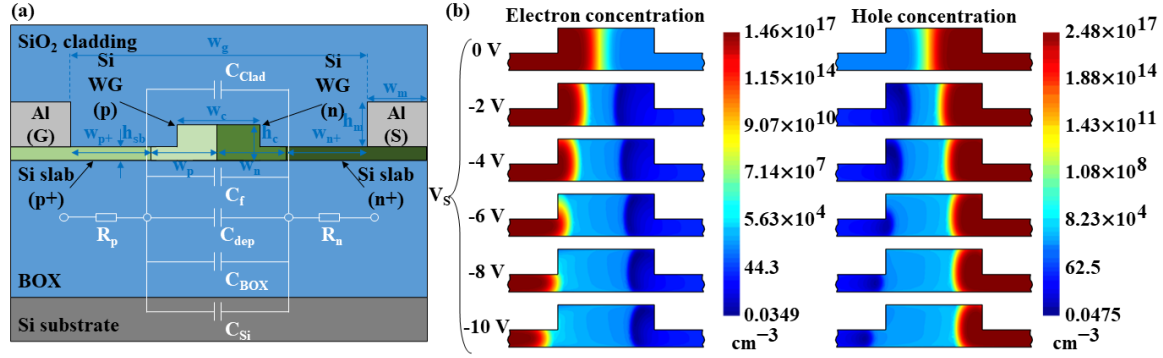


Figure 2.2: (a) Cross-section of a lateral PN diode rib waveguide-based optical phase modulator with an equivalent circuit. (b) Change of electron and hole concentrations in the waveguide region because of different applied reverse bias voltages. For Lumerical Device simulations $w_c = 500$ nm, $w_p = w_n = 1.3$ μm , $w_{p+} = w_{n+} = 3.15$ μm , $w_g = 8.4$ μm , $w_m = 7.3$ μm , $h_c = 220$ nm, $h_{\text{sb}} = 90$ nm and $h_m = 750$ nm are considered.

The external voltage or the RF modulation signal is fed to the waveguide using highly doped silicon slabs which introduce resistive components (R_p contributed by the P+ doped slab and R_n contributed by the N+ doped slab) in the cross-sectional equivalent circuit of the modulator shown in Fig. 2.2a. The PN junction capacitance or the depletion region capacitance and the capacitance due the fringing electrical fields at the waveguide corners are expressed by the terms C_{dep} and C_f respectively. The other capacitances parallel to the junction capacitance are the capacitances contributed by the upper silicon dioxide cladding, C_{clad} , by the BOX layer below the silicon device layer, C_{BOX} , and by the silicon substrate layer, C_{Si} . All these resistances and capacitances together limit the 3-dB bandwidth ($f_{3\text{dB}}$) of the modulator to ~ 41 GHz [17] because of the RC time constant ($f_{3\text{dB}} \propto 1/2\pi R_{\text{eq}} C_{\text{eq}}$) where $R_{\text{eq}} = R_p + R_n$ and $C_{\text{eq}} = C_{\text{dep}} + C_f + C_{\text{clad}} + C_{\text{BOX}} + C_{\text{Si}}$ [52]. Since the the PN diode is driven through ohmic contacts or through the highly doped silicon striploads to modulate the optical mode, we can address this modulator as the resistively coupled silicon (RC-Si) modulator.

2.1.4 Capacitively Coupled Silicon (CC-Si) Modulator

In Chapter 1 I have already discussed about the bandwidth enhancement of a conventional SOH (RC-SOH) modulator by replacing the highly doped silicon striploads or slabs with a capacitively material [12, 22, 23, 24]. Fig 1.1d and 1.1e show the cross-sectional views as well as the cross-sectional RF equivalent circuits of the RC-SOH and CC-SOH modulators. The conventional PN diode-based silicon modulator has a very similar cross-sectional equivalent circuit like the RC-SOH modulators which is depicted in Fig. 2.2a. Similar to the RC-SOH modulator, the RC-Si modulator has a limited bandwidth up to 41 GHz because of a non-zero RC time constant [17]. Therefore, a new modulator structure is proposed where the highly doped silicon slabs & the silicon dioxide

cladding will be replaced by a capacitive cladding and the metal layer is placed directly on the BOX layer as depicted in Fig. 2.3a. For capacitive cladding I propose a material with very high dielectric constant (high-k material) to ensure high electrical field confinement inside the waveguide region compared to the cladding region and with low refractive index at 1550 nm wavelength to ensure optical mode confinement in the waveguide core. Barium titanate (BaTiO_3 or BTO) is a material that has such properties with dielectric constant, $\epsilon_{\text{BTO}} \approx 40$ in the RF regime and with refractive index, $n_{\text{BTO}} \approx 1.85$ near 1550 nm wavelength [22]. The waveguide looks like a strip waveguide with a lateral PN diode which has a fundamental TE mode like Fig. 2.1b. In the cross-sectional equivalent circuit of this modulator instead of R_n & R_p we have $C_{\text{hk},s}$ which accounts for the series capacitance and instead of C_{clad} we have $C_{\text{hk},p}$ which accounts for the parallel capacitance due to the BTO cladding. Since this modulator does not have any resistance in its cross-sectional circuit, it fundamentally does not have any RC time limitation. Thus, the $f_{3\text{dB}}$ is only limited by the microwave propagation loss, impedance mismatch between the modulator & the external RF signal driver, and the velocity mismatch between the optical & RF modes.

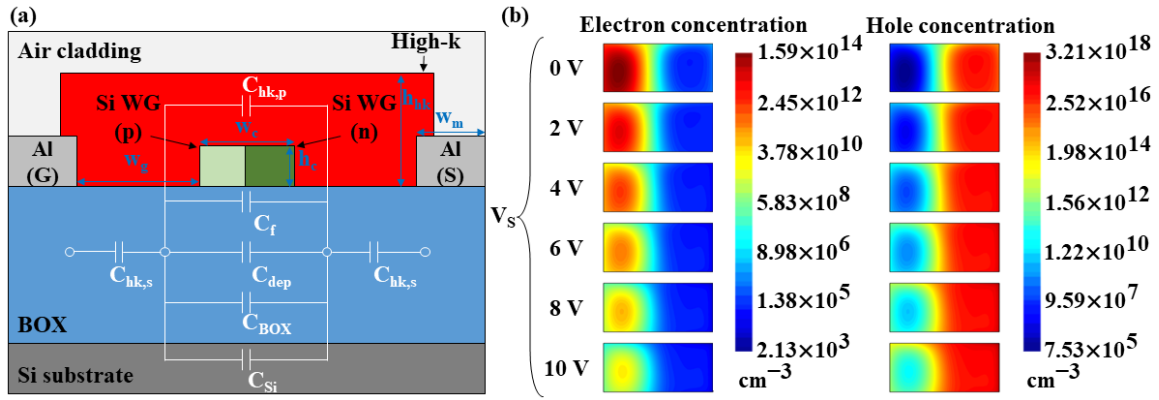


Figure 2.3: (a) Cross-section of a CC-Si optical phase modulator with an equivalent circuit. (b) Change of electron and hole concentrations in the waveguide region because of different applied reverse bias voltages. For Lumerical Device simulations $w_c = 500 \text{ nm}$, $w_g = 3.95 \mu\text{m}$, $w_g = 7.3 \mu\text{m}$, $h_c = 220 \text{ nm}$, $h_{\text{hk}} = 400 \text{ nm}$, and $h_m = 750 \text{ nm}$ are considered.

To have a quantitative idea of the change of n_{eff} and the change of optical loss (α_{opt}) for different reverse bias voltages, electrostatic simulations were done in Lumerical Device for a device structure shown in Fig. 2.3a. Almost all the dimensions and the doping concentrations remain the same as the simulation which was done for the RC-Si modulator shown in Fig. 2.2a apart from the thickness of the BTO ($h_{\text{hk}} = 400 \text{ nm}$). The change of electron and hole concentrations are shown in Fig. 2.3b when the reverse bias signal voltage (V_s) is swept from 0 to 10 V. Fig. 2.4a and 2.4b show the change of effective refractive index, Δn_{eff} and the optical loss of the waveguide due to free carriers, α_{opt} respectively with respect to the reverse bias signal voltage, V_s for both RC-Si and CC-Si modulators. Although higher bandwidth is expected for the CC-Si modulators compared to the RC-Si modulators, the effective refractive index changes at a much lower rate with respect to V_s in case of CC-Si modulator compared to the RC-Si modulator. As a result, whereas a $V_{\pi}L$ of $\sim 4.8 \text{ V.cm}$ is calculated for a straight waveguide RC-Si phase modulator, a $V_{\pi}L$ as high as $\sim 17.7 \text{ V.cm}$ is expected for a CC-Si phase modulator. But a lower $V_{\pi}L$ is achievable by considering a lower gap between the metal layer and the waveguide, w_g that does not significantly compromise the optical

loss due to metal absorption.

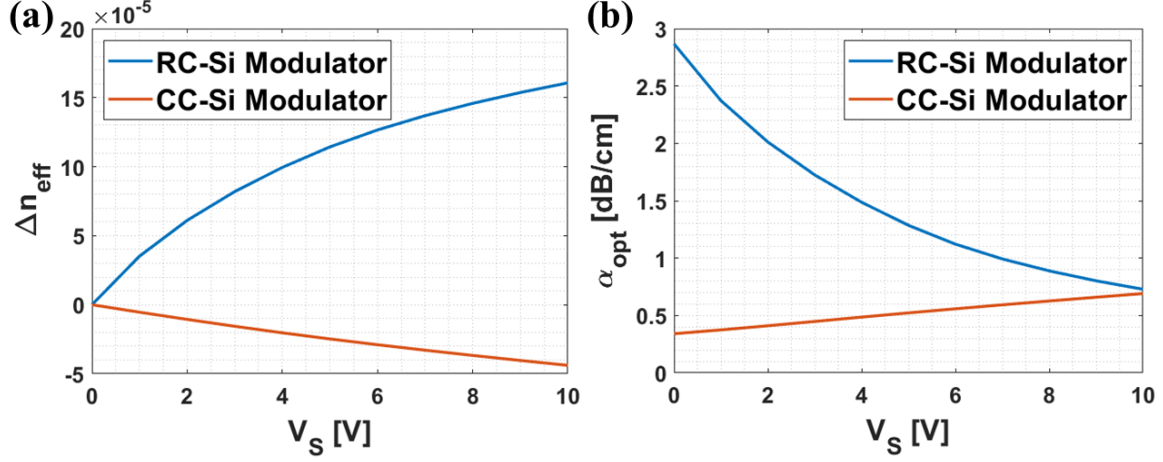


Figure 2.4: The change of (a) Δn_{eff} and (b) α_{opt} with respect to the change of reverse bias signal voltage, V_S .

2.1.5 MZI-Based Silicon Modulators

Section 2.1.3 and 2.1.4 describe how a straight waveguide can work as an optical phase shifter. These phase shifters are placed in one or both arms of an MZI to achieve intensity modulation. Therefore, this section provides a concise overview of the silicon modulator based on the MZI configuration. An MZI uses an optical coupler to combine one or more input lights waves into two optical waves and uses another optical coupler to recombine these two optical waves to one or more output ports. Depending on the number of input and output waves an $M \times N$ MZI is defined where M is the number of input waves and N is the number of output waves. We will focus more on a 1×2 MZI like Fig. 2.5a since this configuration was used in our experiments. We will use the transfer matrix model (TMM) where each sub-component of the 1×2 MZI can be represented by a matrix and the final output waves are determined by multiplying the individual matrices. In a 1×2 MZI the input optical wave, E_I is split into two waves, E_{A1} (through Arm-1) and E_{A2} (through Arm-2). Thus, the matrix formulation of E_{A1} and E_{A2} becomes,

$$\begin{bmatrix} E_{A1} \\ E_{A2} \end{bmatrix} = \begin{bmatrix} \sqrt{\epsilon_1} \\ \sqrt{\epsilon_2} \end{bmatrix} \begin{bmatrix} E_I \end{bmatrix} \quad (2.14)$$

Here, ϵ_1 and ϵ_2 are the output power ratio of Arm-1 and Arm-2 respectively. Generally, equal power splitting is preferred and for such lossless power splitters $\epsilon_1 = \epsilon_2 = 0.5$. In our devices I have used a 1×2 multimode interferometers (MMI) coupler to split the optical power equally in two arms. The propagation in the arms of the MZI is modelled by modifying the amplitude and the phase of E_{A1} and E_{A2} .

$$\begin{bmatrix} E'_{A1} \\ E'_{A2} \end{bmatrix} = \begin{bmatrix} \exp(-j\phi_1 - \frac{\alpha_1}{2}L_1) & 0 \\ 0 & \exp(-j\phi_2 - \frac{\alpha_2}{2}L_2) \end{bmatrix} \begin{bmatrix} E_{A1} \\ E_{A2} \end{bmatrix} \quad (2.15)$$

Here, φ_1 & φ_2 are the total phase shifts in Arm-1 and Arm-2 respectively, α_1 & α_2 are the optical propagation loss in the respective arms, L_1 & L_2 are the total lengths of each arm, and E'_{A1} & E'_{A2} are the electrical fields after the phase shifting region of Arm-1 & Arm-2 respectively.

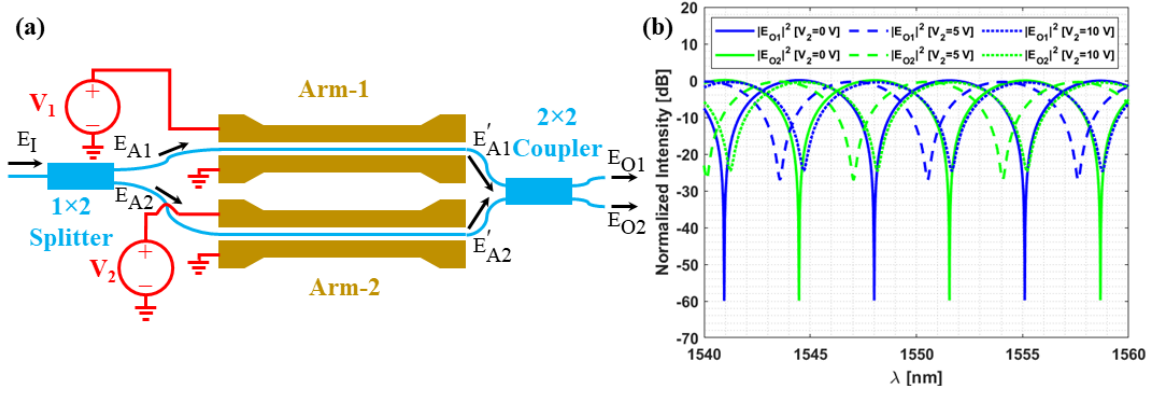


Figure 2.5: (a) A 1×2 unbalanced MZI modulator where the Arm-1 and Arm-2 of the interferometer are driven by the external modulation direct current (DC) voltage sources V_1 and V_2 respectively. (b) Output optical intensity spectra ($I_{O1} = |E_{O1}|^2$ & $I_{O2} = |E_{O2}|^2$) of the 1×2 unbalanced MZI when only one arm of the interferometer is driven by different direct current (DC) bias for $L_{\text{active1}} = 5$ mm).

A relative phase differences between the two paths of a silicon MZI can be introduced by exploiting the plasma dispersion effect. Thus, the total phase shift for Arm-1 and Arm-2 can be expressed as,

$$\varphi_1 = \frac{2\pi}{\lambda} [n_{\text{eff}} L_{\text{nm1}} + n_{\text{eff}}(V_1) L_{\text{active1}}] + j \frac{\alpha_{\text{eff}}(V_1)}{2} L_{\text{active1}} \quad (2.16)$$

$$\varphi_2 = \frac{2\pi}{\lambda} [n_{\text{eff}} L_{\text{nm2}} + n_{\text{eff}}(V_2) L_{\text{active2}}] + j \frac{\alpha_{\text{eff}}(V_2)}{2} L_{\text{active2}} \quad (2.17)$$

Here, the total lengths of each arm, L_1 & L_2 can be broken down in to two parts – L_{nm1} & L_{nm2} which are the lengths of the passive parts of each arm, and L_{active1} & L_{active2} which are the lengths of the active parts of each arm where modulations take place. The imaginary parts of the phase shifts in two arms originate from amplitude-phase coupling which causes unintentional modulation of loss, α_{eff} because of the modulation voltages, V_1 and V_2 in the active waveguide regions of Arm-1 and Arm-2 respectively. For an unbalanced MZI L_{nm1} and L_{nm2} are intentionally kept different ($\Delta L = L_2 - L_1 = L_{\text{nm2}} - L_{\text{nm1}}$) and L_{active1} & L_{active2} are kept equal to create a measurable free spectral range (FSR) which facilitates phase shift measurements. Finally, the optical signals from two arms, E'_{A1} and E'_{A2} are combined together using a 2×2 optical coupler and gives the output signals,

$$\begin{bmatrix} E_{O1} \\ E_{O2} \end{bmatrix} = \begin{bmatrix} \sqrt{1-\epsilon} & j\sqrt{\epsilon} \\ j\sqrt{\epsilon} & \sqrt{1-\epsilon} \end{bmatrix} \begin{bmatrix} E'_{A1} \\ E'_{A2} \end{bmatrix} \quad (2.18)$$

Where, ϵ is the power splitting ratio between the two arms of the optical coupler. The imaginary unit j represents the $\frac{\pi}{2}$ phase shift between the direct and cross coupled inputs. Again, generally equal coupling is used in such optical couplers i.e., $\epsilon = 0.5$. In our devices I have used a 2×2 MMI coupler to combine the light in two arms after phase shifting. Fig. 2.5b shows the intensity spectrum for both outputs when the optical phase of only Arm-2 is modulated by different reverse bias voltages

while maintaining zero bias voltage at Arm-1 ($V_1 = 0$ V). The V_π of the phase shifter arm (Arm-2) can be determined from Fig. 2.5b by considering the voltage raise to drive the MZI from the maximum output optical power to minimum output optical power or vice-versa.

2.1.6 Driving Configurations of Silicon Modulators

For fast EO response the silicon modulators need to operate in reverse bias. Therefore, to maintain the reverse bias condition of the PN diode-based silicon MZI modulators the N-type terminal needs to maintain higher potential compared to the compared to the P-type terminal. But even if no direct current (DC) bias is applied, for an RF sinusoidal signal at least half-period of a complete wavelength the PN diode will be in the reverse bias regime. In such case, the overall bandwidth of the modulator will be less than a modulator biased with a DC reverse bias [17]. Moreover, unlike the lithium niobate modulators, the silicon modulator can generate only negative phase shift in an optical signal. Therefore, if both arms of an MZI modulator needs to be driven, a differential phase shift needs to be applied between two arms of an MZI. Thus, there are three main configurations by which the MZI-based silicon modulator can be driven – single drive, dual drive, and series push-pull configurations [53].

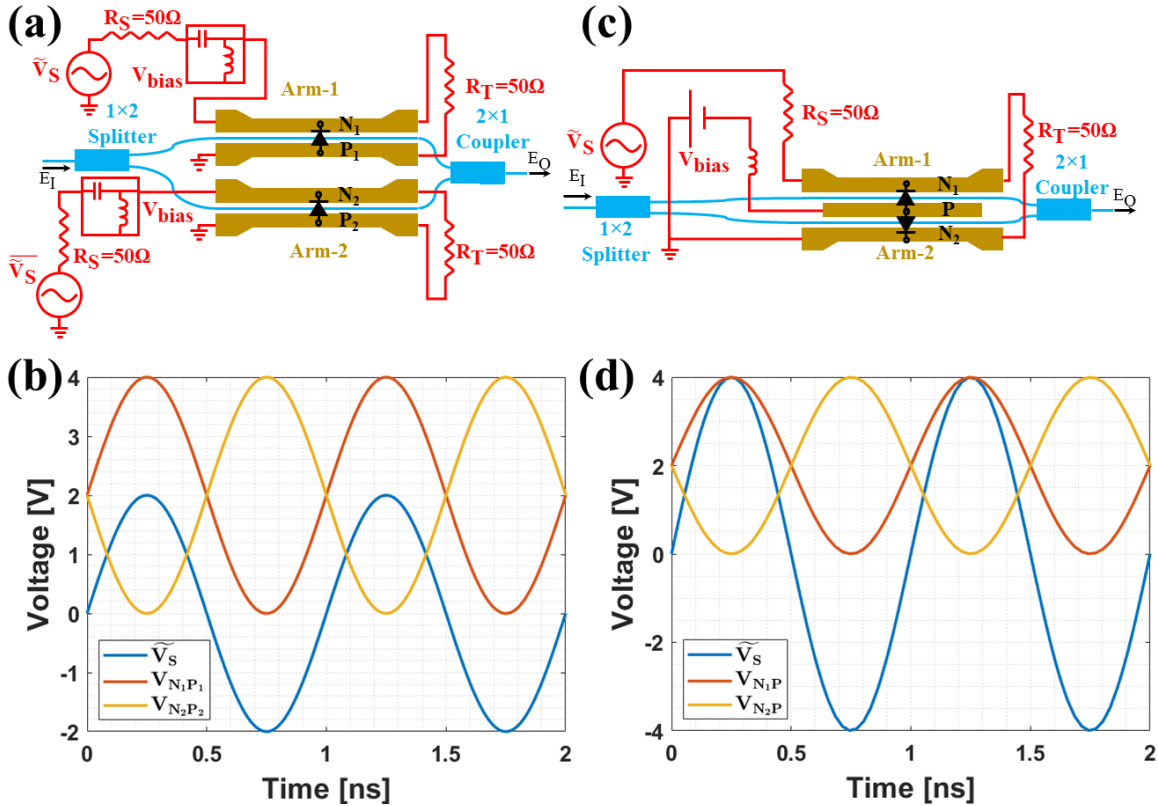


Figure 2.6: (a) Schematic diagram of a dual drive modulator. (b) Waveform of the potential difference between the upper (driven) electrode and the lower electrode (ground) of each arm of the MZI in a dual drive connection for a source RF source voltage with 4 V amplitude and 4 V DC bias. (c) Schematic diagram of a SPP modulator. (d) Waveform of potential differences across the first and second arms of the MZI in an SPP connection for a source RF source voltage with 4 V amplitude and 2 V DC bias.

2.1.6.1 Single Drive and Dual Drive Modulators

Fig. 2.6a shows a schematic diagram of a dual drive modulator. In such case, a bias tee is required to connect the DC bias in series with the RF signal. This is how by applying proper DC bias voltage the N-type terminal of the modulator can always be set at a higher potential. If a source RF signal, $\tilde{V}_S$ is applied at one arm of the MZI, a complementary RF signal, $\tilde{V}_S$ is applied at the other arm. Thus, compared to one arm (with $\tilde{V}_S$) a negative effective refractive index change can be realized at the other arm (with $\tilde{V}_S$). Figure 2.6b illustrates the waveform of the source RF voltage, denoted as $\tilde{V}_S$. It also depicts the potential differences between the upper and lower electrodes of the upper arm ($V_{N_1P_1}$) and the lower arm ($V_{N_2P_2}$) of the MZI, respectively. In case of a single drive modulator only one arm of the MZI is driven.

2.1.6.2 Series Push-Pull Modulators

Fig. 2.6b shows a schematic diagram of an SPP modulator. In such case, the P-type terminals of two arms of the MZI are connected and the RF source signal, $\tilde{V}_S$ is applied between the N-terminals of the two MZI arms. The common P-terminal is biased with a negative DC voltage. Thus, a differential push-pull configuration is realized with only a single driving voltage. Typically, the DC bias amplitude is half of the RF signal amplitude. The waveform of the source RF voltage $\tilde{V}_S$, potential differences across the first arm of the MZI, V_{N_1P} and the second arm of the MZI, V_{N_2P} are shown in Fig. 2.6d.

2.1.7 MZI-Based Modulator Characterization

It has been already mentioned in Chapter 1 that the three major performance parameters of a modulator are the halfwave voltage (V_π), the optical propagation loss, and the 3-dB EO bandwidth (f_{3dB}). In this section, we will specially focus on the theories behind the V_π estimation method by measuring optical sideband power using an optical spectrum analyzer (OSA) and the f_{3dB} estimation method by measuring the electrical S-parameter using a vector network analyzer (VNA).

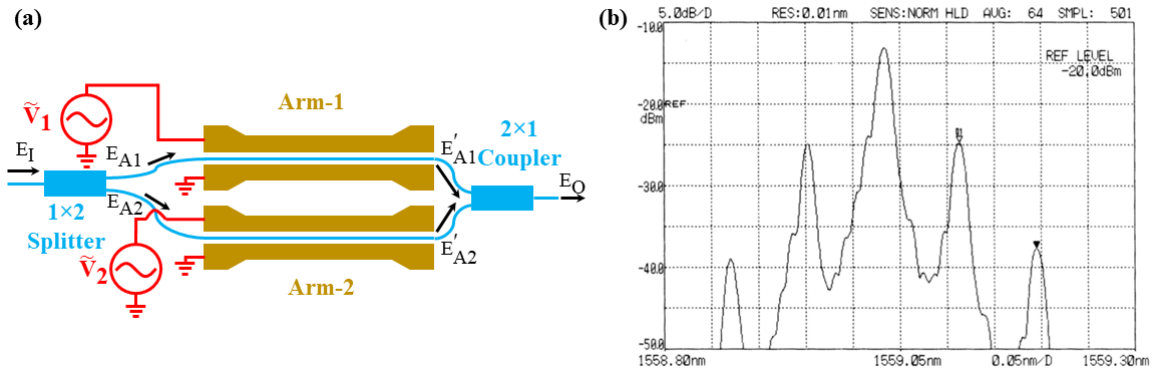


Figure 2.7: (a) A 1×1 unbalanced MZI modulator where the Arm-1 and Arm-2 of the interferometer are driven by the external modulation RF voltage sources $\tilde{V}_1$ and $\tilde{V}_2$ respectively. (b) Output optical intensity spectrum ($I_O = |E_O|^2$) of the 1×1 unbalanced MZI when only one arm of the interferometer is driven by an RF signal [54].

2.1.7.1 Half-Wave Voltage (V_π) Estimation

In order to do understand the V_π measurement method using an OSA we will consider 1×1 unbalance MZI modulator like Fig. 2.7a. If we consider the input electrical field as E_I , then the output electrical field, E_O of such an MZI modulator can be expressed as [54],

$$E_O = \frac{E_I}{2} \exp(j\omega_0 t) [\exp(jm_1 \cos \omega_m t) + \exp(jm_2 \cos \omega_m t + j\phi)] \quad (2.19)$$

Here, ω_0 is the optical carrier frequency, which is usually near 193 THz (or 1550 nm wavelength), ω_m is the frequency of the applied external modulation signal, m_1 & m_2 (modulation indices) are phase modulation amplitudes in MZI Arm-1 & Arm-2 respectively, and ϕ is the constant phase difference between the two MZI arms because of the length imbalance between Arm-1 & Arm-2. After propagating through the 1×1 MZI modulator the driving RF signal perturb both the optical intensity and the phase of the output optical wave. In frequency domain, Eq. 2.19 can be expressed by the following equation using the Jacobian expansion,

$$E_O = \frac{E_I}{2} \exp(j\omega_0 t) \sum_{k=-\infty}^{\infty} (j)^k [J_k(m_1) + J_k(m_2) \exp(j\phi)] \exp(jk\omega_m t) \quad (2.20)$$

Here, J_k is the k^{th} order Bessel function of first kind. Eq. 2.20 shows that frequency response of the output optical signal has a carrier frequency of ω_0 which is accompanied by infinite number of sidebands with $\omega_0 + k\omega_m$ frequencies. The generalized equation of the intensity of any sideband at the output of such MZI can be expressed by,

$$I_O(\omega_0 + k\omega_m) = \frac{I_I}{4} [J_k^2(m_1) + J_k^2(m_2) + 2J_k(m_1)J_k(m_2)\cos\phi] \quad (2.21)$$

Here, $I_I = |E_I|^2$ and $I_O = |E_O|^2$ are the intensities of the input and output optical signals of the unbalanced MZI modulator. If we drive only one arm of the MZI i.e., $m_1 = m$ & $m_2 = 0$ then the expression of sideband intensity becomes,

$$I_O(\omega_0 + k\omega_m) = \frac{I_I}{4} \begin{cases} [1 + J_0^2(m) + 2J_0(m)\cos\phi] & \text{for } k=0 \\ J_k^2(m) & \text{otherwise} \end{cases} \quad (2.22)$$

By selecting a carrier frequency, ω_0 to operate the MZI modulator in quadrature point we can make $\cos\phi = 0$. But to avoid error the intensity ratio between the first and second order sideband intensity ratio is considered instead of the carrier and first order sideband ration to extract the numerical value of the modulation index, m . This ratio is expressed by the term $R_{2,1}$ (instead of $R_{1,0}$) and by the following equation,

$$R_{2,1} = \frac{J_2^2(m)}{J_1^2(m)} = \frac{I_O(\omega_0 + 2\omega_m)}{I_O(\omega_0 + \omega_m)} \quad (2.23)$$

Fig. 2.7b shows the intensity spectrum of the output optical beam of the unbalanced MZI modulator when only one arm of the interferometer is driven by the external RF voltage source with $f_m = \omega_m/2\pi = 10$ GHz. The peak intensities at the carrier frequency, $I_O(\omega_0)$, at the first order sidebands, $I_O(\omega_0 \pm \omega_0)$, and at the second order sidebands, $I_O(\omega_0 \pm 2\omega_0)$ are clearly visible in Fig. 2.7b [54]. However, in many cases the first order sideband power is very low, and the second order side-

band power cannot be detected by the spectrum analyzer. In such cases, $R_{1,0}$ (shown in Eq. 2.24) is used and ϕ needs to be estimated from the modulator's transmission spectra.

$$R_{1,0} = \frac{J_1^2(m)}{1 + J_0^2(m) + 2J_0(m) \cos\phi} = \frac{I_O(\omega_0 + \omega_m)}{I_O(\omega_0)} \quad (2.24)$$

After the calculation of m , if the peak amplitude of the external modulation is V_m , then the V_π of the device can be calculated using the following equation,

$$V_m = \frac{m}{\pi} V_\pi \quad (2.25)$$

2.1.7.2 Bandwidth (f_{3dB}) Estimation

The total of phase shift experienced by an optical wave in the active region of one MZI arm is directly proportional to the voltage applied across the silicon waveguide with a proportionality constant containing different parameters like the optical carrier wavelength, waveguide dimensions, refractive index, overlap between the optical and the RF modes, and change of effective refractive index due to plasma dispersion effect. The external electrical modulation signal, $\tilde{V}_m = V_m \cos \omega_m t$ that modulates the carrier concentration i.e., the effective refractive index of the silicon waveguide is fed through the electrical transmission lines. The most used transmission lines are the coplanar striplines or the ground-signal (GS) transmission lines and the coplanar waveguides or the ground-signal-ground (GSG) transmission lines. For kHz or MHz range $f_m = \omega_m/2\pi$ the transmission line can be considered as a lumped element. But for GHz range f_m the applied voltage is of traveling-wave nature which is governed by the Telegrapher's equations [55]. In this case, the total amount of phase shift achieved by the optical wave is the cumulative of all phase shifts induced by the instantaneous voltage seen by the optical wave at a specific position on the transmission line. Fig. 2.8a shows the equivalent circuit model a travelling wave modulator and Fig. 2.8b shows the equivalent circuit model of the transmission line part of the travelling wave electrode. The values of R , L , G , and C depend on the geometry of the transmission line and the electrical properties of the conductors and dielectrics. The shunt conductance per unit length, G and the shunt capacitance per unit length, C originate from the cross-sectional equivalent circuit shown in Fig. 2.2a. The series line resistance per unit length, R mainly comes from the skin effect & the ohmic loss of the conductor, and the series line inductance per unit length, L mainly comes from the magnetic field around the conductors & self-inductance.

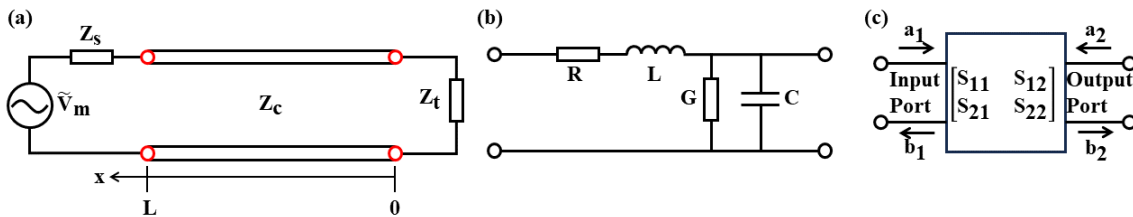


Figure 2.8: The equivalent circuit models of (a) a travelling-wave modulator with a length, L & a characteristic impedance, Z_c , and (b) the transmission line with per unit length impedances R , L , G & C . (c) Two port network equivalent of a transmission line.

The instantaneous voltage at a particular point on the transmission line is expressed by the following

set of equations [56],

$$V(x, \omega_m) = \frac{V_m}{2} (1 + \rho_1) \exp(-jB_o L) \left[\frac{\exp\{-j(B_e - B_o)x\} + \rho_1 \exp\{j(B_o + B_e)x\}}{\exp(-jB_e L) + \rho_1 \rho_2 \exp(jB_e L)} \right] \quad (2.26)$$

$$\rho_1 = \frac{Z_c - Z_s}{Z_c + Z_s} \quad (2.27)$$

$$\rho_2 = \frac{Z_t - Z_c}{Z_t + Z_c} \quad (2.28)$$

$$B_o = \frac{2\pi f_m}{c} n_{g,opt} \quad (2.29)$$

$$B_e = \frac{2\pi f_m}{c} n_{e,RF} + j\alpha_m \quad (2.30)$$

In these relations, Z_c is the characteristic impedance of the transmission line, Z_s & Z_t are the source & termination impedances respectively, $n_{e,RF}$ & α_m denote the amplitude attenuation constant & the effective index of the RF modulation signal, and $n_{g,opt}$ is the optical effective group refractive index. To simplify the analysis, we considered the average voltage along the length, L of the modulator [56],

$$\begin{aligned} V_{avg}(\omega_m) &= \frac{1}{L} \int_0^L V(x, \omega_m) dx \\ &= \frac{V_m (1 + \rho_1) \exp(-jB_o L)}{2 [(e(-jB_e L) + \rho_1 \rho_2 \exp(-jB_e L))] (U_+ + \rho_2 U_-)} \end{aligned} \quad (2.31)$$

Here,

$$V_{\pm} = \exp(\pm j\phi_{\pm}) \frac{\sin(\phi_{\pm})}{\phi_{\pm}} \quad \& \quad \phi_{\pm} = \frac{(B_e \mp B_o) L}{2} \quad (2.32)$$

It is worth mentioning that Eq. 2.32 accounts for the impact of the impedance mismatch of the modulator transmission line at the source end & at the termination end, the RF propagation loss, as well as the velocity mismatch between the optical and the RF waves of the traveling-wave modulator. The EO frequency response of the modulator can be written as,

$$m(\omega_m) = \left| \frac{V_{avg}(\omega_m)}{V_{avg}(0)} \right| \quad (2.33)$$

Here, $V_{avg}(0) = V_m/2$ is the DC potential difference across the transmission line. The expressions like $n_{e,RF}$, α_m , Z_c etc. can be estimated from the electrical S-parameter of the transmission line of the modulator which can be easily measured using a VNA. The transmission line can be considered as a two port networks like Fig. 2.8c and the microwave frequency dependent S-parameter of this network can be expressed by the following equations,

$$S_{11} = \frac{b_1}{a_1} \Big|_{a_2=0} \quad S_{12} = \frac{b_1}{a_2} \Big|_{a_1=0} \quad S_{21} = \frac{b_2}{a_1} \Big|_{a_2=0} \quad S_{22} = \frac{b_2}{a_2} \Big|_{a_1=0} \quad (2.34)$$

Here, a_1 & b_1 are the transmitted and reflected waves seen at the input port and a_2 & b_2 are the transmitted and reflected waves seen at the output port. For a symmetric structure like the transmission line of the modulator, the complex reflection coefficients, $S_{11} = S_{22}$, and the complex transmission coefficients, $S_{21} = S_{12}$. The S-parameter of such symmetrical structure can be converted

to ABCD parameters which ultimately lead to following set of equations [57],

$$Z_c = Z_0 \sqrt{\frac{(1 + S_{11})^2 - S_{21}^2}{(1 - S_{11})^2 - S_{21}^2}} \quad (2.35)$$

$$\gamma = \frac{1}{L} \ln \left(\frac{1 - S_{11}^2 + S_{21}^2}{2S_{21}} \pm K \right)^2 = \alpha_m + j\beta \quad (2.36)$$

$$\beta = \frac{2\pi f_m}{c} n_{e,RF} \quad (2.37)$$

$$K = \left\{ \frac{(S_{11}^2 - S_{21}^2 + 1)^2 - (2S_{11})^2}{(2S_{21})^2} \right\}^{\frac{1}{2}} \quad (2.38)$$

In these relations, Z_0 can be considered as 50Ω if proper calibration is performed for the desired microwave frequency range to account for the losses generated by the VNA and the connecting coaxial cables, γ is the complex propagation constant which determines the microwave propagation loss, α_m and the microwave effective refractive index, $n_{e,RF}$. The relations in Eq. 2.34-2.38 can be used to estimate the analytic EO frequency response, $m(\omega_m)$ of the modulator mentioned in Eq. 2.33.

2.1.8 Commercial SOI Foundry Platform

To demonstrate a silicon modulator, we not only require the active waveguide region with the lateral PN diode structure, but also, we require a wide variety of different active and passive structures like the bent strip waveguides, strip-to-rib waveguide mode converters, heaters for phase shifting, MMI, directional couplers, Y-junctions etc. Hence, like the integrated circuit (IC) foundries in the electronics industry, silicon photonic foundries such as, Advanced Micro Foundry (AMF) in Singapore, American Institute for Manufacturing Integrated (AIM) Photonics & Global Foundries in the U.S., Information Technology Electronics Laboratory (CEA Leti) in France, Interuniversity Microelectronics Centre (IMEC) in Belgium etc. have emerged for wafer-scale prototyping and

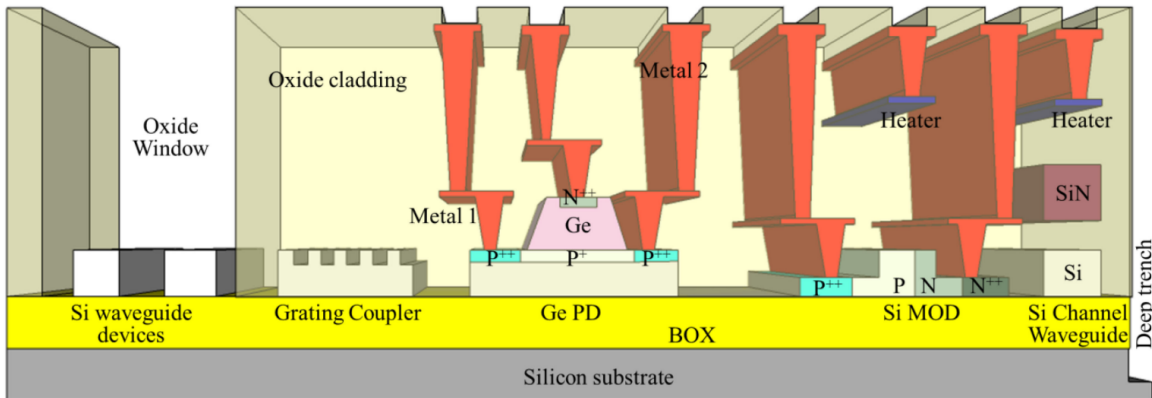


Figure 2.9: A cross-sectional view of the standard MPW run of Advanced Micro Foundry (AMF). Some common elements are like heater-based phase shifters, modulators, photodiodes, grating couplers etc. are shown. The figure was reproduced from Ref. [20].

manufacturing. These foundries often offer MPW runs for multiple users on the same 300 mm SOI wafer to minimize the cost. To reduce the number MPW runs for to optimize a PIC for a specific purpose or a particular device these foundries also offer process design kits (PDK) which include the optimized designs of some commonly used active and passive devices. For some passive devices like the wavelength division multiplexers (WDM) a lower refractive index contrast is desired which brings silicon nitride (SiN) in the picture to develop the SiN-on-SOI platform. SiN also offers solutions for passive devices in the visible wavelengths where silicon has higher optical loss. Also, since as a material silicon is not very good for photodetectors, epitaxially lattice-matched germanium (Ge) photodiodes are also offered on this platform. A cross-sectional view of such platform offered by AMF is shown in Fig. 2.9 [20]. Although SiN and Ge layers can add many nice features to the SOI-based PICs, these layers were not used while fabricating the modulators.

2.2 Experimental Demonstrations of the CC-Si Modulators

To demonstrate the proof of concept of the CC-Si modulator the devices were fabricated in the MPW run of AMF. A CC-Si modulator with silicon dioxide cladding was fabricated which is fully compatible with the available processes offered by the foundry. I also have fabricated some strip waveguides with lateral PN diode structures and fabricated the CC-Si modulator with BTO cladding after several BEOL post-fabrication processes. In this section, I will also discuss about the BEOL post-fabrication processes of CC-Si modulator with BTO cladding and the characterization methods of both modulators.

2.2.1 Single Drive CC-Si Modulator with Silicon Dioxide Cladding

A CC-Si modulator with silicon dioxide cladding was fabricated on the standard SOI platform with 220 nm thick silicon device layer in the MPW run of AMF. A similar type of device with a different structure was reported before [58]. To the best of my knowledge, this is the first experimental demonstration of a CC-Si modulator or a silicon modulator using a strip waveguide. This section is started by describing the device structure and modulation mechanism of our proposed structure. In a proof-of-concept experiment, the plasma dispersion effects were observed in the proposed waveguide structure by applying sinusoidal RF signals of different frequencies at one arm of an MZI structure and generated sidebands up to 50 MHz around the CW optical carrier signal. A $V_{\pi}L$ of ~ 9.4 V.cm is estimated for an electrical modulation signal of 20 MHz using a self-heterodyne measurement system [59]. I also demonstrated the EO measurements for high frequency RF signals. I was able to generate first order sidebands up to 68 GHz and observed an almost flat frequency response [60]. Later, an f_{3dB} of ~ 9.3 GHz has been estimated from the measured electrical S-parameters of the modulator transmission line. By careful optimization of different parameters, the f_{3dB} of this device can be enhanced beyond 40 GHz [61].

2.2.1.1 Device Structure and Electro-Optic Simulation

The cross-sectional view of a CC-Si modulator with silicon dioxide cladding is shown in Fig. 2.10a. A 650 nm wide lateral lightly doped silicon PN diode is used as a strip waveguide which supports only the fundamental TE optical mode. The PN junction is in the middle of the waveguide. A 90

nm (h_{Slab}) highly P-doped (P+) silicon slab was placed 500 nm away from the P side of the strip waveguide and a highly N-doped (N+) silicon slab with the same h_{Slab} was placed 500 nm away from the N side of the strip waveguide. A 500 nm gap between the strip waveguide and the slabs was used to avoid excess optical loss due to the high doping of the silicon slabs. Two aluminium metal layers Mt1 and Mt2 layers were used for routing both DC bias signals and RF signals. The two aluminium vias Via1 and Via2 establish a connection between the silicon slab and the Mt1 layers and between the Mt1 and the Mt2 layers respectively. All the layers are encapsulated in a $\sim 3 \mu\text{m}$ (h_{Clad}) silicon dioxide cladding layer apart from the bond pad opening layer areas which expose the Mt2 layer to air for applying DC or RF signals using picoprobes.

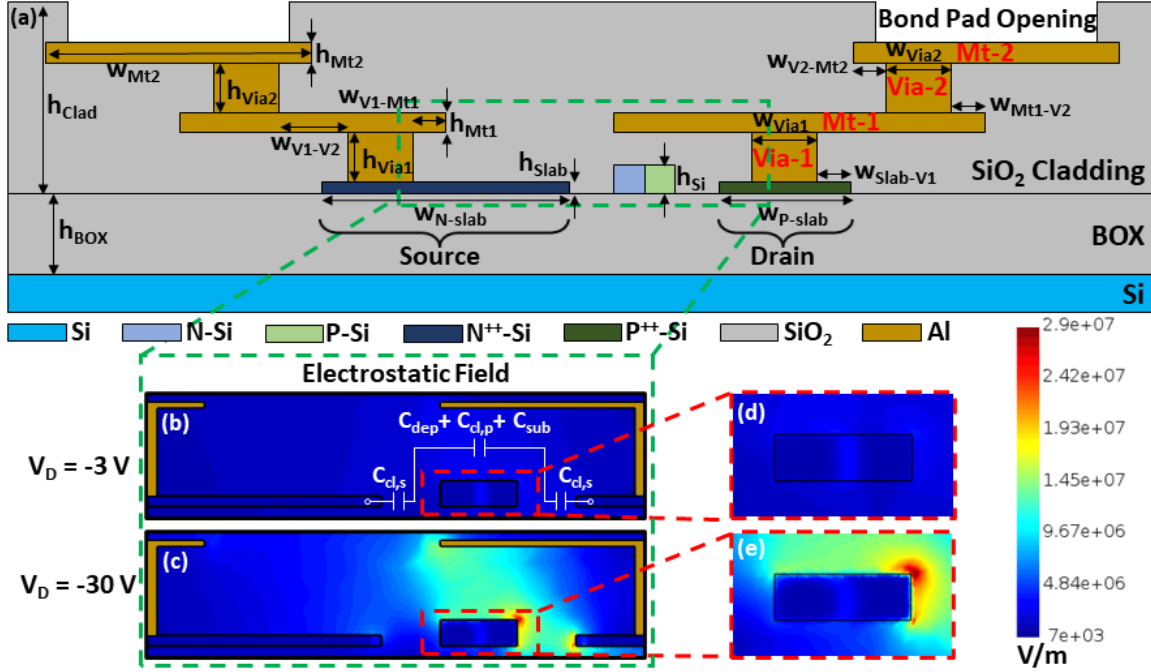


Figure 2.10: (a) Cross-sectional view of a PN diode strip waveguide-based silicon modulator. Lumerical Device simulations for the static electric field distribution for drain voltage, V_D of (b) -3 V and (c) -30 V respectively. C_{dep} , $C_{\text{cl,s}}$, $C_{\text{cl,p}}$ and C_{sub} of the RF equivalent circuit shown in (b) represent the depletion capacitance of the PN diode, the series component of the cladding capacitance, the parallel component of the cladding capacitance and the substrate capacitance respectively. Electrostatic field distribution close to the waveguide regions for V_D of (d) -3 V and (e) -30 V respectively.

For the purpose of simulation, I used layer thicknesses as: Mt1 thickness, $h_{\text{Mt1}} = 50 \text{ nm}$, Mt2 thickness, $h_{\text{Mt2}} = 100 \text{ nm}$, Via-1 depth, $h_{\text{Via1}} = 750 \text{ nm}$, and Via-2 depth, $h_{\text{Via2}} = 2 \mu\text{m}$. I also considered the P, N, P+ and N+ doping concentrations to be $\sim 3.5 \times 10^{17} \text{ cm}^{-3}$, $\sim 5 \times 10^{17} \text{ cm}^{-3}$, $\sim 1.8 \times 10^{20} \text{ cm}^{-3}$ and $\sim 8.7 \times 10^{20} \text{ cm}^{-3}$ respectively. The silicon slabs, metal layers and metal vias together form the GS transmission line for the electrical drive signal (DC or RF) to modulate the carrier concentrations in the strip waveguide. The N and P-type sides of the transmission line are considered as the source and the drain respectively. I used minimum feature size offset widths set by the foundry. These were $w_{\text{Via1}} = 1.5 \mu\text{m}$ (minimum width of Via1 layer), $w_{\text{Via2}} = 2 \mu\text{m}$ (minimum width of Via2 layer), $w_{\text{Slab-V1}} = 500 \text{ nm}$ (silicon slab-Via1 minimum offset width), $w_{\text{V1-Mt1}} = 400 \text{ nm}$ (Via1-Mt1 minimum offset width), $w_{\text{Mt1-V2}} = 1.5 \mu\text{m}$ (Mt1-Via2 minimum offset width) and $w_{\text{V2-Mt2}} = 1.5 \mu\text{m}$ (Via2-Mt2 minimum offset width). Unlike conventional silicon modulators, there

is no ohmic contact between the silicon waveguide core and the transmission line. Therefore, the carrier concentrations and distributions must be modulated by applying an electric field through a silicon dioxide capacitive barrier. The electric field inside the strip waveguide needs to be enhanced to efficiently modulate the carrier distribution i.e., n_{eff} of the fundamental optical TE mode [9]. To enhance the electric field inside the waveguide the Via1-Via2 offset width in both source and drain sides are considered sufficiently wide ($w_{V1-V2} = 12 \mu\text{m}$), so that the electric field is not confined between the Mt2 layers of the source and the drain. Additionally, the Mt1 layer of the drain side is extended above the strip waveguide and the Vial layer of the source side is pushed to the left sufficiently so that the electric field is not confined between the Mt1 layers of the source and the drain as well. Thus, I end up with a drain (P+) silicon slab width ($w_{P\text{-Slab}} = w_{V1} + 2w_{\text{Slab-V1}}$) of $2.5 \mu\text{m}$ and a wide source (N+) silicon slab width ($w_{N\text{-Slab}}$) of $3.9 \mu\text{m}$. Using a commercial microwave simulation tool CST Studio Suite, the Mt2 layer width (w_{Mt2}) is determined to be $99.5 \mu\text{m}$ to optimize the transmission line for 50Ω .

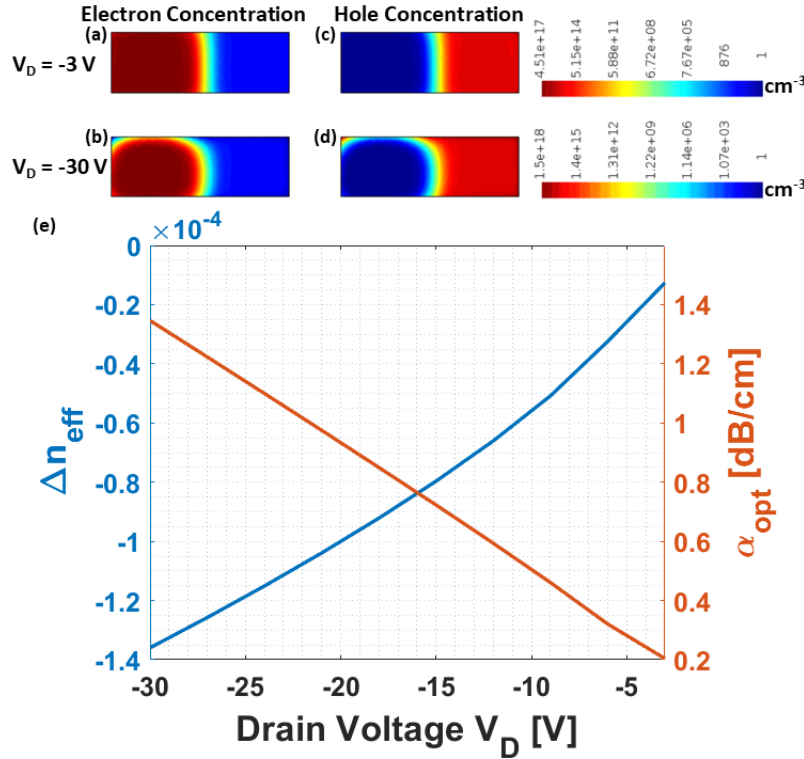


Figure 2.11: Electron concentrations for V_D of (a) -3 V and (b) -30 V respectively. Hole concentrations for V_D of (c) -3 V and (d) -30 V respectively. (e) Change of effective index (Δn_{eff}) and optical mode loss (α_{opt}) with respect to the DC bias voltage applied at the drain (V_D) while the source is grounded ($V_D = 0 \text{ V}$) calculated from the simulation results carried out in Lumerical Device and Mode Solutions.

For electrostatic simulations Lumerical Device was used. To ensure the reverse bias operation a negative DC bias voltage V_D is applied at the drain side whereas the source is always connected to the ground ($V_S = 0 \text{ V}$). Fig. 2.10b and 2.10c show electrical field distributions for V_D of -3 V and -30 V respectively. A better zoomed view of the electrical field distributions inside the waveguide for each V_D are shown in Fig. 2.10d and 2.10e respectively. From the electrostatic simulations we can

see that the change in electrical field inside the waveguide is quite small compared to the surrounding silicon dioxide cladding region since the dielectric constant of silicon dioxide (~ 3.9) [62] is significantly lower than that of silicon (~ 11.7) [63]. Fig. 2.11a & 2.11b show the electron concentrations and Fig. 2.11c & 2.11d show the hole concentrations in the waveguide region for the abovementioned V_D respectively. Since the change of electric field in the waveguide with respect to V_D is small, the change of electron and hole concentrations are also small in the waveguide. Lumerical Mode Solution tool has been used to quantify the change in effective refractive index (Δn_{eff}) of the fundamental optical TE mode because of the change of electron and hole concentrations in the waveguide i.e., the change of applied DC bias voltage at the drain. I exploited the Soref and Bennett Model [9] to calculate the Δn_{eff} and the optical mode loss (α_{opt}) with respect to V_D which is depicted in Fig. 2.11e. When V_D is reduced from -3 V to -30 V, we get Δn_{eff} of -1.23×10^{-4} which refers to a $V_{\pi}L$ of ~ 17 V.cm for a straight waveguide phase modulator.

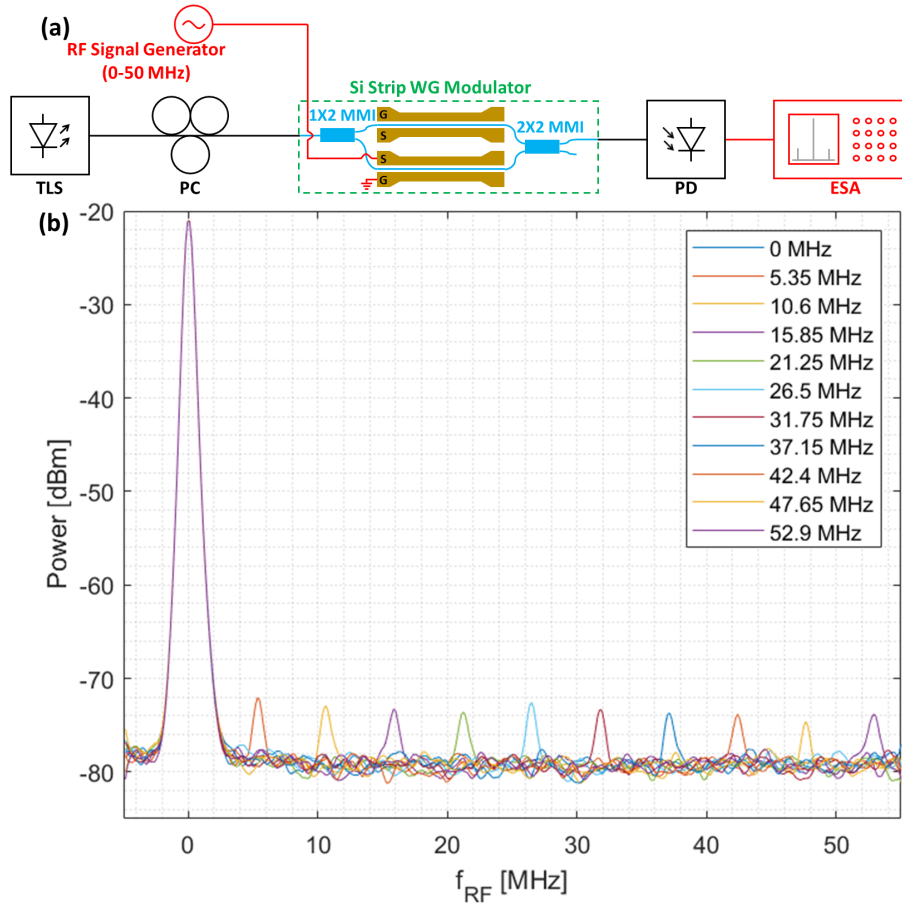


Figure 2.12: (a) Experimental setup for observing the EO response for the proposed modulator. (b) Power measured at the electrical spectrum analyzer (ESA) for modulation signal in the range of 5-50 MHz with an increment of 5 MHz for $V_{\text{pp}} = 5$ V.

2.2.1.2 Low Frequency Sideband Generation

To observe the carrier dispersion phenomena in a silicon strip waveguide-based optical modulator or the CC-Si modulator with silicon dioxide cladding an unbalanced MZI structure was used. The

experimental setup is shown in Fig. 2.12a. Both arms of the MZI can be separately biased by applying a modulation signal using a ground-signal-signal-ground (GSSG) RF picoprobe. The active length of both arms was 2 mm. Optical signals are coupled in and out of the chip using the grating couplers. A 1×2 MMI coupler is used to equally split the optical signal and after modulation a 2×2 MMI is used to combine the modulated optical signals from both MZI arms. I applied a modulation signal to one arm which was generated from a signal generator with a peak-to-peak voltage (V_{pp}) of 5 V. Since the maximum frequency of the modulation signal that I used was 50 MHz, a 2 mm device can be considered as a lumped element. Therefore, the other side of the arm was not terminated. I used a tunable laser source (TLS) to generate a CW optical carrier signal at a wavelength of 1552.95 nm with 0 dBm power to operate the modulator at the quadrature point. The optical carrier signal enters the chip after passing through a polarization controller (PC). Since a 50 MHz sideband cannot be seen in an OSA because of limited resolution, I detected the modulated optical signal at the output of the device using a high-speed photodiode (PD) followed by an electrical spectrum analyzer (ESA). I could generate sidebands up to 50 MHz starting from 5 MHz with a step size of 5 MHz. Fig. 2.12b shows all the generated sidebands using the setup shown in Fig. 2.12a. This proves the modulation in the silicon strip waveguide by coupling the RF electrical field in the waveguide core region capacitively. The sideband power was very low, and I could not observe any second order sidebands because of high optical coupling loss (~ 30 dB at $\lambda = 1552.95$ nm) and low RF signal power at the contact pads of the modulator. Also, the DC response of the ESA was very noisy. Hence, I could not use this measurement to accurately estimate the V_π of this device from the carrier-sideband power ratio.

2.2.1.3 Half-Wave Voltage (V_π) Estimation

In order to estimate the V_π of this device I modified the setup (Fig. 2.13a) to move the spectrum from DC regime to 200 MHz. The power of the input CW carrier signal is split using a 50/50 power splitter. One output of the power splitter goes to the PC followed by the fabricated device. Other output of the power splitter goes to an acousto-optic modulator (AOM). To maximize the output optical power the AOM is operated at 200 MHz with $V_{pp} = 50$ mV generated from an RF signal generator followed by an RF amplifier. The modulated optical signals from the proposed modulator and the AOM are combined using a 50/50 power combiner. and modulated optical signals beat with each other. The optical signal generated by this self-heterodyne system [64] is detected using a PD followed by an ESA. We can observe the carrier signal at 200 MHz and the sidebands will be around 200 MHz. Because of the high optical loss of the (~ 7 dB) the AOM the carrier signal power at 200 MHz was quite low and the power of the sidebands were very low and hard to detect. Therefore, I amplified the modulation sinusoidal RF signal of 20 MHz and $V_{pp} = 1$ V generated from another signal generator using an RF amplifier. After amplification the RF signal had a V_{pp} of 35.2 V which is close to what I considered in the simulations. The amplified 20 MHz modulation RF signal resulted in first order sidebands at 180 MHz & 220 MHz (± 20 MHz of 200 MHz) and second order sidebands at 160 MHz & 240 MHz (± 40 MHz of 200 MHz) which are shown in Fig. 2.13b. From the carrier-sideband power ratio V_π of a push-pull MZI modulator can be calculated [22, 24]. But, when only one arm of the MZI is modulated a more intuitive way to estimate V_π is to consider the power ratio of the second and the first order sidebands [54]. I used Eq. 2.23 & 2.25 to calculate the modulation index, m and the V_π of the modulator. In these relations, $V_m =$

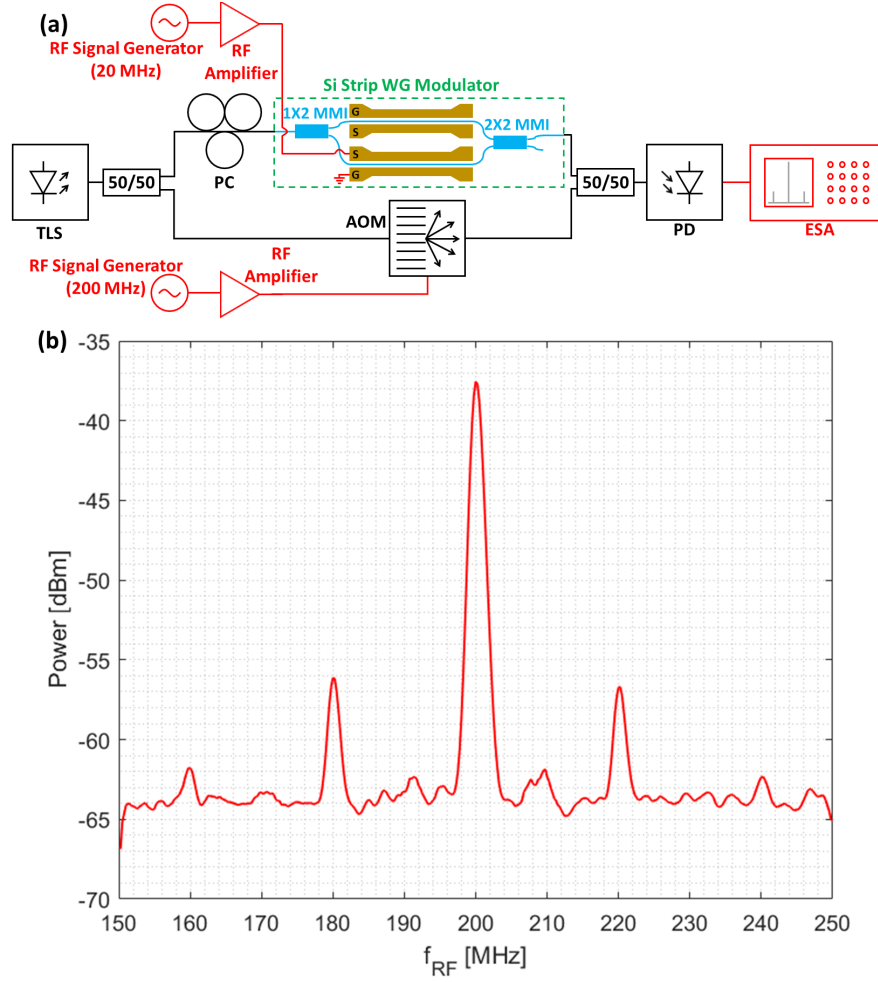


Figure 2.13: (a) Experimental setup for V_π estimation. (b) Carrier power and first & second order sideband power measurement at the ESA for modulation signal of 20 MHz.

17.6 V (half of $V_{pp} = 35.2$ V) is the amplitude of the modulation signal after amplification. Using Eq. 2.23 & 2.25 I estimated a V_π of ~ 46.95 V which results in a $V_\pi L$ of ~ 9.4 V.cm. This value is different from our simulation results. This happens because, I considered doping concentrations in simulations in such a way that they are well-matched with the sheet resistances of different silicon doping levels and thicknesses. But, in reality the doping concentrations can be completely different and cannot be disclosed due to confidentiality.

2.2.1.4 Electro-Optic (EO) Measurement

After the V_π measurement, I carried out the EO measurement for higher frequency RF signals. To demonstrate the EO response of the CC-Si modulator I built an experimental setup like Fig. 2.14a. Optical signals were coupled in and out of the chip using grating couplers. This time I used an RF signal generator which can generate RF sinusoids up to 70 GHz with maximum +25 dBm power. The high-power sinusoidal RF signals are applied to the top arm of the MZI. The other end of the modulator's transmission line was terminated with a $50\ \Omega$ broadband (50 GHz) termination load. A TLS was used to generate a CW optical carrier signal of 1549.86 nm wavelength with +9 dBm power

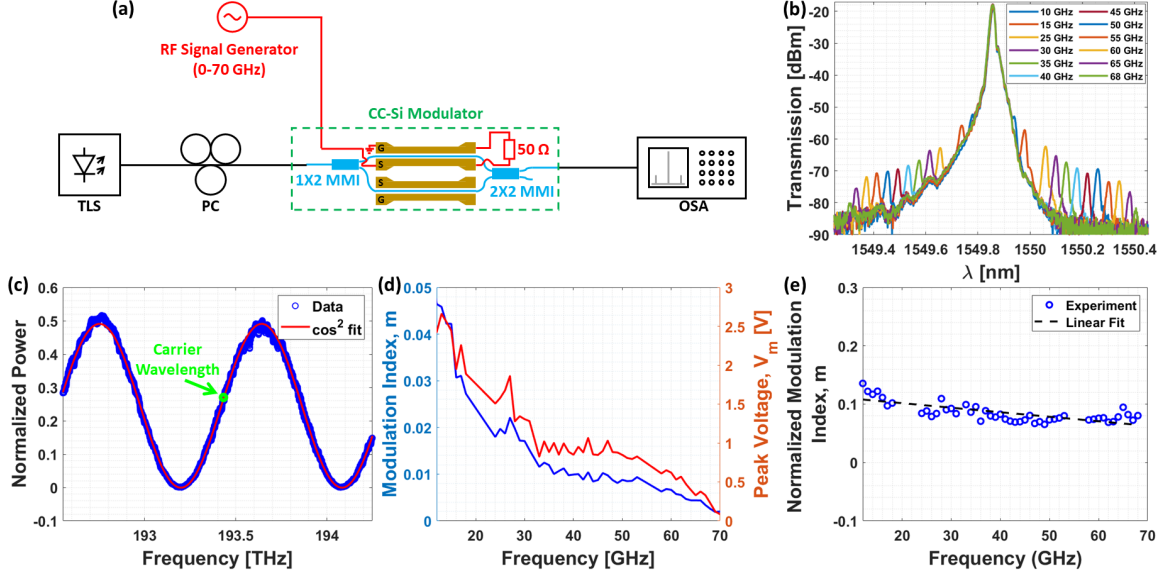


Figure 2.14: (a) Experimental setup for EO measurement. (b) Transmission spectra of the modulated optical signal for RF signal of different frequencies. (c) Measured intensity spectra of an unbalanced MZI CC-Si modulator (blue circles) and associated fit (red line) according to Eq. 2.39. The carrier wavelength is marked with a green circle. (d) Frequency dependent $m(\omega_m)$ (blue line) and associated $V(\omega_m)$ (red line). (e) Normalized $m(\omega_m)$ and associated linear fit.

to operate the modulator near the quadrature point. The optical carrier signal enters the chip after passing through a polarization controller. The modulated optical signals were detected by a high-resolution OSA. Thus, I could generate sidebands up to 68 GHz which can be clearly seen from Fig. 2.14b. These are the first order sidebands, and the second order sidebands could not be detected since their powers are extremely low. Therefore, in this case I calculated the frequency dependent modulation indices from the carrier-sideband power ratio expressed by Eq. 2.24 [54]. In such case, carrier nulling is difficult because of noise and distortions of the $\cos^2$ function of an unbalanced MZI CC-Si modulator. Therefore, to reduce the associated uncertainties, I measured the transmission spectrum of the unbalanced MZI CC-Si modulator over an extended spectral range which includes the current operating optical carrier to estimate the exact phase shift ϕ . At this point, I fit the measured transmission spectrum with a model function of an unbalanced MZI modulator which is shown in Eq. 2.39.

$$T_{\text{UMZI}}(\omega) = A_0 + A_1 \cos^2 \left(\frac{1}{2} (B_0 + B_1 (\omega - \omega_c) + B_2 (\omega - \omega_c)^2) \right) \quad (2.39)$$

In Eq. 2.39, ω_c is the center frequency of the considered frequency range of the unbalanced MZI spectrum, $B_0 = \beta^{(0)} \Delta L$, $B_1 = \beta^{(1)} \Delta L$, and $B_2 = \frac{1}{2} \beta^{(2)} \Delta L$ where, ΔL is the path length difference between two arms of the MZI CC-Si modulator. $\beta^{(0)}$ is the propagation constant, and $\beta^{(1)}$ & $\beta^{(2)}$ are the first & the second order derivative of the propagation constant respectively. I determined fitting parameters A_0 , A_1 , B_0 , B_1 , and B_2 by least-square fit method to the measured transmission spectrum. The phase shift ϕ of an unbalanced MZI modulator can be calculated from Eq. 2.40.

$$\phi(\omega_0) = \frac{1}{2} (B_0 + B_1 (\omega_0 - \omega_c) + B_2 (\omega_0 - \omega_c)^2) \quad (2.40)$$

The measured transmission spectrum (blue circles) after eliminating the grating coupler effect and the fit of the measured data with Eq. 2.39 (red line) are shown in Fig. 2.14c. The operating optical carrier is marked with a green circle in Fig. 2.14c. The phase shift value, ϕ and the carrier to sideband ratios, $R_{1,0}$ are used to calculate the frequency dependent modulation indices, $m(\omega_m)$ which is depicted (blue line) in Fig. 2.14d. Although, the signal generator generates equal power (+25 dBm) for the microwave sinusoid of all frequencies, the losses of the RF cable and the RF probes are frequency dependent. As a result, the RF power which is applied at the modulator terminal is frequency dependent. The RF powers for different frequencies are measured using an RF power meter and the relevant frequency dependent peak voltages, $V(\omega_m)$ are plotted in Fig. 2.14d (red line). It is observed from Fig. 2.14d that the curve of $m(\omega_m)$ follows the curve of $V(\omega_m)$. Therefore, we can assume that the sideband optical power changes linearly with the applied RF peak voltage. Later the $m(\omega_m)$ curve was normalized with respect to a frequency independent +2 dBm RF power which is plotted in Fig. 2.14e. A linear fit of the normalized $m(\omega_m)$ is also shown in Fig. 2.14e which shows that the frequency response of the CC-Si modulator is almost flat with respect to the frequency of the RF signal. This CC-Si modulator is expected to have even flatter frequency response if a negative DC bias is applied. But for the safety of the signal generator and to avoid further RF loss I did not apply any DC bias through any bias T.

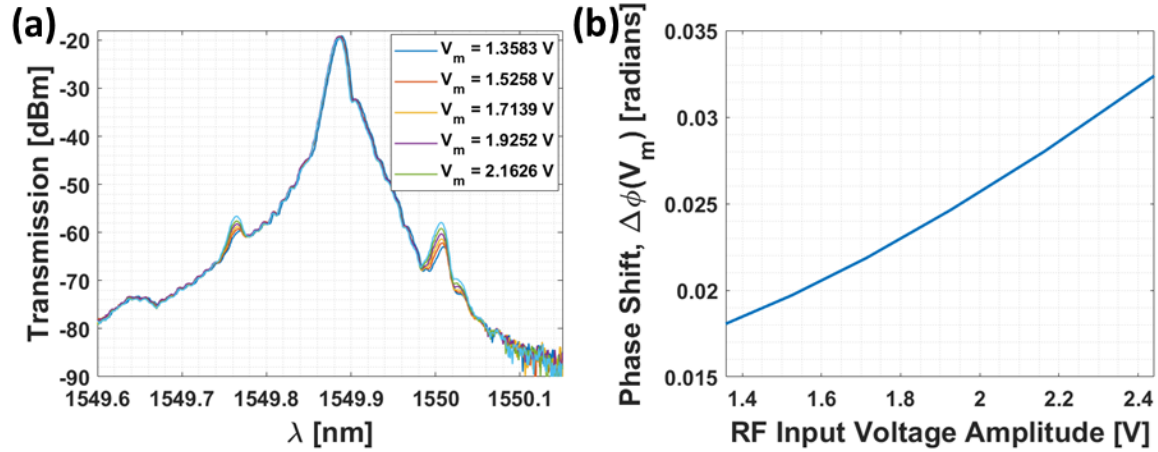


Figure 2.15: (a) Transmission spectra of modulated optical signal for different RF power at 15 GHz. (b) Voltage dependent phase shift, $\phi(V_m)$ at 15 GHz.

Further, I measured the transmission spectra of the modulated optical signals for different RF power i.e., different peak voltage, V_m at 15 GHz which are shown in Fig. 2.15a. The power at the modulator terminal was measured using the RF power meter before measuring these transmission spectra. I calculated the voltage dependent phase shift, $\phi(V_m)$ for different V_m and plotted it in Fig. 2.15b. As expected, for a device like this the change of phase with respect to the applied voltage is low.

2.2.1.5 Modulation Bandwidth (f_{3dB}) Estimation

The EO measurements show that the CC-Si modulator has high bandwidth. The exact f_{3dB} cannot be estimated because of the low sideband power and resolution limitation of the OSA. To have a reasonable estimation of the modulation bandwidth I measured the electrical S-parameters of the GS transmission line of the modulator. The experimental setup for measuring the S-parameters is

shown in Fig. 2.16a. RF signals from 10 MHz to 40 GHz were generated from the first port of a VNA and applied at the input (left) side of the transmission line on one arm of the MZI using a 40 GHz RF GSSG picoprobe. The output electrical signal was collected at the second port of the VNA from the output end of the transmission line of the same arm of the MZI, using another 40 GHz RF GSSG probe. The reflection ($|S_{11}|^2$) and forward transmission coefficients ($|S_{21}|^2$) from the input side of the transmission line are shown in Fig. 2.16b. Although the transmission line has good transmission, it also has a big reflection in the higher frequencies (>15 GHz). The complex propagation constant, γ , the characteristic impedance of the transmission line, Z_c , and the effective index of the microwave mode, $n_{e,RF}$ can be calculated using Eq. 2.35-2.38 [57].

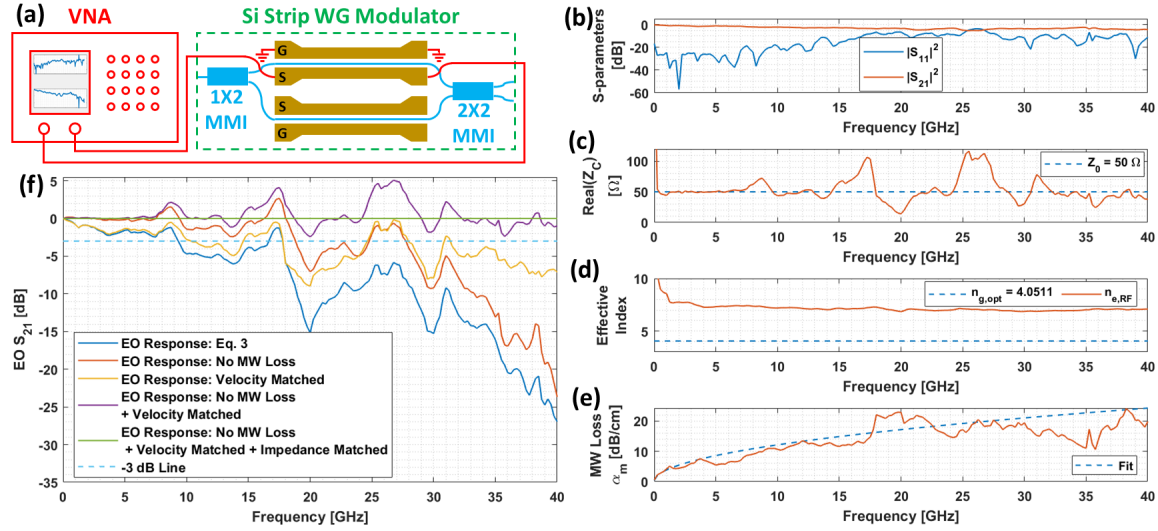


Figure 2.16: (a) Experimental setup for electrical S-parameter measurement. (b) Measured electrical transmission power ($|S_{21}|^2$) and reflected power ($|S_{11}|^2$) spectra of the modulator's GS transmission line. (c) Real part of characteristic impedance, (Z_c) of the GS transmission line calculated from Eq. 2.35. (d) Optical group refractive index, ($n_{g,opt}$) around 1550 nm and the effective refractive index of the RF mode, $n_{e,RF}$ which has been calculated using Eq. 2.36. (e) Calculated microwave propagation loss, (α_m) using Eq. 2.35. (f) EO frequency response, (EO S_{21}) of the modulator for various conditions.

The characteristic impedance of the transmission line, Z_c (Fig. 2.16c) is very close to 50Ω at lower frequencies (<7 GHz), but the impedance starts to mismatch in the higher frequency range. This behaviour aligns with the measured high $|S_{11}|^2$ at higher frequencies. Fig. 2.16d shows the $n_{e,RF}$ and the optical group refractive index, $n_{g,opt}$ which clearly show the velocity mismatch between the RF and optical modes (Fig. 2.16d-inset). Lastly, the RF propagation loss is shown in Fig. 2.16e which is also quite large because of the highly doped silicon slabs of the transmission line. All these parameters were used to estimate the EO frequency response ($|S_{21}|^2$) of the modulator using Eq. 2.27-2.33 [12]. Since a VNA calibration was performed before carrying out the measurements we can consider both source impedance, Z_s and termination impedance, Z_t to be 50Ω . Considering all these values, we can calculate the EO S_{21} of the modulator for various conditions which are depicted in Fig. 2.16f. The direct implementation of Eq. 2.31 is shown using the blue line which estimates a 3-dB EO bandwidth (f_{3dB}) of ~ 9.3 GHz. The actual estimated bandwidth can be enhanced by further dimensional optimization. For example, the yellow line in Fig. 2.16f estimates an f_{3dB} of ~ 10.4 GHz after considering a velocity matched modulator ($n_{e,RF} = n_{g,opt} + 0.01$) and the red line

estimates an $f_{3\text{dB}}$ of ~ 18.9 GHz when the transmission line of the modulator has no microwave loss ($\alpha_m = 0$ dB/cm). If we consider both $n_{e,\text{RF}} = n_{g,\text{opt}} + 0.01$ and $\alpha_m = 0$ dB/cm the 3-dB bandwidth of the modulator goes beyond 40 GHz (purple line in Fig. 2.16f). On top of that if the impedance mismatch is minimized throughout the whole RF band, then we can expect a flat EO frequency response which is shown with the green line in Fig. 2.16f. Because of the high V_π the direct EO measurement of the modulator was not possible. It is worth mentioning that the VNA calibration was performed without the 40 GHz RF probes. Hence, the actual 3-dB bandwidths in all previous cases would be even higher if the VNA is calibrated with the RF probes.

2.2.2 Series Push-Pull CC-Si Modulator with Silicon Dioxide Cladding

The previous demonstration was for single drive CC-Si MZI modulator. I demonstrate a CC-Si MZI modulator in with silicon dioxide cladding in SPP configuration as well which is fabricated in the same MPW run of AMF. The EO measurement of the modulator can generate sidebands up to 68 GHz around a C-band CW optical carrier. A flat EO response has been observed up to 68 GHz. A $V_\pi L$ of ~ 87.8 V.cm was also estimated by comparing the carrier wavelength power with the sideband power for different amplitudes of a 15 GHz RF signal.

2.2.2.1 Device Structure

Fig. 2.17a shows the cross-sectional view of a CC-Si MZI modulator in SPP configuration. Each arm of the MZI has a 650 nm wide silicon strip waveguide ensures the single mode condition for fundamental TE optical mode. The lightly doped lateral PN junction is exactly at the middle of the waveguide's cross-section. A P+ silicon slab and a N+ silicon slab with 90 nm thickness (h_{Slab}) are placed 500 nm away from the P side of the strip waveguide and the N side of the strip waveguide respectively. Lumerical Device is used to perform electrostatic simulations by applying different DC bias voltage, V_{DC} between -3 V and -30 V maintaining the signal terminal voltage, $V_{\text{sig}} = 0$ V. All the layer thicknesses and the doping concentration remain the same as I did for the single drive CC-Si modulator with silicon dioxide cladding. Fig. 2.17b & 2.17c show the electron concentrations and Fig. 2.17d & 2.17e show the hole concentrations in the waveguide region of the left arm of the MZI for the abovementioned V_{DC} respectively. We notice a change of carrier concentrations when V_{DC} is reduced from -3 V to -30 V. As a result, we get a change of effective refractive index, Δn_{eff} i.e., the modulation of optical signal.

2.2.2.2 Experimental Demonstration

To demonstrate the EO response of the CC-Si MZI modulator in SPP configuration I built an experimental setup like Fig. 2.17f. A high-power RF signal generator is used which can generate sinusoidal signal up to 70 GHz with $+25$ dBm power. The RF signals are applied between the signal and ground terminal of the MZI. A 50Ω termination load is used to terminate the other end of the RF transmission line of the modulator. Generally, a negative DC bias should be applied to the DC bias contact. But the RF signal generator is very sensitive to the DC voltages and therefore, for the safety of the instrument I avoid applying any DC bias. I use a TLS to generate a $+9$ dBm CW optical carrier at 1550.31 nm wavelength which near the quadrature point of the MZI. The optical carrier is coupled to the chip using a grating coupler after passing through a PC. The coupled optical

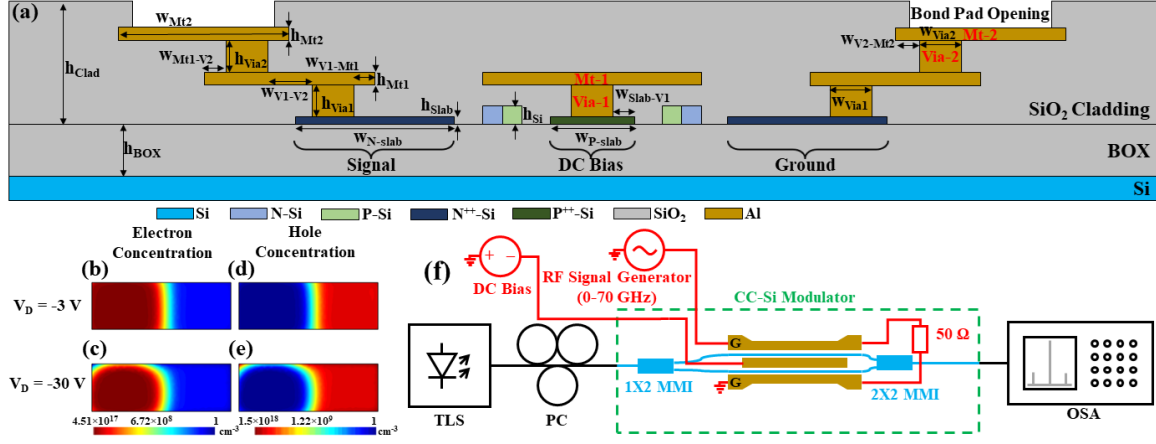


Figure 2.17: (a) Cross-sectional view of a CC-Si MZI modulator with silicon dioxide cladding in SPP configuration. Lumerical Device simulations for the electron concentrations for V_{DC} of (b) -3 V & (c) -30 V respectively and for the hole concentrations for V_{DC} of (d) -3 V & (e) -30 V respectively. (f) Experimental setup for EO measurement of the CC-Si MZI modulator in SPP configuration.

carrier signal is split into two MZI arms using a 1×2 MMI coupler. After phase modulation optical signals from both arms are combined using a 2×2 MMI coupler. These two output modulated optical signals are coupled out of the chip using two other grating couplers. One of the modulate signals is detected by a high-resolution OSA. In this way, I am able to generate sidebands up to 68 GHz which can be clearly seen from Fig. 2.18a. The frequency dependent modulation indices, $m(\omega_m)$ is calculated from the carrier-sideband power ratio expressed by Eq. 2.24 [54]. Again, carrier nulling is difficult to achieve in this case due to the distortions of the $\cos^2$ function of an unbalanced MZI CC-Si modulator and noise. To reduce the relevant uncertainties, the normalized transmission spectrum of the unbalanced MZI modulator is measured over a wider spectral range that includes the current operating optical carrier. In order to estimate the exact phase shift, ϕ , a fit of a model function of an unbalanced MZI modulator shown in Eq. 2.39 is performed with the experimental data. Thus, the phase shift, ϕ of an unbalanced CC-Si MZI modulator with silicon dioxide cladding in SPP configuration can be calculated from Eq. 2.40. After eliminating the effect of the grating couplers, the measured transmission spectrum looks like the blue circles in Fig. 2.18b. The fit to the measured data exploiting Eq. 2.39 is shown in the red line of Fig. 2.18b and the operating optical carrier is pointed out with a green circle in Fig. 2.18b. The phase shift value, ϕ , and the carrier to sideband ratio, $R_{1,0}$ are used to calculate the $m(\omega_m)$ using Eq. 2.24 which is shown in the blue line of Fig. 2.18c. The signal generator can generate a power of +25 dBm for all frequencies. However, the losses of the RF probes and the RF cable are frequency dependent. Hence, the RF power which is actually applied at the modulator terminal is microwave frequency dependent. I measure the RF power for different frequencies using an RF power meter. The associated frequency dependent peak voltages, $V(\omega_m)$ based on these RF power measurements are plotted in the red line of Fig. 2.18c. It is observed that the curve of $m(\omega_m)$ follow the curve of $V(\omega_m)$. The $m(\omega_m)$ curve is normalized with respect to a frequency independent +2 dBm RF power which is plotted in Fig. 2.18d. A linear fit of the normalized $m(\omega_m)$ is also shown in Fig. 2.18d which shows that the CC-Si MZI modulator with silicon dioxide cladding in SPP configuration has a very flat frequency response.

Followed by the EO measurement, I measure transmission spectra of different modulated optical

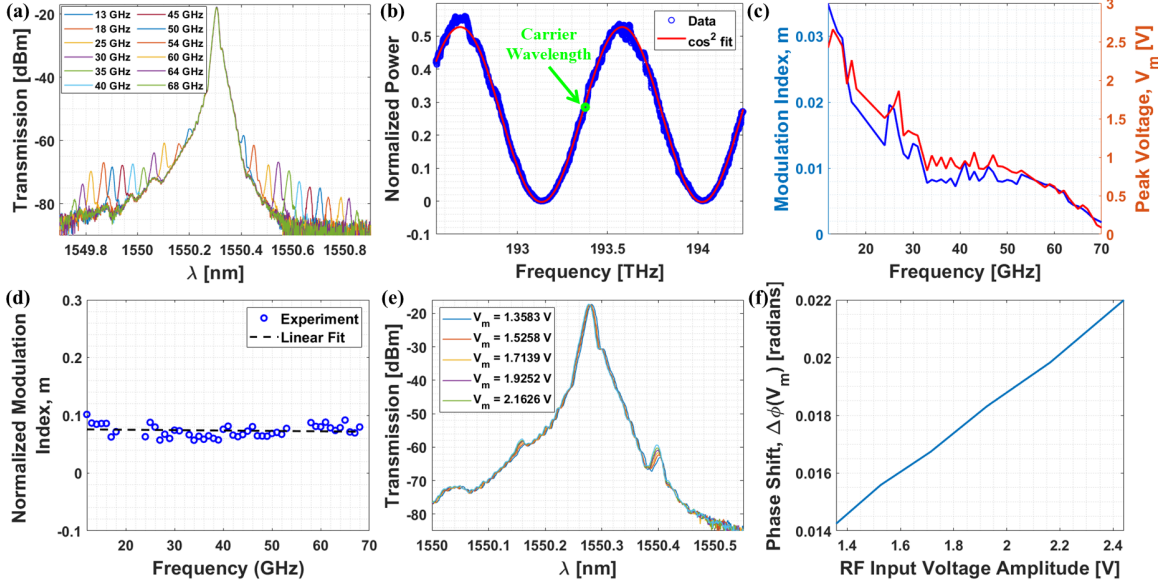


Figure 2.18: (a) Transmission spectra of the modulated optical signal for RF signal of different frequencies. (b) Measured intensity spectra of an unbalanced MZI CC-Si modulator with silicon dioxide cladding (blue circles) in SPP configuration and associated fit (red line) according to Eq. 2.39. The carrier wavelength is pointed out with a green circle. (c) Frequency dependent $m(\omega_m)$ (blue line) and associated $V(\omega_m)$ (red line). (d) Normalized $m(\omega_m)$ and associated linear fit. (e) Transmission spectra of modulated optical signal for different RF power at 15 GHz. (f) Voltage dependent phase shift, $\phi(V_m)$ at 15 GHz.

signals for 15 GHz RF sinusoidal signals with different powers i.e., different peak voltage, V_m . The measured transmission spectra are shown in Fig. 2.18e. Before applying the 15 GHz RF signals, I measure the powers of different RF signals using an RF power meter. Thus, I determine the phase shift, $\phi(V_m)$ for 15 GHz RF signal which is voltage dependent. Fig. 2.18f shows the phase shifts for different RF powers i.e., different V_m of 15 GHz signals. Considering a linear relationship between V_m and $\phi(V_m)$ I estimate a $V_\pi L$ of ~ 87.8 V.cm for a 2 mm long CC-Si modulator. These devices are expected to have higher V_π , since I use silicon dioxide cladding which reduces the electrical field confinements within the waveguide region [59].

2.2.3 Series Push-Pull CC-Si Modulator with Barium Titanate Cladding

A CC-Si modulator with high-k material, amorphous BTO cladding in SPP configuration has also been demonstrated. I measured the EO response of the CC-Si modulator with BTO cladding up to 67 GHz and observed a flat frequency response. For an SPP configuration MZI structure I further estimated a $V_\pi L$ of ~ 2 V.m at 15 GHz of RF signal without any DC bias.

2.2.3.1 Device Structure and Fabrication

To demonstrate the concept of the CC-Si modulator with high-k material cladding, I fabricated a 2 mm long CC-Si MZI modulator in SPP configuration with BTO as high-k dielectric. A schematic diagram of the MZI in SPP configuration is depicted in Fig. 2.19a, and the cross-sectional view of the device is shown in Fig. 2.19b. The CC-Si modulator is partially fabricated on a standard 300

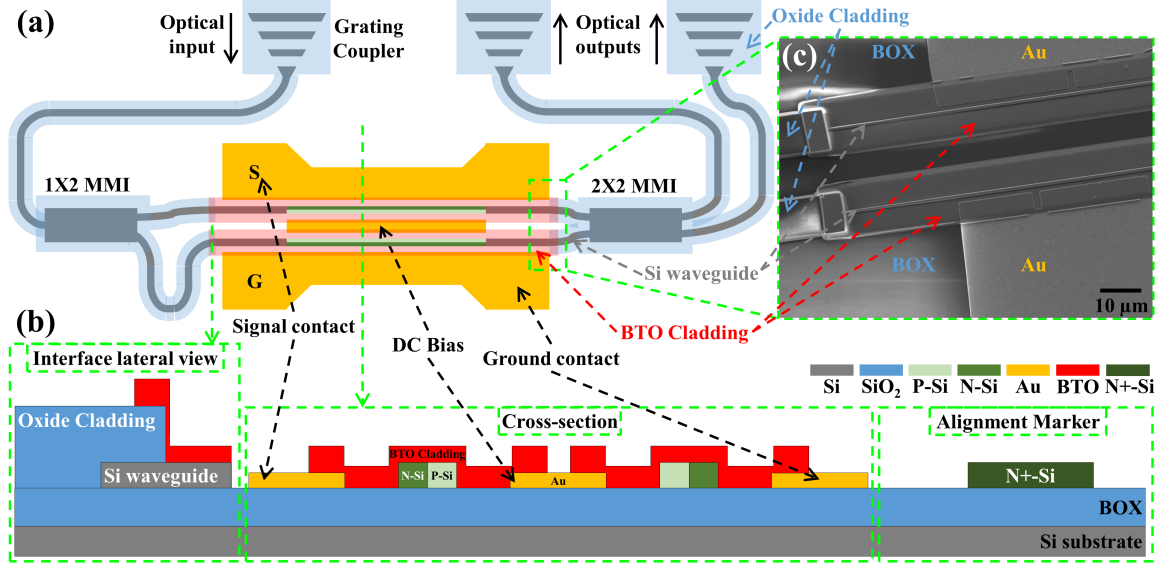


Figure 2.19: (a) Schematic diagram of a CC-Si MZI modulator in SPP configuration. (b) Stack of all layers to realize the CC-Si modulator, the silicon dioxide-barium titanate cladding interface, and the alignment marker. (c) scanning electron microscope (SEM) image near the silicon dioxide-barium titanate cladding interface region of a CC-Si MZI modulator in SPP configuration.

mm SOI substrate at an MPW run of a commercial foundry, AMF [20]. Essential passive devices used in the device like silicon waveguides for routing of light, grating couplers, MMI couplers etc. are adopted from the PDK of AMF, and they require silicon dioxide cladding for optimum performance. Therefore, only these devices of the whole chip are covered with silicon dioxide cladding, and everywhere else we have oxide cladding opening. Thus, we observe abrupt transition of the optical mode where the silicon dioxide cladding ends and BTO cladding starts or vice versa because of the minimum feature size limitation of the oxide opening layer. Fig. 2.19b also shows the lateral view of such an interface. For coupling light in and out of the chip I used grating couplers. The input optical signal is split into two arms using a 1×2 MMI coupler and the waveguides in each arm face an abrupt transition of optical mode while entering from silicon dioxide cladding region to BTO cladding region. The active section of each of the arm has a lateral PN diode waveguide EO phase shifter of 2 mm length. After the active region the optical modes in both arms face another abrupt transition when light returns from the BTO cladding region to silicon dioxide cladding region. Finally, the optical signals from both arms are combined together using a 2×2 MMI coupler and the output lights are collected from the other side of the 2×2 MMI. For electrical contact chromium (Cr) followed by gold (Au) layers are used. The RF signal is applied between the electrical contacts adjacent to the N-type sides of two MZI arms. In the middle of two MZI arms and adjacent to the P-type sides of both waveguides, there is another electrical contact that is placed to apply an appropriate MZI bias for push-pull operation. The gaps between two consecutive electrical contacts are filled with BTO cladding. For fabricated device I use a gap of $1 \mu\text{m}$ between the electrical contact and the waveguide ($w_g = 1 \mu\text{m}$) for reasonable alignment, low optical loss and decent V_π . I use alignment markers of highly N-doped silicon (N+-Si) layer fabricated by the foundry to align the gold and BTO layers as depicted in Fig. 2.19b. A scanning electron microscope (SEM) image in Fig. 2.19c shows the silicon dioxide-barium titanate cladding interface section as well as a top view

of one end of the fabricated device.

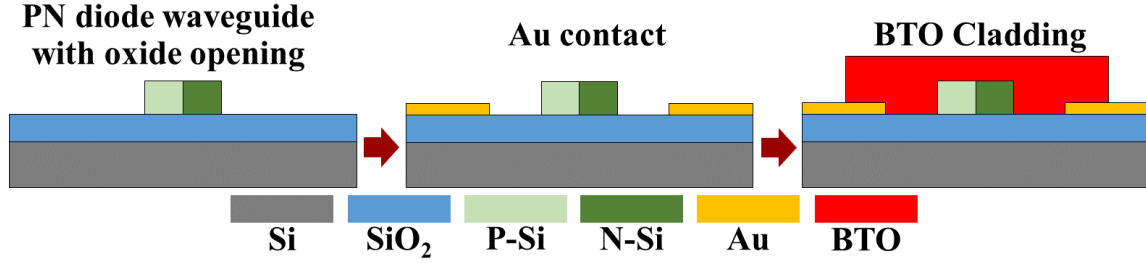


Figure 2.20: BEOL process of the CC-Si modulator. First lithographic step: deposition of 10 nm chromium and 140 nm gold via e-beam evaporation followed by a lift-off of a polymethyl methacrylate (PMMA) bilayer. Second lithographic step: deposition of ~ 250 nm BTO via RF magnetron sputtering followed by another lift-off of a polymethyl methacrylate (PMMA) bilayer.

The CC-Si MZI modulator is partially fabricated in an MPW run of AMF. The passive devices from the PDK and the PN diode waveguide of the phase shifter section are fabricated on a standard SOI substrate having a silicon device layer thickness (h_{Si}) of 220 nm, and a BOX layer thickness (h_{BOX}) of 2 μm . The electrical contact and the BTO cladding are fabricated in BEOL processing in house on top of the chip fabricated at the foundry. The BEOL processing consists of a two-step lithographic process using electron-beam (e-beam) lithography. In the first lithographic step, electrical contacts are fabricated directly on the BOX layer via e-beam evaporation of 10 nm of chromium and 140 nm gold followed by a lift-off process of a polymethyl methacrylate (PMMA) bilayer. In the second lithographic step, BTO cladding is fabricated via RF magnetron sputtering of ~ 250 nm BTO thin film using a stoichiometric BTO target followed by another lift-off process of a PMMA bilayer. The BTO deposition was carried out at a base pressure of $\sim 10^{-7}$ torr in presence of argon (Ar) gas with 30 standard cubic centimeter per minute (sccm) flowrate. The BEOL post-fabrication process flow has been shown in Fig. 2.20.

2.2.3.2 Optical Characterization

Since the device is partially fabricated in an MPW run of a commercial foundry and uses the optimized structures of grating couplers, MMIs and strip waveguides from the PDK of AMF, the optical characteristics of those structures are already well defined. I measure the optical characteristics of only the structures which are fabricated during the BEOL processing. I measure the optical transmission spectrum of a straight waveguide using a pair of grating couplers back-to-back. Thus, the response of a pair of grating couplers is determined (shown in Fig. 2.21a). Later, I measure the transmission spectra of the unbalanced CC-Si MZI (shown in Fig. 2.21a) and the grating coupler response is removed from the CC-Si MZI by a simple subtraction. From the cutback structure measurements, I determine a mode propagation loss of ~ 3.35 dB/mm in the active region of the modulator and a silicon dioxide-barium titanate interface loss of ~ 0.38 dB per facet from the linear fittings shown in Fig. 2.21b and Fig. 2.21c respectively at 1551.91 nm wavelength. The high optical propagation loss comes from the imperfect lift-off process and the misalignment of the metal contacts.

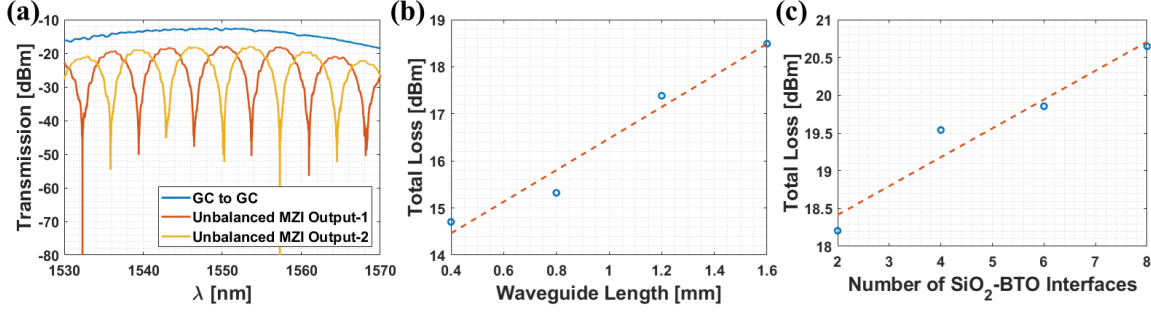


Figure 2.21: (a) Transmission spectra of a back-to-back connection between a pair of grating couplers and the two outputs of an unbalanced MZI. Cutback structure measurement to determine (b) the propagation loss and (c) silicon dioxide-barium titanate interface loss from the linear fit of the experimental data at 1551.91 nm wavelength.

2.2.3.3 Electro-optic (EO) Measurement

I build an experimental setup like Fig. 2.22a to demonstrate the EO response of the CC-Si MZI modulator in SPP configuration. I use an RF signal generator which can generate RF sinusoids up to 70 GHz with maximum +25 dBm power. The high-power sinusoidal RF signals are applied across the signal-ground (SG) terminal of the MZI. The other end of the modulator's transmission line is terminated with a 50 Ω broadband (50 GHz) termination load. To avoid the damage of the RF signal generator no external DC bias was applied. A TLS is used to generate a CW optical carrier signal of 1551.91 nm wavelength with +9 dBm power to operate the modulator near the quadrature point. The optical carrier signal enters the chip via a grating coupler after passing through a polarization controller. The modulated optical signals are coupled out of the chip using another grating coupler and are detected by a high-resolution OSA. Thus, I can generate sidebands up to 67 GHz which can be clearly seen from Fig. 2.22b. I calculate the frequency dependent modulation indices from the carrier-sideband power ratio expressed by Eq. 2.24 [54]. Carrier nulling is difficult in this case because of distortions of the $\cos^2$ function of an unbalanced MZI CC-Si modulator and noise. To reduce the associated uncertainties, I measure the transmission spectrum of the unbalanced MZI CC-Si modulator over an extended spectral range which includes the current operating optical carrier in order to estimate the exact phase shift ϕ . I fit the measured transmission spectrum with a model function of an unbalanced MZI modulator which is shown in Eq. 2.39. By least-square fit method to the measured transmission spectrum I determine the fitting parameters estimated the phase shift ϕ of the unbalanced MZI modulator using Eq. 2.40.

The measured transmission spectrum (blue circles) after eliminating the effect of grating couplers and the fit to the measured data exploiting Eq. 2.39 (red line) are shown in Fig. 2.22c. The operating optical carrier is pointed out with a green circle in Fig. 2.22c. The carrier to sideband ratio, $R_{1,0}$ and the phase shift value, ϕ and are used to calculate the frequency dependent modulation indices, $m(\omega_m)$ using Eq. 2.24 which is depicted (blue line) in Fig. 2.22d. The signal generator is able to generate an equal power of +25 dBm for microwave sinusoidal signals of all frequencies. But the losses of the cable and the probes are microwave frequency dependent. Thus, the power which is actually applied at the modulator terminal is microwave frequency dependent. Therefore, I measure the power for different frequencies using a power meter. The relevant frequency dependent peak

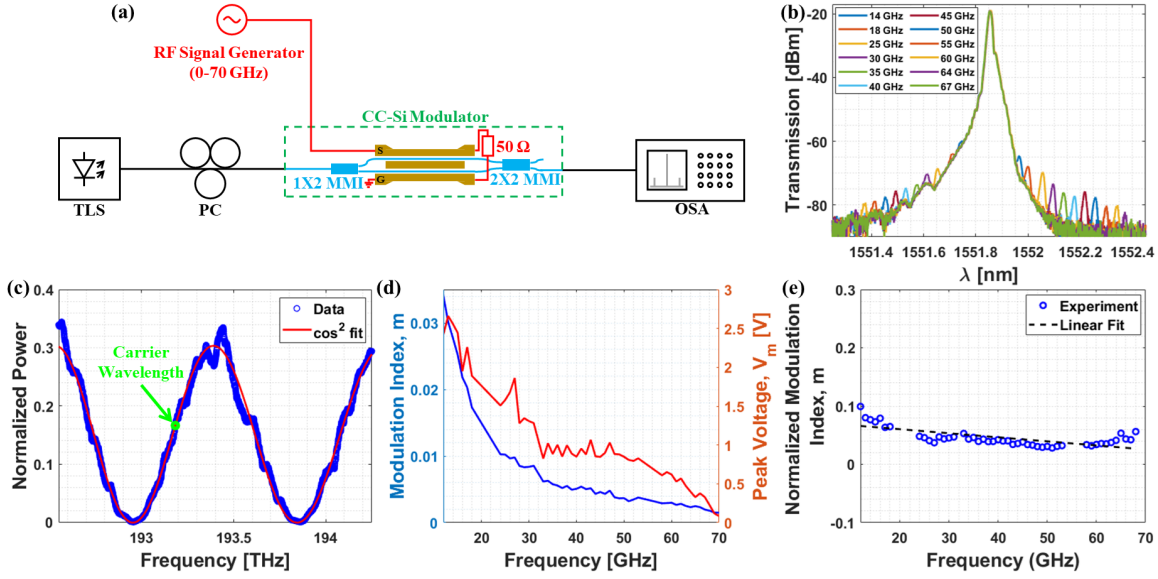


Figure 2.22: (a) Experimental setup for EO measurement. (b) Transmission spectra of the modulated optical signal for RF signals of different frequencies. (c) Measured intensity spectra of an unbalanced MZI CC-Si modulator (blue circles) and associated fit (red line) according to Eq. 2.39. The carrier wavelength is pointed out with a green circle. (d) Frequency dependent $m(\omega_m)$ (blue line) and associated $V(\omega_m)$ (red line). (e) Normalized $m(\omega_m)$ and associated linear fit.

voltages, $V(\omega_m)$ based on these power measurements are plotted in Fig. 2.22d (red line). It is observed from Fig. 2.22d that the curve of $m(\omega_m)$ follow the curve of $V(\omega_m)$. Hence, we can assume that the sideband optical power varies linearly with $V(\omega_m)$. I normalize the $m(\omega_m)$ curve with respect to a frequency independent +2 dBm power which is plotted in Fig. 2.22e. A linear fit of the normalized $m(\omega_m)$ is also shown in Fig. 2.22e which proves the validity of flat frequency response of the CC-Si MZI modulator in SPP configuration. Eq. 2.40.

I also measure the modulated optical signal transmission spectra for different power i.e., different peak voltage, $m(\omega_m)$ at 15 GHz which are shown in Fig. 2.23a. The power at the modulator terminal is measured using the power meter before measuring these transmission spectra. From these measurements I determine the voltage dependent phase shift, $\phi(V_m)$ at 15 GHz which is plotted in Fig. 2.23b. There might be two main reasons behind low modulation index. Firstly, a DC bias needs to be applied at the central electrode to realize a push-pull operation. To ensure the safety of the high-power signal generator no DC bias is applied at the DC contact. Thus, the electric field is distributed across the waveguides in both MZI arms. As a result, I get negative effective refractive index change in both MZI arms, i.e., $\Delta n_{\text{eff,arm-1}} \neq \Delta n_{\text{eff,arm-2}}$, $\Delta n_{\text{eff,arm-1}} < 0$, and $\Delta n_{\text{eff,arm-2}} < 0$. Thus, instead of push-pull operation I get a chirp which in the end result in a lower differential phase shift between the two MZI arms. Secondly, we are limited by our fabrication capabilities. To achieve higher modulation index, we need to place the electrical contacts close to the waveguide (lower w_g) and the thickness of BTO should be higher (higher h_{BTO}). I use doped silicon alignment markers to align the metal contact and the BTO cladding layers. Due to charging effect of the doped silicon markers it is very difficult to do proper alignments. Thus, achieving $w_g < 1 \mu\text{m}$ is very difficult. Also, in our case the thickness of BTO, h_{BTO} is limited to ~ 250 nm of deposition

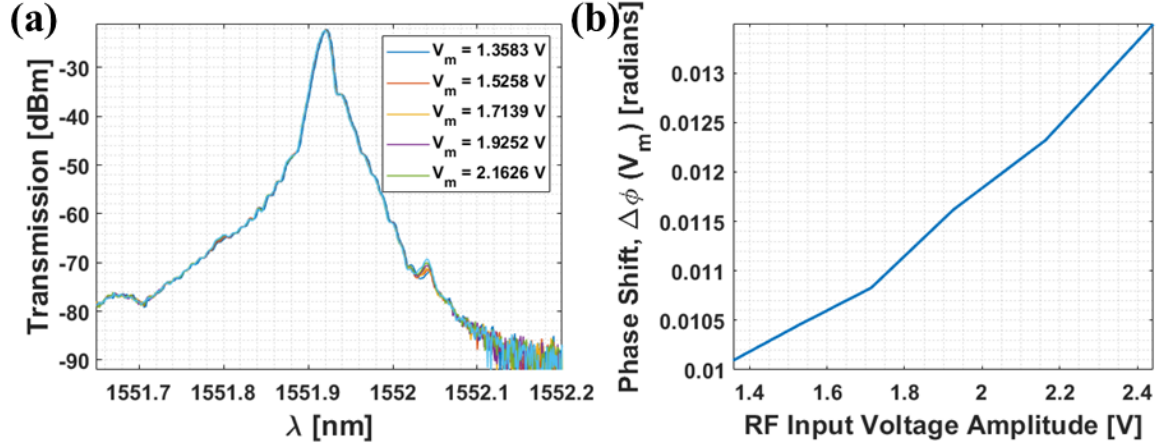


Figure 2.23: (a) Experimental setup for EO measurement. (b) Transmission spectra of the modulated optical signal for RF signals of different frequencies. (c) Measured intensity spectra of an unbalanced MZI CC-Si modulator (blue circles) and associated fit (red line) according to Eq. 2.39. The carrier wavelength is pointed out with a green circle. (d) Frequency dependent $m(\omega_m)$ (blue line) and associated $V(\omega_m)$ (red line). (e) Normalized $m(\omega_m)$ and associated linear fit.

thickness due to low deposition rate and limited capability of PMMA lift-off process. Also, I keep the metal thickness, h_m is limited to 150 nm for easier lift-off process of PMMA bilayer. For lower V_π higher values of h_{BTO} and h_m are helpful.

2.2.3.4 Device Optimization

I do further simulations to observe the effect of different dimensions of the CC-Si modulator on $V_\pi L$. I consider a straight waveguide phase modulator structure like Fig. 2.3(a). I do electrostatic simulations for different modulation voltages as well as do separate mode simulations for each modulation voltage in Lumerical Device and Mode modules respectively. Thus, exploiting the Soref and Bennet model I calculated the $V_\pi L$ for different w_g , h_{BTO} and h_m which are shown in Fig. 2.24a, 2.24b and 2.24c respectively. If the fabrication process can be optimized for lower w_g , higher h_{BTO} and higher h_m , then a lower $V_\pi L$ value compared to the demonstrated device can be achieved.

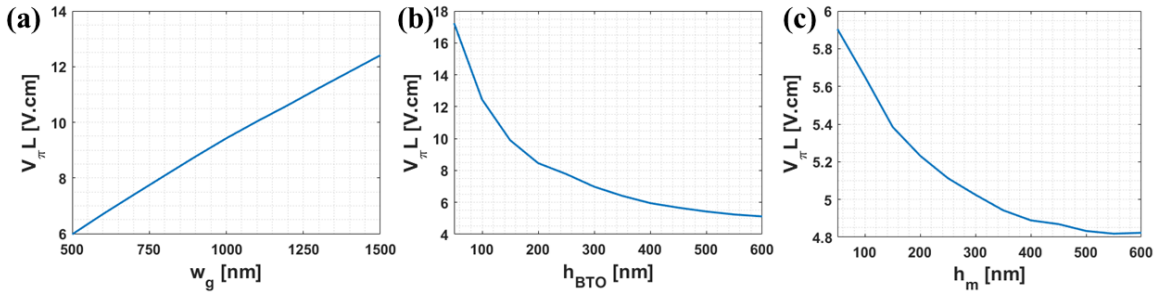


Figure 2.24: (a) The change of $V_\pi L$ for various (a) gaps between the waveguide and the metal contact, w_g , (b) various BTO cladding thicknesses, h_{BTO} , and (c) different metal contact thicknesses, h_m .

2.3 Conclusion

In this chapter, I demonstrated three different types of CC-Si modulator – two devices with silicon dioxide cladding and one device with BTO cladding.

The CC-Si modulators with silicon dioxide cladding were demonstrated in both single drive and SPP configurations and are completely foundry compatible. I exploited the CC-Si modulator with silicon dioxide cladding in single drive configuration to observe the plasma dispersion effects in a silicon strip waveguide. For this device I generated sidebands from 5 MHz to 50 MHz with a step size of 5 MHz around a CW optical carrier of 1552.95 nm using an unbalanced MZI structure. For the same device I also estimated a V_π of ~ 46.95 V for a 2 mm device using a self-heterodyne system which refers to $V_\pi L$ of ~ 9.4 V.cm. The EO measurement of the device shows that the first order sidebands can be generated up to 68 GHz of RF signals and an almost flat frequency response was observed up to 68 GHz. I also measured the electrical S-parameters of the transmission line of the modulator and estimated a 3-dB bandwidth of ~ 9.3 GHz. I also showed that the modulator can achieve more than 40 GHz of 3-dB bandwidth if we can reduce the microwave loss of the modulator transmission line and match the group velocity of the optical modes with the phase velocity of the microwave modes. Later, I demonstrated the CC-Si modulator with silicon dioxide cladding in SPP configuration. I performed the EO measurement of this device and observe a flat EO frequency response up to 68 GHz again which proves the broadband nature of this device. For a 2 mm device a $V_\pi L$ of ~ 87.8 V.cm is estimated when 15 GHz RF sinusoidal signals with different powers applied to the modulator.

I was able to demonstrate the CC-Si MZI modulator with BTO cladding in SPP configuration only. The EO measurement of this modulator is carried out for RF sinusoidal signals up to 67 GHz which leads to a very flat frequency response which proves the validity of capacitive coupling for bandwidth enhancement. For a 2 mm device a $V_\pi L$ of ~ 2 V.m is calculated for an RF signal with 15 GHz frequency with different voltage amplitudes.

Chapter 3

Silicon Microring Resonator Based Optical Phase Shifter

This chapter addresses the essential principles of optical modulators based on microring resonators (MRR), with a particular focus on their application within a silicon platform. The organization of this chapter is as follows. Section 3.1 delves into the theoretical aspects of add-drop MRRs. It also covers the forward bias operation of a PN diode-based MRR and its application in optical phase and intensity modulation. Furthermore, a brief overview of advanced modulation formats, including quadrature-amplitude modulation (QAM), phase shift keying (phase shift keying (PSK)), and amplitude-phase shift keying (APSK), utilizing optical phase shifters and intensity modulators is provided. In Section 3.2, we present our proposed concept of optical phase modulation employing silicon MRRs. This section also includes discussions on experimental demonstrations of MRR-based optical phase shifters, measurement techniques for intensity and phase spectrum extraction, as well as the implementation of phase shifters in higher-order modulation formats. The chapter concludes in Section 3.3.

3.1 Theoretical Background

The intensity and phase of an optical carrier can be modulated by tuning the resonance of a silicon MRR structure exploiting the plasma dispersion effect. We have already discussed the plasma dispersion effect as well as reverse bias operation of a conventional lateral PN diode-based silicon modulator in Chapter 2. In this section, we will explain the higher-order modulation formats using optical phase shifters and the fundamentals of an add-drop MRR. We further discussed the forward bias operation of the lateral PN diode-based silicon phase modulator and its use to perform intensity or phase modulation by tuning the effective refractive index in the ring.

3.1.1 Higher Order Modulation Formats

In the past century, intensity modulation or amplitude modulation was the most popular form of modulation format due to simplicity. Many silicon-based amplitude modulators have been successfully demonstrated so far [65, 66]. In the beginning of this century, a rapid shift from the

amplitude-based communications to phase encoded modulation formats has been noticed in the optical communication technologies due to the exponential increase of internet bandwidth requirements. Because of an extra degree of freedom, i.e., the optical phase modulation in addition to the intensity modulation higher order modulation formats such as, QAM, PSK, APSK etc. become quite popular. In this section, we will discuss how the optical phase shifters can generate the constellation diagram of different higher order modulation formats.

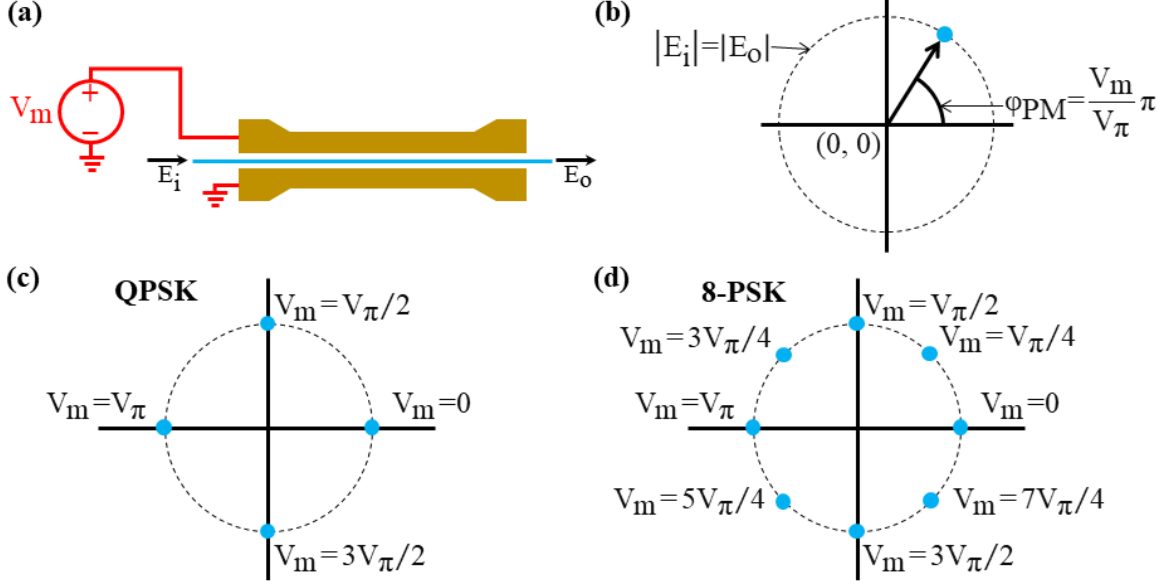


Figure 3.1: (a) Schematic diagram of straight waveguide phase modulator. (b) Phase modulated optical wave with an arbitrary phase shift mapped on a polar constellation diagram. Constellation diagrams of (c) quadrature phase shift keying (QPSK) and (d) 8-PSK modulated optical signals respectively.

Fig. 3.1a shows the schematic diagram of a simple optical phase shifter. If the input optical wave in a straight waveguide phase shifter is, E_i the output wave, E_o of the phase shifter is,

$$E_o = E_i \exp [j\phi_{PM} (V_m)] \quad (3.1)$$

Here, $\phi_{PM} (V_m) = \Delta n_{eff} (V_m) k_0$ is the cumulative phase shift and Δn_{eff} is the change of waveguide effective refractive index due to the applied modulation voltage, V_m . $k_0 (= 2\pi/\lambda)$ is the propagation constant and λ is the wavelength of the input optical wave in free space. Assuming the phase shift proportional to the applied voltage, V_m the transfer function, T_{PM} of the phase modulator can be expressed by,

$$T_{PM} = \frac{E_o}{E_i} = \exp \left[j \frac{V_m}{V_\pi} \pi \right] \quad (3.2)$$

Here, V_π is the half-wave voltage of the phase modulator. Phase modulators like this can be used to generate PSK signals where the degree of freedom is only the optical phase, i.e. the amplitude or the intensity of the optical wave remains the same. Fig. 3.1b shows the modulated optical signal plotted on a polar (r, θ) coordinate which is also known as the constellation diagram where radius, r and θ represent, $|E_o|$ the amplitude and phase shift, $\phi_{PM} (V_m)$ of the output optical wave. We can see that, for a pure phase modulator like this only the ϕ_{PM} is a function of V_m which makes

$|E_o| = |E_i|$. Fig. 3.1c and 3.1d show constellation diagrams of two different PSK modulation formats – the quadrature phase shift keying (QPSK) and the 8-PSK modulation formats respectively.

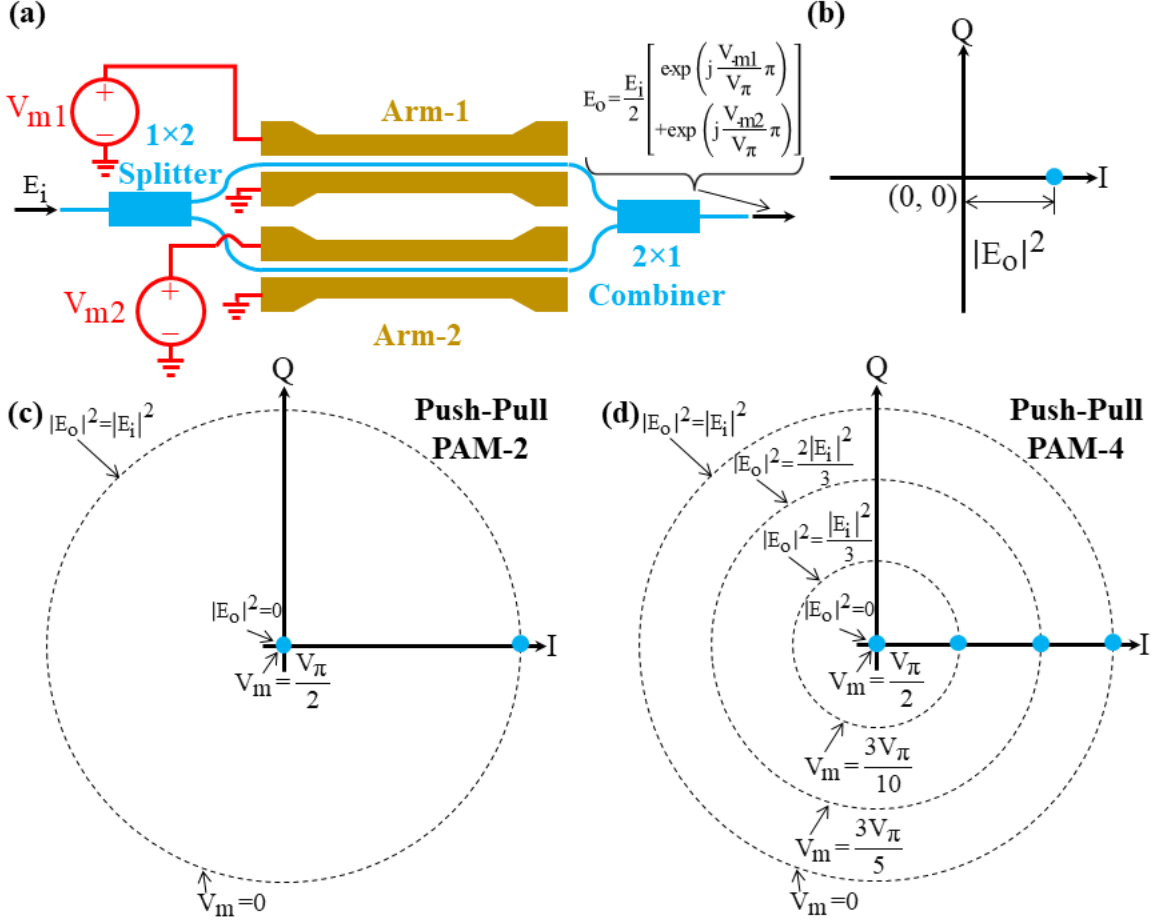


Figure 3.2: (a) Schematic diagram of MZI-based intensity modulator. (b) Intensity modulated optical wave with an arbitrary intensity mapped on a one-dimensional cartesian constellation diagram. Constellation diagrams of the optical signals modulated with (c) PAM-2 format and (d) PAM-4 format respectively.

Two identical phase modulators can be used in two arms of an MZI to realize an intensity modulator as depicted in Fig. 3.2a. If V_{m1} and V_{m2} are the modulation voltages applied in two arms of the MZI then the transfer function of the MZI, T_{MZI} becomes,

$$T_{\text{MZI}} = \frac{E_o}{E_i} = \frac{1}{2} \left[\exp \left(j \frac{V_{m1}}{V_\pi} \pi \right) + \exp \left(j \frac{V_{m2}}{V_\pi} \pi \right) \right] \quad (3.3)$$

For a push-pull operation of the MZI we consider $V_m = V_{m1} = -V_{m2}$. Therefore, the transfer function of the push-pull type MZI, $T_{\text{MZI,PP}}$ becomes,

$$T_{\text{MZI,PP}} = \frac{E_o}{E_i} = \frac{1}{2} \left[\exp \left(j \frac{V_m}{V_\pi} \pi \right) + \exp \left(-j \frac{V_m}{V_\pi} \pi \right) \right] = \cos \left(\frac{V_m}{V_\pi} \pi \right) \quad (3.4)$$

Fig. 3.2c and 3.2d show constellation diagrams of two different intensity modulation (PAM) formats – the 2-level PAM (PAM-2) and the 4-level PAM (PAM-4) modulation formats respectively.

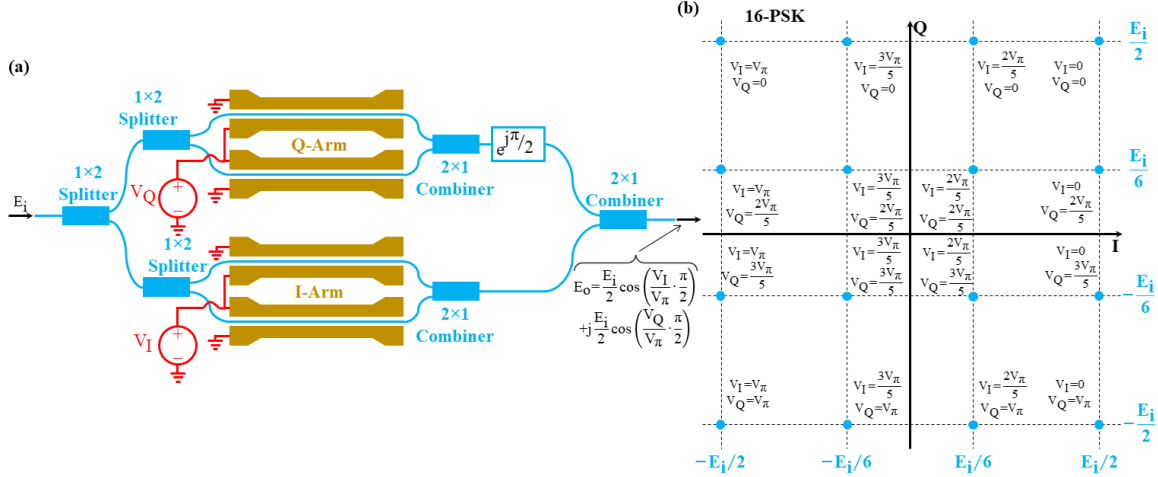


Figure 3.3: (a) Schematic diagram of push-pull MZI-based IQ modulator. (b) Constellation diagram of the optical signals modulated with 16-QAM format.

Both these modulation formats only use only one degree i.e., either phase or amplitude of the optical signal for modulation. Higher order modulation formats like QAM and APSK modulation formats use both phase and amplitude or intensity of the optical signal to enhance the modulation capacity. Fig. 3.3a shows a schematic diagram of an IQ modulator which consists of two push-pull MZI modulators arranged in two arms of another MZI. To utilize both the real and imaginary axis of a complex cartesian constellation diagram a permanent $\pi/2$ phase shift is performed in one arm of the IQ modulator which is known as the quadrature arm (Q-arm) whereas the other arm is known as the in-phase arm (I-arm). The final transfer function of the IQ modulator can be described by the following equation,

$$T_{IQ} = \frac{E_o}{E_i} = \frac{1}{2} \cos\left(\frac{V_I}{V_\pi} \cdot \frac{\pi}{2}\right) + j \frac{1}{2} \cos\left(\frac{V_Q}{V_\pi} \cdot \frac{\pi}{2}\right) \quad (3.5)$$

Here, V_I and V_Q are the modulation voltages applied at the push-pull MZI modulators placed in the I and the Q-arm of the IQ modulator respectively. Fig. 3.3b shows the constellation diagram of 16-level QAM (16-QAM) format.

Another way to exploit both degrees of freedom i.e., the phase and the intensity of the optical signal is to use a push-pull MZI and a straight waveguide phase modulator in series to realize an APSK signal as depicted in Fig. 3.14a. An APSK signal utilizes the complex polar constellation diagram where the magnitude denotes the intensity of the optical signal and argument denotes the phase of the optical wave. The transfer function, T_{APSK} of such arrangement is,

$$T_{APSK} = \frac{E_o}{E_i} = \cos\left(\frac{V_{IM}}{V_\pi} \pi\right) \exp\left(j \frac{V_{PM}}{V_\pi} \pi\right) \quad (3.6)$$

Here, V_{IM} and V_{PM} are the modulation voltages applied at the push-pull MZI modulator and the straight waveguide phase modulator shown in Fig 3.4a respectively. A 16-level APSK (16-APSK) modulation format is shown in Fig 3.4b.

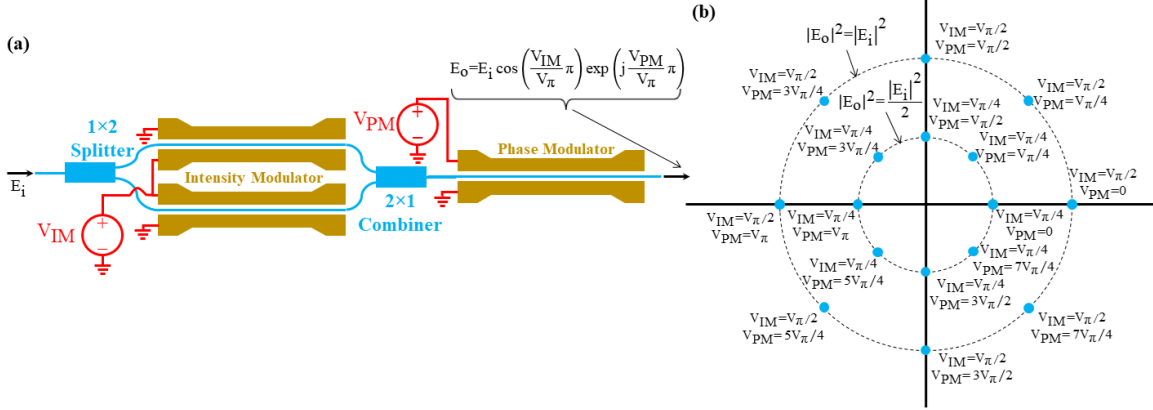


Figure 3.4: (a) Schematic diagram of push-pull MZI modulator in series with a phase modulator to realize an APSK signal. (b) Constellation diagram of the optical signals modulated with 16-APSK modulation format.

3.1.2 Silicon Add-Drop Microring Resonator

A silicon add-drop MRR is an optical device which has four ports optical device labeled as input, through, drop, and add ports. Such a ring resonator is shown in Fig 3.5a. Initially we assume that a monochromatic optical signal with frequency, ω is applied to the input port. The input waveguide is evanescently coupled to the ring resonator by a coupling coefficient κ_1 , and a transmission coefficient, τ_1 . Similarly, the output bus waveguide is evanescently coupled to the ring resonator by a coupling coefficient κ_2 , and a transmission coefficient, τ_2 . For a lossless coupling, $\kappa_i^2 + \tau_i^2 = 1$, where $i=1$ and $i=2$ for input and output waveguides respectively. Depending on the values of κ_1 and τ_1 a portion of the optical wave enters the microring and the rest of the portion of the optical wave goes to the through port. While the optical wave is half-way through the ring a portion of the optical wave in the microring couples to the output bus waveguide which is seen at the drop port and rest of the portion continues to travel in the microring depending on the values of κ_2 and τ_2 . After a round trip the optical wave inside the microring the again couples back to the input bus waveguide which interferes with the pre-existing input optical wave at the input bus waveguide. As a result, a resonant behavior is observed both at the through and the drop ports.

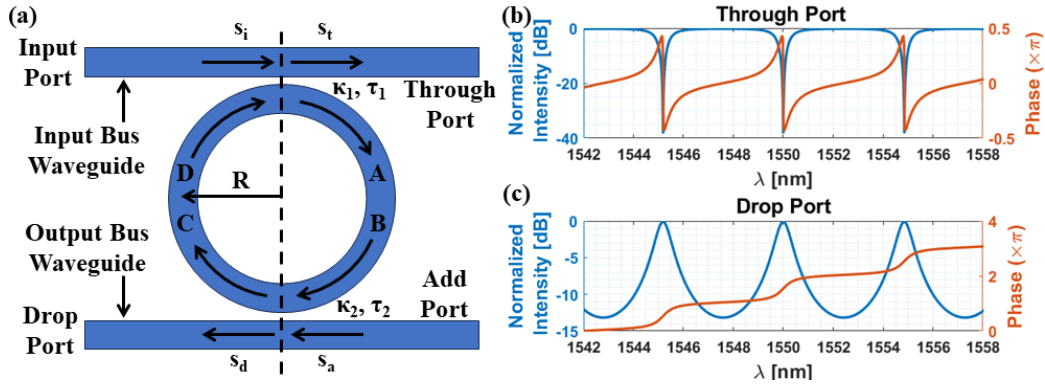


Figure 3.5: (a) Schematic diagram of an add-drop MRR. Analytical simulation of the intensity and phase spectra of the optical waves seen at (b) the through and (c) the drop ports respectively.

We label s_i & s_t as the optical signals just before & after coupling region in the input bus waveguide respectively and s_a & s_d as the optical signals just before & after coupling region in the output bus waveguide respectively as depicted in Fig. 3.5a. If we label the optical signals just before & after the input coupling region inside the microring as A & D respectively and the optical signals just before & after the output coupling region inside the microring as B & C then we get the following relations using the transfer matrix model of a point directional coupler,

$$\begin{bmatrix} s_t \\ A \end{bmatrix} = \begin{bmatrix} \tau_1 & -j\kappa_1 \\ -j\kappa_1 & \tau_1 \end{bmatrix} \begin{bmatrix} s_i \\ D \end{bmatrix} \quad (3.7)$$

$$\begin{bmatrix} s_d \\ C \end{bmatrix} = \begin{bmatrix} \tau_2 & -j\kappa_2 \\ -j\kappa_2 & \tau_2 \end{bmatrix} \begin{bmatrix} s_a \\ B \end{bmatrix} \quad (3.8)$$

The optical waves, B and D are related to the waves, A and C, respectively by,

$$B = A \exp(-j\gamma_r \pi R) = A \sqrt{a_{rt}} \exp\left(-j \frac{\phi_{rt}}{2}\right) \quad (3.9)$$

$$D = C \exp(-j\gamma_r \pi R) = C \sqrt{a_{rt}} \exp\left(-j \frac{\phi_{rt}}{2}\right) \quad (3.10)$$

Here, $\gamma_r (= \beta_r - j\alpha_r/2)$ is the complex propagation constant of the bent waveguide in the microring, $\phi_{rt} (= 2\pi R\beta_r)$ is the round-trip phase shift accumulated by the optical wave inside the microring, R is the radius of the microring, $\beta_r (= n_{\text{eff},r}k_0)$ is the real part of the complex propagation constant, $n_{\text{eff},r}$ is the effective refractive index of the mode confined in the bent waveguide of the microring, $a_{rt} (= e^{-\alpha_r \pi R})$ is the round-trip field attenuation factor in the microring, and α_r is the optical propagation loss coefficient inside the microring. Another popular expression for free space propagation constant is $k_0 = \omega/c$ where c is speed of light in free space. Considering no signal input at the drop port ($s_a = 0$) we can use Eq. 3.7-3.7 to solve for the transfer function at the through and the drop ports [67].

$$H_t(\phi_{rt}) \equiv \frac{s_t}{s_i} = \frac{\tau_1 - \tau_2 a_{rt} e^{-j\phi_{rt}}}{1 - \tau_1 \tau_2 a_{rt} e^{-j\phi_{rt}}} \quad (3.11)$$

$$H_d(\phi_{rt}) \equiv \frac{s_d}{s_i} = -\frac{\kappa_1 \kappa_2 \sqrt{a_{rt}} e^{-j \frac{\phi_{rt}}{2}}}{1 - \tau_1 \tau_2 a_{rt} e^{-j\phi_{rt}}} \quad (3.12)$$

The expressions of the transfer functions of the through and the drop ports in Eq. 3.11 and 3.12 respectively can be expressed as a function of the wavelength.

$$H_t(\lambda) = \frac{\tau_1 - \tau_2 e^{-\alpha_r \pi R} e^{-j \frac{4\pi^2 R n_{\text{eff},r}}{\lambda}}}{1 - \tau_1 \tau_2 e^{-\alpha_r \pi R} e^{-j \frac{4\pi^2 R n_{\text{eff},r}}{\lambda}}} \quad (3.13)$$

$$H_d(\lambda) = -\frac{\kappa_1 \kappa_2 e^{-\alpha_r \pi R} e^{-j \frac{2\pi^2 R n_{\text{eff},r}}{\lambda}}}{1 - \tau_1 \tau_2 e^{-\alpha_r \pi R} e^{-j \frac{4\pi^2 R n_{\text{eff},r}}{\lambda}}} \quad (3.14)$$

Fig. 3.5b and 3.5c show the normalized intensity and phase spectra for the through and the drop ports respectively where we considered for $n_{\text{eff},r} \approx 2.56602$ at 1550 nm wavelength from Lumerical

Mode simulations for bent rib waveguides with lateral PN diode considering $R \approx 20\mu m$. Optical loss, α_r has two main components – the scattering loss due to sidewall roughness which was collected from ref. [68], and optical loss due to free carriers considered from the simulated values from Lumerical Device & Mode simulations which gave a value of $\alpha_r \approx 3.82$ dB/cm or 87.9 Np/m at 1550 nm wavelength when no external bias voltage was applied. Additionally, we considered dispersion of the waveguide inside the microring i.e., $n_{eff,r}$ and α_r both are wavelength dependent which we got from the wavelength sweeps of Lumerical Mode simulations. We also considered wavelength independent $\tau_1 = \tau_2 = 0.8$ and a lossless coupling region i.e., $\tau_1^2 + \kappa_1^2 = 1$ and $\tau_2^2 + \kappa_2^2 = 1$. We can see that for some wavelengths the intensity becomes close to zero which are known as the resonant wavelengths for this microring. At the same time, we can see that exactly at the resonant wavelengths the at drop port have maximum intensity.

3.1.3 Forward Bias Operation of Lateral PN Diode-Based Silicon MRR Modulator

To utilize the properties of an MRR for intensity or phase modulation we need to tune the effective refractive index, $n_{eff,r}$ of the microring waveguide as a function of external modulation voltage. This can be by exploiting the plasma dispersion effect in the lateral PN diode-based silicon rib waveguide. We have discussed the reverse bias operation of the conventional silicon modulators in Section 2.1.5. In this section, we will discuss about the forward bias operation of the same structure considered in Fig. 2.2a. Fig. 3.6a shows the change of electron and hole concentrations with the change of external forward bias modulation voltage simulated in Lumerical Device. As a result, the effective refractive index of the waveguide, $n_{eff,r}$ as well as the optical loss, α_r inside the microring change which can be calculated using the Soref and Bennett model [9] and is depicted in Fig. 3.6b.

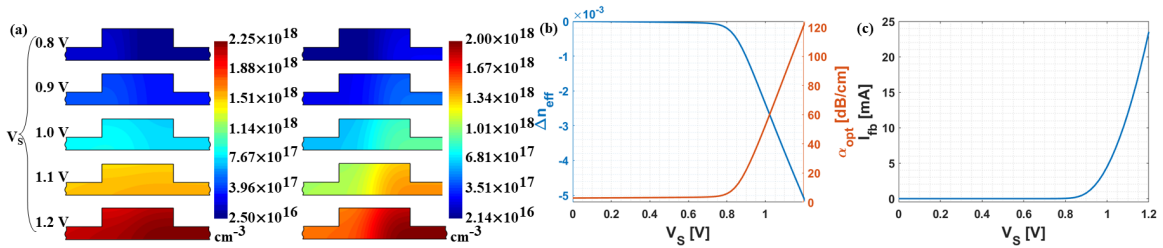


Figure 3.6: (a) Change of electron and hole concentrations in the waveguide region because of different applied forward bias voltages. (b) The change of Δn_{eff} and α_{opt} with respect to the change of forward bias signal voltage, V_S . Lumerical Device and Mode simulations were performed considering the structure shown in Fig. 2.2a.

Unlike reverse bias operation, forward bias operation involves current flow, I_{fb} through the PN diode which increases exponentially with applied voltage (Fig. 3.2c). From Fig. 3.6c we can see that, the knee voltage of the PN diode is around 0.9 V. After this voltage the current flow increases rapidly. We can also notice that, the change of $n_{eff,r}$ is follow almost the same trend as the current flow i.e., the $\Delta n_{eff,r} \propto I_{fb}$. The change in effective index in forward bias is much higher than the reverse bias operation. But, because of finite I_{fb} we have finite RC time constant which limits the f_{3dB} of the device to maximum 1 GHz.

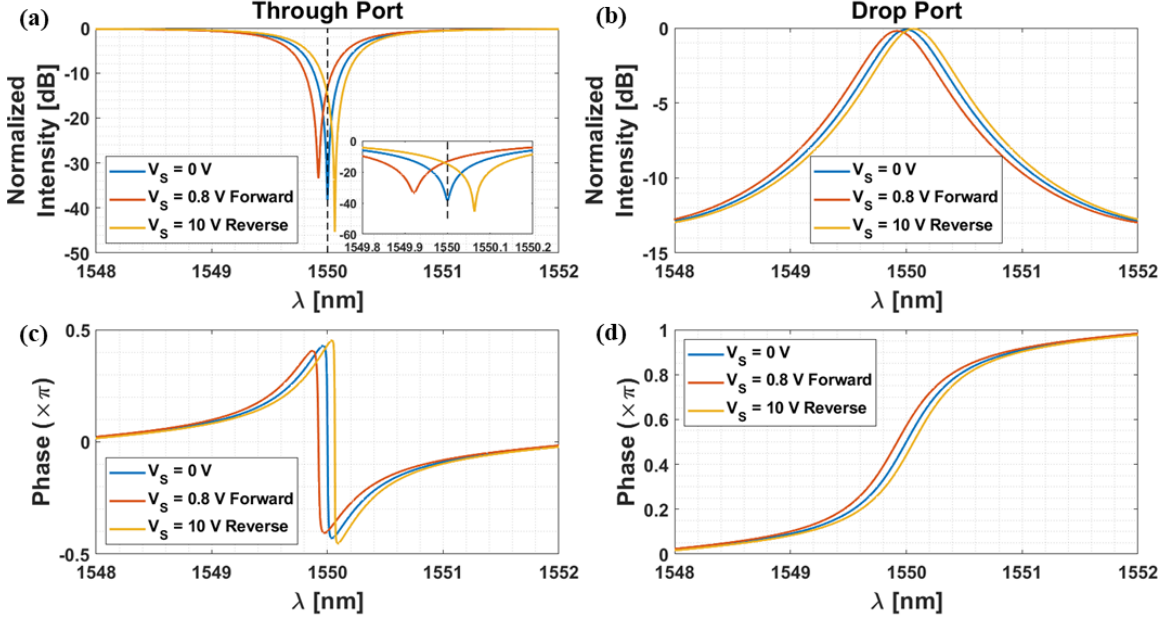


Figure 3.7: Intensity spectra observed at (a) through port, and (b) the drop port of an add-drop MRR respectively for V_S of 0 V, +0.8 V (forward bias) and -10 V (reverse bias). Phase responses at (c) through port, and (d) the drop port for the aforementioned voltages.

Fig. 3.7a and 3.7b show the intensity spectra of through and drop port respectively for different forward and reverse bias modulation voltages. An optical carrier wavelength, λ_c is chosen at the resonance to have very low intensity at the through port when no modulation voltage is applied. If the modulation voltage, V_S is changed then the resonant wavelength of the MRR also changes and a higher intensity is observed at the through port. The inset of Fig 3.7a clearly shows that, for $V_S = +0.8$ V (forward bias) the resonance moves to the left by ~ 0.08 nm which gives a resonance wavelength tunability of ~ 0.01 nm/V and the carrier wavelength intensity is increased by ~ 24.8 dB. We can also see that, for $V_S = -10$ V (reverse bias) the resonance moves to the right by 0.06 nm which gives a resonance much lower wavelength tunability (0.006 nm/V) compared to the forward bias operation and the carrier wavelength intensity is increased by ~ 23.8 dB. This is how the intensity modulation can be performed using an MRR in forward and reverse bias regime. Although reverse bias operation shows much lower resonant wavelength tunability, this operation is dominated by the carrier drift which makes it much faster than the forward bias operation which is dominated by the carrier diffusion. Therefore, for high-speed intensity modulation MRR-based modulators are operated in reverse bias despite having lower resonant wavelength tunability i.e., higher V_π . However, in neither case the phase responses shown in Fig. 3.7c and 3.7d can be used effectively to make a phase modulator since the intensity of the through and drop ports also change sharply around the resonance.

We proposed a novel silicon MRR-based modulator which can work as a phase modulator with minimum intensity variation. The proposal and the experimental demonstrations will be discussed in the following sections in this chapter.

3.2 Silicon Add-Drop MRR Modulator-Based Optical Phase Shifter

In Section 3.1.2 we have seen that, enough phase shift is observed at the optical outputs at the through and drop ports when we cross the resonance. But, because of high intensity variation use of this structure as a phase shifter is impractical. Our proposed concept of optical phase shifting using an add-drop silicon MRR modulator is shown schematically in Fig. 3.8a. The through and drop ports are combined to realize a phase shift in the signal. In some earlier studies, optical phase shifters have been demonstrated using over-coupled MRRs [28, 29, 30]. Although a low extinction ratio can be achieved using over-coupled MRRs, they require comparatively high modulation voltages to achieve a full 2π phase shift because of wide resonance regions in the intensity spectra. In our approach, we combine the signals from the through port with the drop port of the MRR. The result allows us to make use of the intensity from the drop port, via constructive interference. Hence, a critically coupled MRR which has a narrower resonance feature can be used to achieve a full 2π phase shift with lower modulation voltage. The transfer function, $H_{CO}(\lambda)$ of such device can be defined by the following equation,

$$H_{CO}(\lambda) = \frac{1}{\sqrt{2}} [H_t(\lambda) + H_d(\lambda) e^{j\theta}] \quad (3.15)$$

Here, θ is the phase difference between the through arm and the drop arm before combining which arises from any undesired optical path difference or fabrication inaccuracies. We get $H_t(\lambda)$ and $H_d(\lambda)$ from Eq. 3.13 and 3.14 respectively. Fig. 3.8b, 3.8c and 3.8d show the intensity spectra (blue lines) and the phase responses (red lines) of observed at the through, drop and the combined ports respectively. We can see that, at the combined port $\sim 2\pi$ phase shift is observed if we cross the resonance once with very low intensity variation. Also, we notice that one of each two resonances disappear at the combined port which happens because of the presence of the $\phi_{rt}/2$ term in the numerator of the transfer function of the drop port.

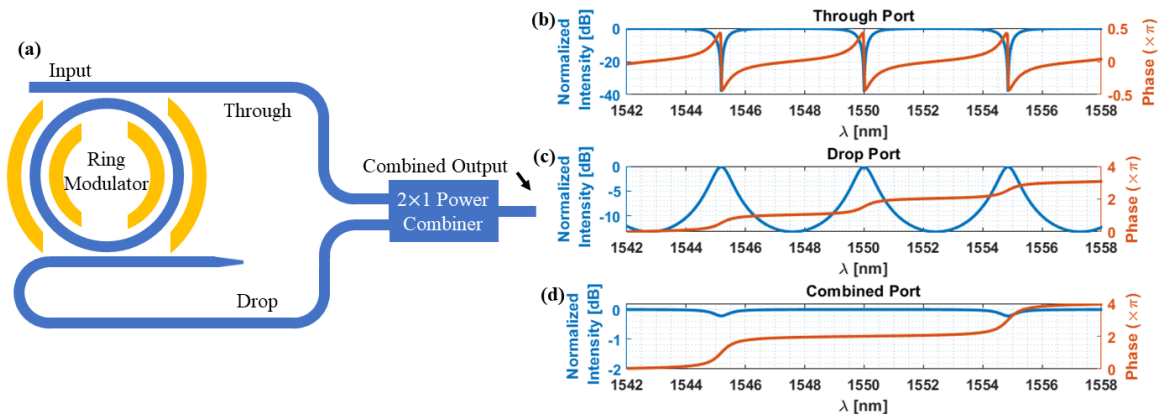


Figure 3.8: (a) Schematic diagram of the concept of silicon MRR modulator-based phase shifter. Analytical simulations of the intensity spectra (blue lines) and phase responses (red lines) observed at (b) the through, (c) the drop, and (d) the combined ports of an add-drop ring modulator. Simulations were done using the dimensions and parameters considered the structure shown in Fig. 3.5a.

In this section, we will discuss our experimental demonstration of our proposed optical phase shifter

for a full 2π phase shift with very low intensity variation [69]. We also demonstrated a method of extracting the phase response of the of our proposed phase shifter by placing the phase shifter in one arm of an MZI and exploiting different output intensity spectra of that MZI structure. Later, using an optical vector analyze (OVA) we verified the linear phase response of the same optical phase shifter [70].

3.2.1 Measurement Results of the Phase Shifts

Our proposed concept of silicon MRR modulator-based optical phase shifter was fabricated using the MPW run of the AMF. Fig. 3.9a and 3.9b show the schematic diagram and the optical micrograph of the fabricated phase shifter respectively where the through and the drop ports are combined to realize phase shifting in an optical carrier signal. We started characterizing our phase shifter by observing the intensity spectra at different ports. To monitor the intensity spectra at the through and the drop ports two 1×2 MMI couplers have been used. There is an inherent path length difference equivalent to π between the through and drop ports of the MRR. To balance the optical path differences between the through and drop port arms because of this and any other fabrication variations we placed two $150 \mu\text{m}$ long titanium nitride (TiN) heaters after the 1×2 MMI couplers. Using a 2×2 MMI coupler both ports have been combined after the TiN heaters. We used the standard 1×2 & 2×2 MMI couplers, grating couplers and ring modulators optimized by the AMF. In this case we use the transfer matrix of a 2×2 MMI (2×2 optical coupler with 50:50 coupling ratio) to find the transfer function of combined port 1, $H_{CO1}(\lambda)$ and combined port 2, $H_{CO2}(\lambda)$. If the phase difference between the through and drop arms is θ , these transfer functions can be defined by the following expressions,

$$H_{CO1}(\lambda) = \frac{1}{\sqrt{2}} [jH_t(\lambda) + H_d(\lambda) e^{j\theta}] \quad (3.16)$$

$$H_{CO2}(\lambda) = \frac{1}{\sqrt{2}} [H_t(\lambda) + jH_d(\lambda) e^{j\theta}] \quad (3.17)$$

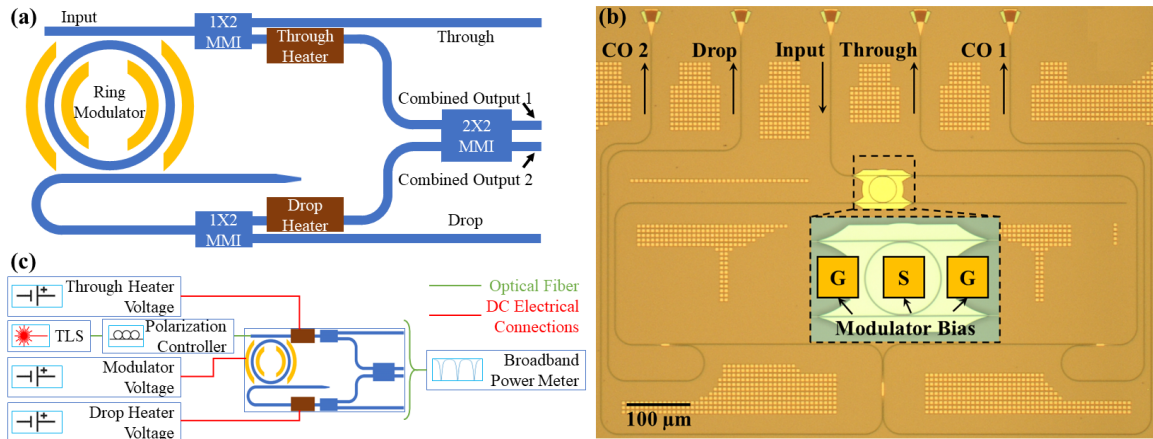


Figure 3.9: (a) The schematic diagram and (b) the microscopic image of the fabricated MRR modulator-based optical phase shifter. (c) Experimental setup for the phase shifter intensity spectra measurement.

The experimental setup for the phase shifter demonstration is shown in Fig. 3.9c. The output from a TLS is end-fire coupled into the chip after passing through a polarization controller (PC). The transmission spectra of optical outputs from the through, drop and combined 1 & 2 ports are measured using a broadband optical power meter. Optical signals are coupled in and out of the chip using the grating couplers. Three precision voltage sources have been used to sweep the voltages of the modulator bias and through & drop arm TiN heaters.

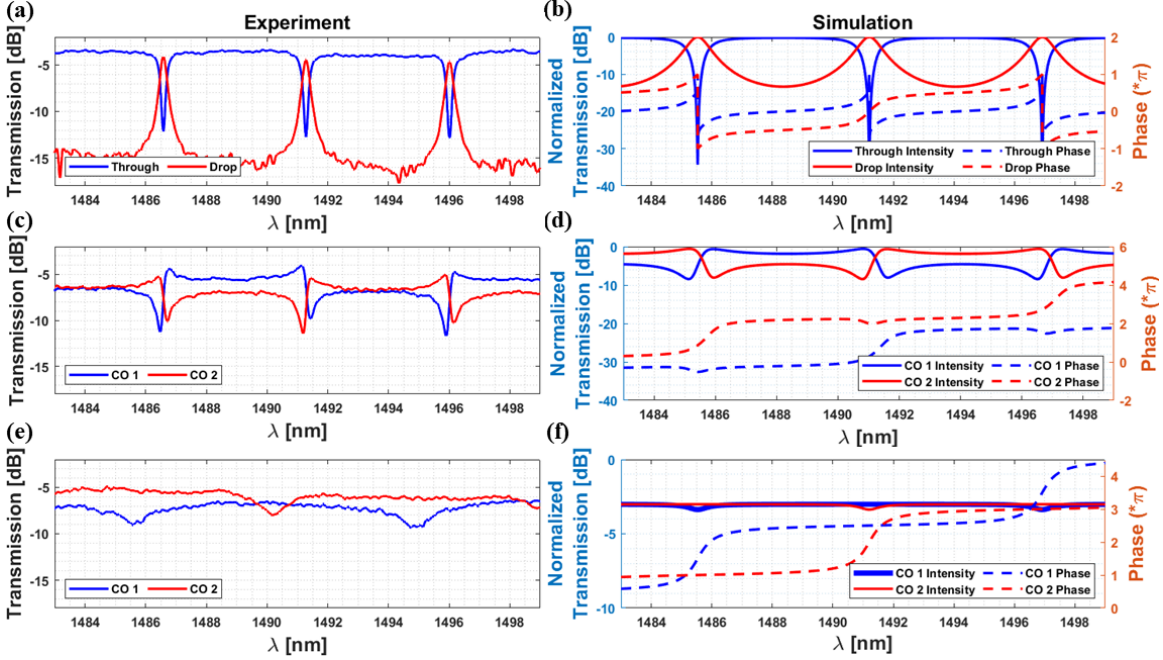


Figure 3.10: (a) Experimentally measured transmission spectra, and (b) analytical simulations of the intensity (solid lines) and phase (broken lines) spectra observed at the through (blue line) and the drop ports (red line). (c) Experimentally measured transmission spectra, and (d) analytical calculations for the intensity and phase response of both combined ports before balancing the optical path lengths of the through and the drop arms before combining. (e) Experimentally measured transmission spectra, and (f) analytical calculations for the intensity and phase response of both combined ports after balancing the optical path lengths of the through and the drop arms before combining.

Initially we started our characterization in short-wavelength band (S-band). Fig. 3.10a shows the measured transmissions spectra at the through (solid blue line) & drop (solid red line) ports which obviously look similar to the analytically simulated intensity spectra of the through (solid blue line) and drop (solid red line) ports shown in Fig. 3.10b. Fig. 3.10c shows the transmission spectra measured at both the combined output 1 & 2 when no voltage is applied at the through and drop arm heaters. These spectra are similar to the simulated intensity spectra shown in Fig. 3.10d considering $\theta = \pi/4$. This means that the optical path lengths of the through and the drop arms are not balanced to achieve the desired intensity spectra yet. To balance these arms, we applied voltage to only the through arm TiN heater. The measured transmission spectra of the combined outputs are shown in Fig. 3.10e while applying 2.52 V to the through arm heater which looks very similar to the simulated intensity spectra shown in Fig. 3.10f considering $\theta = \pi/2$ which looks like our desired intensity spectrum shown in Fig. 3.8d. Therefore, we can consider the path lengths of the through

and the drop arms to be balanced and maintained this through arm heater voltage to maintain a $\pi/2$ phase shift between the through and drop ports for the next experiments. Fig. 3.10b, 3.10d and 3.10e also show the analytically simulated phase responses (broken lines) at the respective ports which can not be measured using the optical power meter.

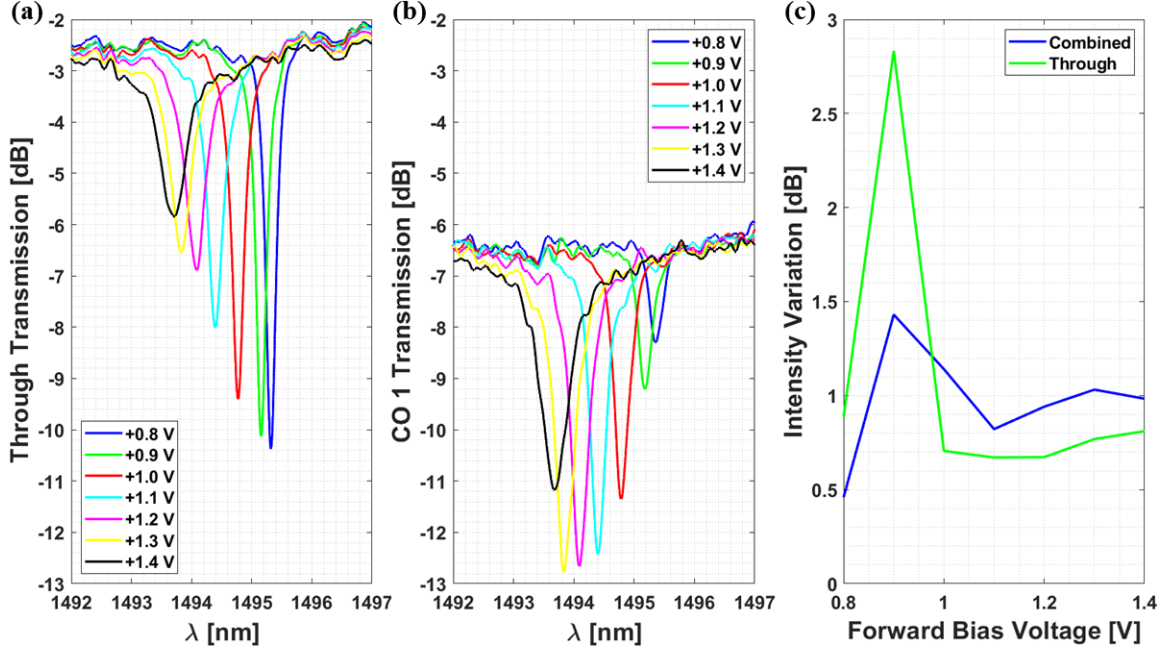


Figure 3.11: Transmission spectra measured at (a) the through port and (b) the combined port 1 for different forward bias voltages applied at the MRR modulator. (c) Transmission variation at the through port and the combined port 1 port for different forward bias voltages for an optical carrier signal 1495.01 nm.

Keeping the through arm heater voltage fixed at 2.52 V, forward bias voltages were applied to the MRR modulator. Fig. 3.11a and 3.11b show the transmission spectra of the through port and the combined output 1 port respectively for forward bias voltages ranging from 0.8 V to 1.4 V with a step size of 0.1 V. Clearly the combined port extinction ratios are much smaller than those of the through ports. An optical carrier wavelength of 1495.01 nm was selected which is close to the left side of the resonance dip. The change of the optical carrier intensities compared to the respective maximum intensities in the whole spectrum in the wavelength range 1492-1497 nm are depicted in Fig. 3.11c for both combined output 1 and through ports. In both cases, the maximum intensity differences are maximum for a forward bias voltage of 0.9 V which indicates that the carrier wavelength (1495.01 nm) becomes the valley of the resonance when the forward bias voltage is ~ 0.9 V. After 1.2 V of bias voltage intensities of combined output 1 and through port starts to return to close to their respective intensities at 0.8 V bias voltage. This indicates that by the time when 1.4 V of bias voltage was applied the resonance dip of the modulator has completely crossed the carrier wavelength from left to right in the spectrum. Hence, we can reach to a conclusion that, the carrier wavelength has been shifted by $\sim 2\pi$ while sweeping the forward bias voltage from 0.8 V to 1.4 V. The maximum intensity variation at the combined output 1 is ~ 0.97 dB whereas the maximum intensity variation at the through port is ~ 2.16 dB.

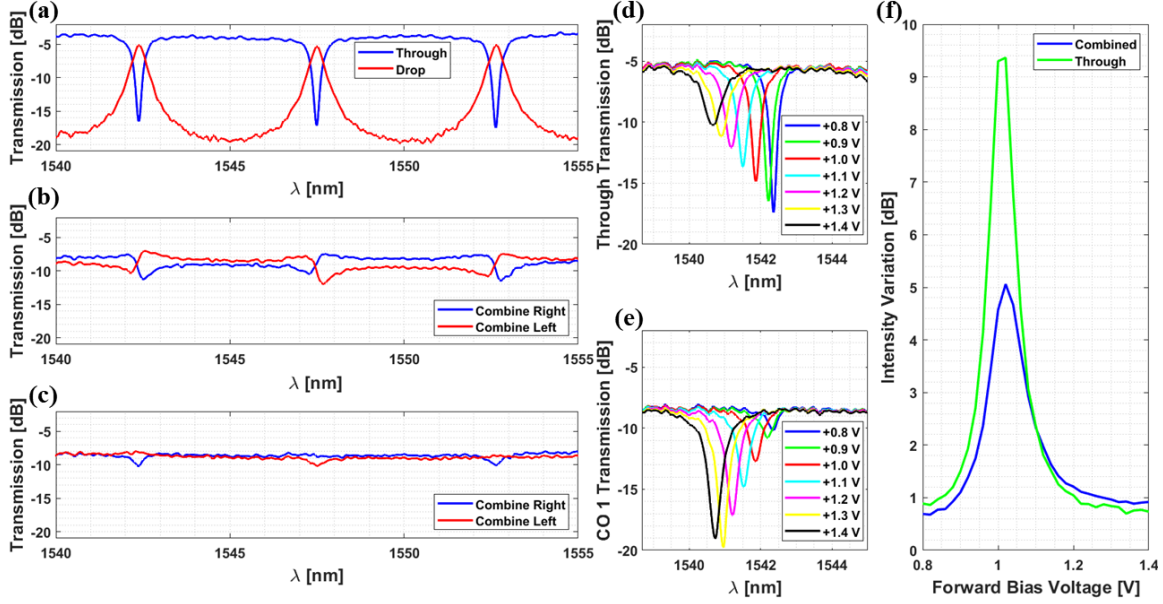


Figure 3.12: Experimentally measured transmission spectra at (a) the through (blue line) & the drop ports (red line), and at (b) the combined ports (b) before and (c) after balancing the optical path lengths of the through and the drop arms before combining. Transmission spectra measured at (d) the through port and (e) the combined port 1 for different forward bias voltages applied at the MRR modulator. (f) Transmission variation at the through port and the combined port 1 port for different forward bias voltages for an optical carrier signal 1541.83 nm.

In Fig. 3.11c we were supposed to get a bell-shaped curve for maximum intensity variation with respect to the change of forward bias modulation voltage. But we noticed slightly different curve because of the step size or the resolution of the forward bias voltage. Therefore, we repeat this experiment with much smaller step size (0.02 V) of forward bias voltage and at the same time, we improved our coupling by adjusting the grating coupler angle to measure the transmission spectra to the conventional-wavelength band (C-band). Fig. 3.12a and shows the experimentally measured transmission spectra at the through (blue line) and the drop ports (red line) in the C-band wavelengths. Fig. 3.12b and 3.12c and show the measured transmission spectra at the combined ports before and after balancing the through arm and drop arm optical path lengths respectively. For balancing the through and drop arms 1.05 V of through arm heater voltage was applied so that $\theta = \pi/2$. We kept the through heater voltage fixed at 1.05 V and swept the modulator bias voltage from 0.8 V to 1.4 V. The transmission spectra of the through port and the combined port 1 in the range of 1540-1555 nm wavelength for the aforementioned bias voltages are shown in Fig. 3.12d and 3.12e respectively. We chose a carrier wavelength of 1541.83 nm and calculated the intensity variation for different bias voltages compared to the MRR modulator biased at 0.8 V for both the through port and combined port 1 which are shown in Fig. 3.12f. The maximum intensity variation for the through port is ~8.67 dB whereas the maximum intensity variation for the combined port 1 is only ~4.38 dB. We can see that, after 1.3 V of bias voltage the intensities of combined port 1 and through port starts to return to close to their respective intensities at 0.8 V bias voltage. That means by the time when 1.4 V of bias voltage was applied the resonance region of the modulator has completely crossed the carrier wavelength (1541.83 nm) from left to right in the spectrum. We

can only assume that the carrier wavelength has been shifted by $\sim 2\pi$ while sweeping the forward bias voltage from 0.8 V to 1.4 V. But we cannot achieve the phase spectrum or actual phase shift while bias voltage is applied from this experiment. We also noticed a larger intensity variation at the combined port 1 (~ 4.38 dB) compared to our previously reported intensity variation (~ 0.97 dB) [71]. This happens because we have more measurement points in the same modulator bias voltage range (0.8-1.4 V) in this experiment compared to the previous one. Also, this time we improved the optical coupling between the chip and the fiber and moved the wavelength range in the C-band from the previously conducted experiment in the S-band wavelength range (1484-1498 nm).

3.2.2 Phase Response Extraction and Constellation Mapping

To measure the phase response of the add-drop MRR-based phase shifter we consider the device shown in Fig. 3.13a. We placed the MRR modulator in one arm of an MZI where the through port response is combined with the drop portion of the add-drop MRR using a 2×2 MMI. Like the previous device (Fig. 3.13a), a $\pi/2$ phase shift is maintained between the through and the drop ports before combining. One output (left output, $(jE_{\text{sig},2})/2$, where $(E_{\text{sig},2} = H_t E_{\text{in}} - H_d E_{\text{in}}$ from this 2×2 MMI is used to monitor the combined output (CO) of the through and drop port and the other output (right output, $(jE_{\text{sig},1})/2$, where $(E_{\text{sig},1} = H_t E_{\text{in}} + H_d E_{\text{in}}$ is used to compare with the other arm (the reference arm) of the MZI using another 2×2 MMI. A third 2×2 MMI is used in the reference arm to monitor the reference signal (right output, $E_{\text{in}}/2$). The other output from the reference arm's 2×2 MMI (left output, $jE_{\text{in}}/2$ is used to measure the phase response of the MRR modulator-based phase shifter. Two outputs from the 2×2 MMI after combining the signal arm with the reference arm are monitored (left output, $E_L = jE_{\text{in}}/2\sqrt{2} + jE_{\text{sig},1}/2\sqrt{2}$ and right output, $E_R = -E_{\text{in}}/2\sqrt{2} + E_{\text{sig},1}/2\sqrt{2}$). We did some analytical calculations to realize the intensity spectra of different optical output signals of this device. Fig. 3.13b shows the intensity spectra (solid lines) of the output signals $jE_{\text{sig},2}/2$ (blue line) and $E_{\text{in}}/2$ (green line). The phase response of $jE_{\text{sig},2}/2$ (broken blue line in Fig. 3.13b) which needs to be extracted after comparing with the flat phase response of $E_{\text{in}}/2$ (broken green line). And Fig. 3.13c shows the intensity spectra of the output signals E_L (red line) and E_R (purple line). The intensity spectra of all these optical outputs can be measured using the device shown in Fig. 3.13a. Light generated from a TLS followed by a PC is coupled to the chip using a grating coupler and all the optical output intensity responses are measured using a broadband optical power meter after coupling out of the chip using 4 grating couplers. A $50 \mu\text{m}$ TiN heater on the through arm is biased at 1.68 V to maintain a $\pi/2$ phase shift between the through and the drop ports. The measured intensity responses shown in Fig. 3.13d and 3.13e are very similar to their simulation counterparts shown in Fig. 3.13b and 3.13c respectively. The measured intensity responses of this device are outlined in Eq. 3.18-3.21. Where the signal intensity (blue box in Fig. 3.13a) is given by:

$$I_{\text{sig},2} = \left| j \frac{E_{\text{sig},2}}{2} \right|^2 = \left| j \frac{(H_t E_{\text{in}} - H_d E_{\text{in}})}{2} \right|^2 \quad (3.18)$$

The reference intensity (green box in Fig. 3.13a) is given by:

$$I_{\text{ref}} = \left| \frac{E_{\text{in}}}{2} \right|^2 \quad (3.19)$$

The left CO intensity (red box in Fig. 3.13a) is given by:

$$I_L = |E_L|^2 = \left| j \frac{E_{in}}{2\sqrt{2}} + j \frac{E_{sig,1}}{2\sqrt{2}} \right|^2 = \frac{|E_{in}|^2}{8} + \frac{|E_{sig,1}|^2}{8} + \frac{|E_{in}| |E_{sig,1}|}{4} \cos \Delta \phi \quad (3.20)$$

$$I_R = |E_R|^2 = \left| -\frac{E_{in}}{2\sqrt{2}} + \frac{E_{sig,1}}{2\sqrt{2}} \right|^2 = \frac{|E_{in}|^2}{8} + \frac{|E_{sig,1}|^2}{8} - \frac{|E_{in}| |E_{sig,1}|}{4} \cos \Delta \phi \quad (3.21)$$

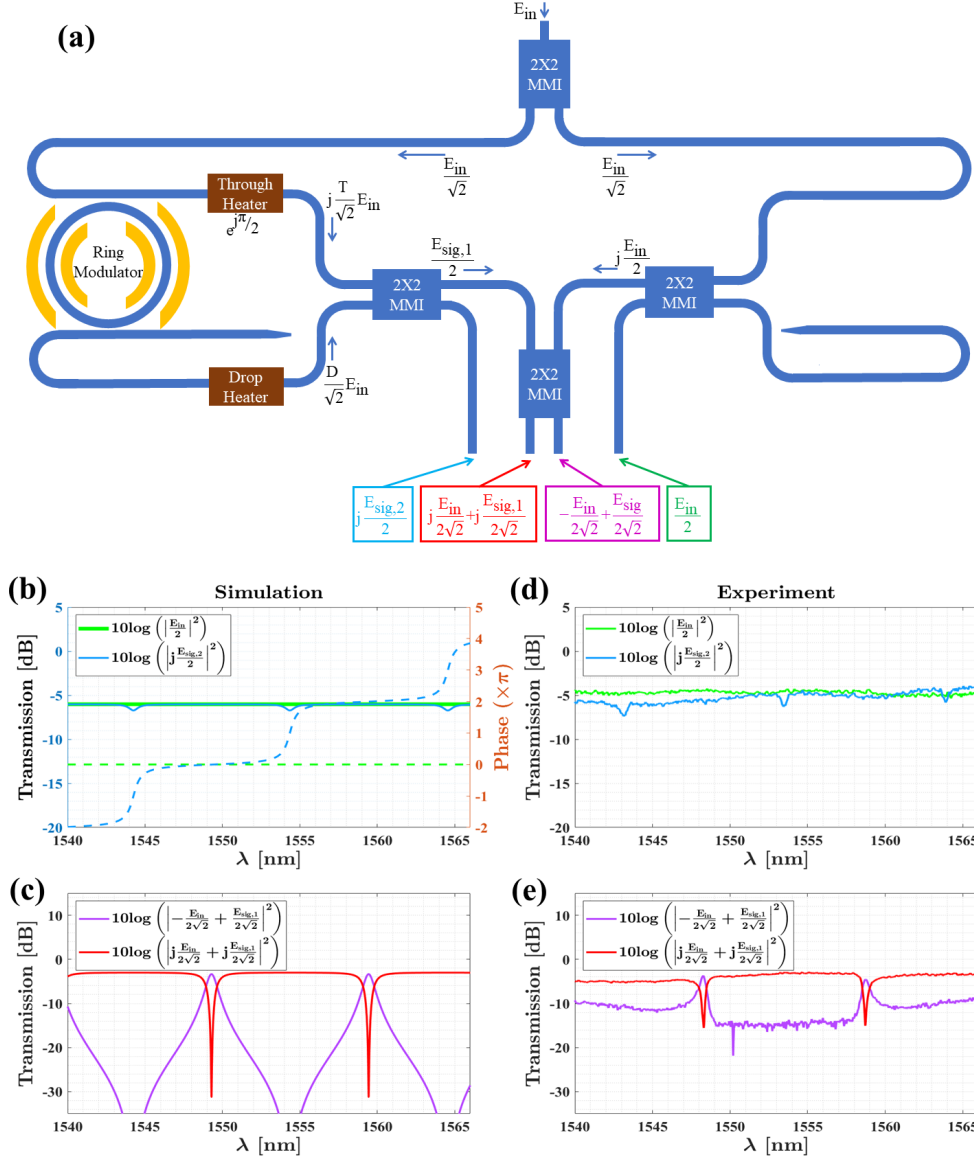


Figure 3.13: (a) The schematic diagram of the silicon add-drop MRR modulator-based phase shifter placed in one arm of an MZI. (b) $E_{in}/2$ (green line) & $jE_{sig,2}/2$ (blue line), and the intensity responses of the output waves (c) E_L (red line) & E_R (purple line). Measured transmission spectra of the optical waves (d) $E_{in}/2$ (green line) & $jE_{sig,2}/2$ (blue line), and (e) E_L (red line) & E_R (purple line).

In Eq. 3.20 and 3.21, $\Delta \phi$ is the phase difference between the input optical signal and the one output

optical signal of our proposed MRR based phase optical shifter i.e., the phase response of the MRR based optical phase shifter. If we subtract the intensity spectrum of the right combined port, I_R of the MZI from the intensity spectrum of the left combined port, I_L we get Eq. 3.22.

$$I_L - I_R = \frac{|E_{in}| |E_{sig,1}|}{2} \cos \Delta \phi \quad (3.22)$$

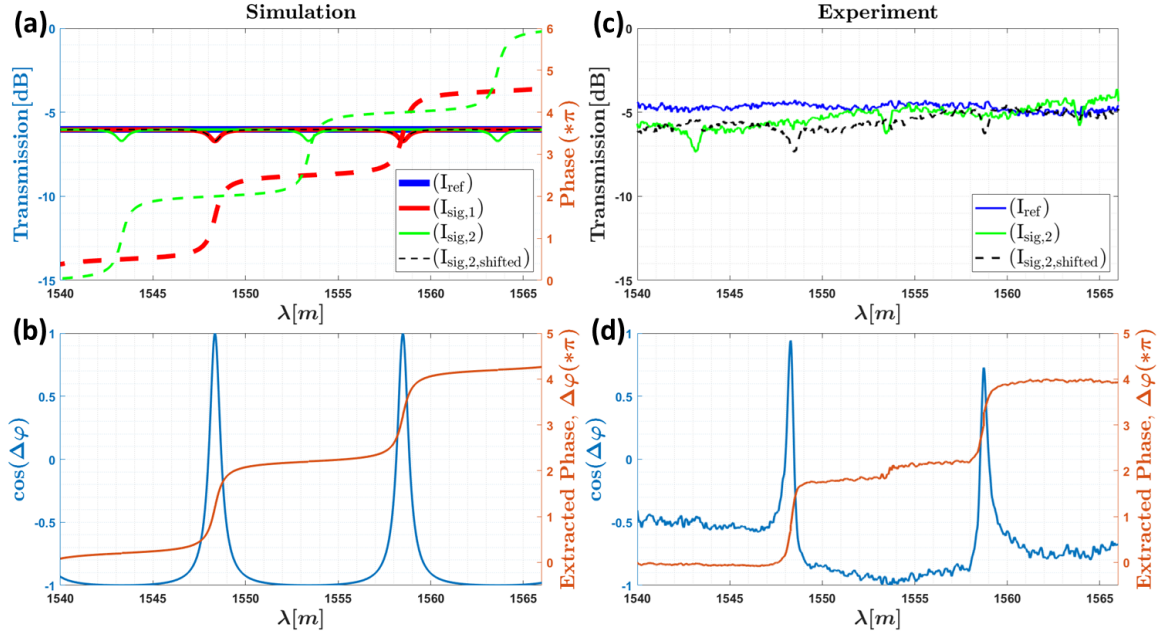


Figure 3.14: (a) Analytical simulation of the intensity spectra, I_{ref} (blue line), $I_{sig,1}$ (red line), $I_{sig,2}$ (green line) & $I_{sig,2,shifted} \approx I_{sig,1}$ (broken black line) and the phase spectra of the signal, $E_{sig,1}/2$ (broken red line) & $(jE_{sig,2}/2)$ (broken green line). (b) Experimental intensity responses, I_{ref} (blue line), $I_{sig,2}$ (green line) & $I_{sig,2,shifted} \approx I_{sig,1}$ (broken black line). (c) Analytical simulation of the extracted cosine of the phase response, $\cos \Delta \phi$ (blue line) and the extracted unwrapped phase response, $\Delta \phi$ (red line) of $E_{sig,1}/2$. (d) Extracted cosine of the phase response, $\cos \Delta \phi$ (blue line) and the extracted unwrapped phase response, $\Delta \phi$ (red line) of $E_{sig,1}/2$ from different measured transmission spectra.

To extract the phase response of the phase shifter we need $|E_{in}|$ and $|E_{sig,1}|$. We can get $|E_{in}|$ from Eq. 3.19 ($|E_{in}| = 2\sqrt{I_{ref}}$). However, we cannot get $|E_{sig,1}|$ ($= 2\sqrt{I_{sig,1}}$) from direct measurement but a good approximation can be found from $I_{sig,2}$. Again, we did some analytical calculations to compare between the intensity responses, $I_{ref} = |E_{in}|^2 / 2$ (blue line), $I_{sig,1} = |E_{sig,1}|^2 / 2$ (red line) and $I_{sig,2} = |E_{sig,2}|^2 / 2$ (green line) which are plotted in Fig. 3.14a. Fig. 3.14a also shows the simulated phase responses of $|E_{sig,1}|/\sqrt{2}$ (broken red line) and $|E_{sig,2}|/\sqrt{2}$ (broken green line). As expected, every alternative resonance out of each two resonances of the MRR disappear in the intensity response, $I_{sig,1}$ and only those disappeared resonances show up in the intensity response, $I_{sig,2}$. If we can shift the intensity spectrum $I_{sig,2}$ by one resonance towards right then the shifted $I_{sig,2}$ ($I_{sig,2,shifted}$) becomes quite similar to $I_{sig,1}$ (broken black line in Fig. 3.14a). Inserting $|E_{sig,1}| = 2\sqrt{I_{sig,1}} = 2\sqrt{I_{sig,2,shifted}}$ and $|E_{in}| = 2\sqrt{I_{ref}}$ in Eq. 3.22 we can extract $\cos \Delta \phi$ (blue line in Fig. 3.14b) i.e., the phase response, $\Delta \phi$ (red line in Fig. 3.14b) of our proposed MRR-based phase

shifter which looks exactly the same as the phase response (broken red line) of $|E_{\text{sig},1}|/\sqrt{2}$ shown in Fig. 3.14a. We carried out the same calculations to extract the phase response of the MRR based phase shifter from the experimental data. Fig. 3.14c shows the experimental intensity responses, $I_{\text{sig},2}$ (green line), $I_{\text{sig},1} \approx I_{\text{sig},2,\text{shifted}}$ (broken black line) and I_{ref} (blue line). As usual, the experimental intensity spectra match the simulations in Fig. 3.14a. The experimental $\cos\Delta\phi$ (blue line) curve and phase response, $\Delta\phi$ (red line) extracted from different intensity spectra are shown in Fig. 3.14d which look very similar to the simulations shown in Fig. 3.14b.

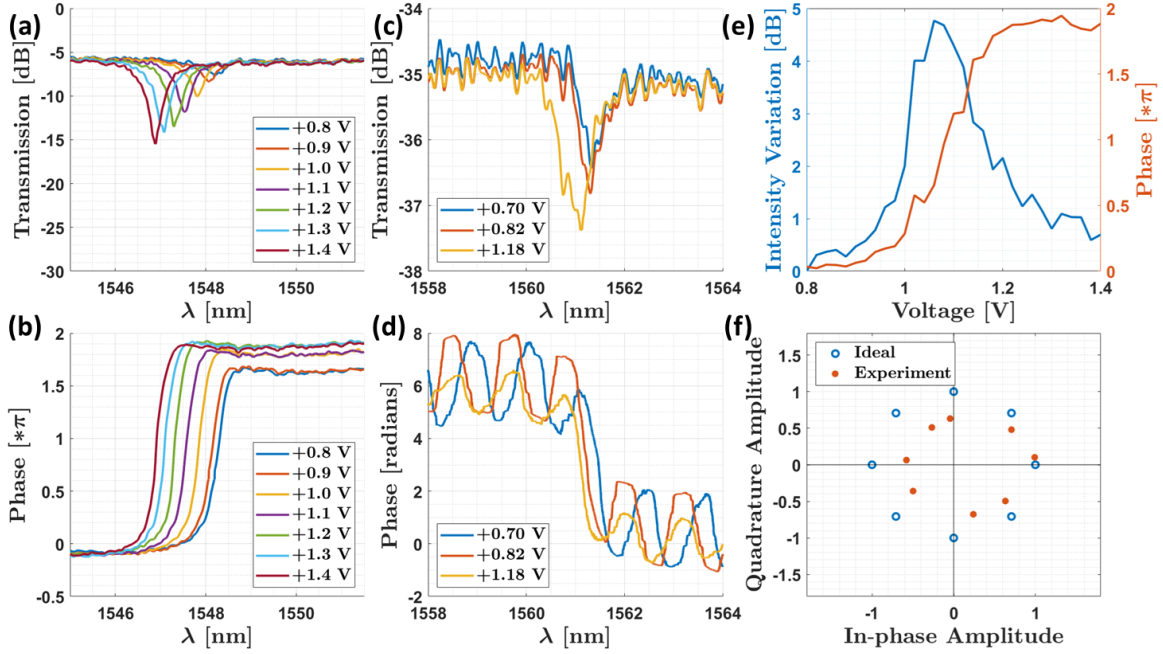


Figure 3.15: (a) Intensity and (b) extracted phase spectra of shifted $E_{\text{sig},1}/2$ for different forward bias modulation voltages in 0.8-1.4 V range. OVA measurement of (c) the intensity and (d) direct phase measurement at the combined output 1 port shown in Fig. 3.9a for forward bias modulation voltage of 0.70 V, 0.82 V and 1.18 V. (e) Transmission variation (blue line) and extracted phase shift (red line) of shifted $E_{\text{sig},1}/2$ for different forward bias voltages for an optical carrier signal 1547.6 nm. (f) Ideal (blue points) and experimentally extracted (red points) modulation points for 8-PSK modulation format.

In addition, we plotted the intensity response, $I_{\text{sig},1} \approx I_{\text{sig},2,\text{shifted}}$ and phase spectra, $\Delta\phi$ for different forward bias modulation voltages in the range 0.8-1.4 V for the wavelength range 1545-1551 nm in Fig. 3.15a and 3.15b respectively. The intensity and phase spectra for different voltages are verified using an OVA in Fig. 3.15c and 3.15d respectively. Although, because of Fabry-Perot effect and the background vibration the OVA phase response measurement shows oscillatory behavior, it clearly shows a 2π phase shift when we move from one side of the resonance to the other side. Considering a carrier wavelength of 1547.6 nm we have plotted the intensity variation as well as the phase of optical carrier for different bias voltages compared to intensity and phase of the MRR modulator when biased at 0.8 V in Fig. 3.15e. We can clearly see that the phase shifter achieves a phase shift of $\sim 1.94\pi$ while the maximum intensity variation is ~ 4.77 dB. This implies that the V_π of our MRR based optical phase shifter is 250 mV at 1547.6 nm carrier wavelength. Finally, we mapped the phase shifts and intensity of the optical phase shifter in a constellation diagram in Fig. 3.15f.

Unlike the regular constellation diagram (cartesian coordinate) used for IQ modulation we used a constellation diagram for polar coordinates. In Fig. 3.15f we have shown a constellation mapping for 8-PSK modulation format. As we expected, because of the residual amplitude modulation the amplitudes of all the modulation points are not the same. Although unlike ideal case (blue points in Fig. 3.15f) the experimental modulation points (red points in Fig. 3.15f) do not stay on the same amplitude circle, the modulation points can still be separated from each other. Also, some experimental modulation points are a little bit off from their original phase position which can be corrected by fine tuning the modulation voltages. By integrating an intensity modulator in series with our proposed phase shifter the degree of freedom can easily be increased to realize higher order modulation formats such as, 16-APSK etc.

3.3 Conclusion

In this chapter, we demonstrated an optical phase shifter fabricated on the standard SOI wafer in a commercial photonic foundry, AMF by combining the through and drop ports of a silicon add-drop MRR modulator. We fabricated two different types of devices – a phase only modulator to compare the maximum intensity variation seen at the through port & the through-drop combined port, and a phase shifter placed in one arm of an MZI to measure the phase spectrum. After measuring the intensity response measurements of the phase only modulator we saw that the through-drop combined port shows a maximum intensity variation of ~ 4.38 dB whereas the through port shows a ~ 8.67 dB of maximum intensity while achieving $\sim 2\pi$ phase shift for a carrier wave of 1540.83 nm wavelength. In the second device we placed our proposed phase shifter in one arm of an MZI to compare the intensity response of the phase shifter with the intensity response of the reference arm to extract the phase response. Using various measured intensity spectra of this device we extracted the phase response of the device which ultimately led us to a net phase shift of ~ 1.94 while having a maximum intensity variation of ~ 4.77 dB for a CW carrier wave of 1547.6 nm wavelength when the forward bias modulation voltage was raised from 0.8 V to 1.4 V. Thus, we achieve a V_π of 250 mV for this device. Finally, we use the phase response measurement to visualize the 8-PSK constellation diagram which can be easily extended to higher order APSK modulation formats using another intensity modulator.

Chapter 4

Lithium Niobate-Barium Titanate Hybrid Modulator

This chapter discusses the fundamentals EO modulators based on second order nonlinearity with emphasis on their implementation in a TFLN or LNOI platform. This chapter is organized as follows. In Section 4.1, the second order nonlinear effect of lithium niobate (LN) and its use in EO modulation on the LNOI platform are discussed. In Section 4.2, we discussed the effect of cladding material on the state-of-the art LNOI platform-based modulators and proposed a new modulator structure with improved half-wave-voltage-length product ($V_\pi L$) which is known as the lithium niobate-barium titanate (LN-BTO) hybrid modulator. We also discussed the detailed dimensional optimization of the LN-BTO hybrid modulator in this section. Finally, this chapter is concluded in Section 4.3.

4.1 Theoretical Background

Unlike crystalline silicon LN does not exhibit a centrosymmetric crystal structure. Therefore, LN is birefringent and shows a large linear EO (Pockels) coefficient and a small non-linear EO (Kerr) coefficient which makes it ideal for linear EO modulation. In this section, we will discuss about the second order nonlinear effects or linear EO effect of LN. In addition, we will discuss about the working principle of the etched TFLN modulators which exploit this effect for EO modulation.

4.1.1 Nonlinear Optics and Pockels Effects

We start our discussion by considering the Maxwell's equations for an electromagnetic wave [71].

$$\text{Coulomb's Law: } \nabla \cdot \vec{D}(\vec{r}, t) = \rho \quad (4.1)$$

$$\text{Gauss's Law for Magnetic Fields: } \nabla \cdot \vec{B}(\vec{r}, t) = 0 \quad (4.2)$$

$$\text{Faraday's Law: } \nabla \times \vec{E}(\vec{r}, t) = -\frac{\partial}{\partial t} \vec{B}(\vec{r}, t) \quad (4.3)$$

$$\text{Ampere's Law: } \nabla \times \vec{H}(\vec{r}, t) = \frac{\partial}{\partial t} \vec{D}(\vec{r}, t) + \vec{J}(\vec{r}, t) \quad (4.4)$$

Here, $\vec{E}$, $\vec{D}$, $\vec{B}$, $\vec{H}$ and $\vec{J}$ are the electric field, the electric flux density, the magnetic flux density, the magnetic field, and the current density respectively inside the matter through which the electromagnetic wave is propagating. ρ is the charge density of the material during the electromagnetic wave propagation. Besides these four equations there are two more associative equations which governs the relationship between the fields and the flux densities.

$$\vec{D}(\vec{r}, t) = \epsilon_r \epsilon_0 \vec{E}(\vec{r}, t) + \vec{P}(\vec{r}, t) \quad (4.5)$$

$$\vec{B}(\vec{r}, t) = \mu_r \mu_0 \vec{H}(\vec{r}, t) \quad (4.6)$$

Here, $\vec{P}$ is the time and space dependent electric polarization which is actually the response of the optical medium as a result of the electromagnetic wave propagation through it. ϵ_0 (8.854×10^{-12} C²N⁻¹m⁻²) and μ_0 ($4\pi \times 10^{-7}$ NA⁻²) are the electric permittivity and magnetic permeability of free space respectively. ϵ_r and μ_r are the relative permittivity and relative permeability of the material respectively compared to the free space permittivity and permeability. For optical waves we can consider, $\rho = 0$, $\vec{J} = 0$ and $\mu_r = 1$. By taking cross product of the Faraday's law and substituting the Ampere's law we can get the wave equation for electric field.

$$\nabla^2 \vec{E}(\vec{r}, t) - \mu_0 \frac{\partial^2}{\partial t^2} (\epsilon_r \epsilon_0 \vec{E}(\vec{r}, t) + \vec{P}(\vec{r}, t)) = 0 \quad (4.7)$$

Here, $\nabla^2 = \nabla \cdot \nabla$ is the spatial Laplacian operator. The relation between the electric field and polarization can be expressed by a Volterra series [72].

$$\begin{aligned} \vec{P}(\vec{r}, t) = & \epsilon_0 \int_0^\infty \underline{\chi}^{(1)}(\vec{r}, t - t') \vec{E}(\vec{r}, t - t') dt' \\ & + \epsilon_0 \int_0^\infty \int_0^\infty \underline{\chi}^{(2)}(\vec{r}, t - t', t - t'') : \vec{E}(\vec{r}, t - t') \vec{E}(\vec{r}, t - t'') dt' dt'' \\ & + \epsilon_0 \int_0^\infty \int_0^\infty \int_0^\infty \underline{\chi}^{(3)}(\vec{r}, t - t', t - t'', t - t''') : \vec{E}(\vec{r}, t - t') \vec{E}(\vec{r}, t - t'') \vec{E}(\vec{r}, t - t''') \\ & dt' dt'' dt''' \\ & + \dots \end{aligned} \quad (4.8)$$

The possible temporal memory effects of the nonlinear optical medium are described by the causal influence function, $\underline{\chi}^{(n)}(t - t', t - t'', \dots, t - t^{(n)})$ which is known as the electric susceptibility tensor of rank $n+1$. Due to causality, $\underline{\chi}^{(n)}(t - t', t - t'', \dots, t - t^{(n)}) = 0$ for $t - t', t - t'', \dots, t - t^{(n)} < 0$ which is also confirmed by the lower boundaries of Eq. 4.8. The electric susceptibility with rank 2, $\underline{\chi}^{(2)}$ is normally several orders of magnitude than the higher order electric susceptibilities. Hence the polarization term can be split into two parts, $\vec{P}(\vec{r}, t)$ can be defined as,

$$\vec{P}(\vec{r}, t) = \vec{P}_L(\vec{r}, t) + \vec{P}_{NL}(\vec{r}, t) \quad (4.9)$$

Here, $\vec{P}_L$ and $\vec{P}_{NL}$ are the linear and nonlinear polarization respectively which take the form of Eq. 4.10 and 4.11 respectively.

$$\vec{P}_L(\vec{r}, t) = \epsilon_0 \int_0^\infty \underline{\chi}^{(1)}(\vec{r}, t - t') \vec{E}(\vec{r}, t - t') dt' \quad (4.10)$$

$$\begin{aligned} \vec{P}_{NL}(\vec{r}, t) = & \epsilon_0 \int_0^\infty \int_0^\infty \underline{\chi}^{(2)}(\vec{r}, t - t', t - t'') : \vec{E}(\vec{r}, t - t') \vec{E}(\vec{r}, t - t'') dt' dt'' \\ & + \epsilon_0 \int_0^\infty \int_0^\infty \int_0^\infty \underline{\chi}^{(3)}(\vec{r}, t - t', t - t'', t - t''') : \vec{E}(\vec{r}, t - t') \vec{E}(\vec{r}, t - t'') \vec{E}(\vec{r}, t - t''') \\ & dt' dt'' dt''' \\ & + \dots \end{aligned} \quad (4.11)$$

The background of the thesis stands on the second order nonlinearity. Therefore, we will limit our discussion to $\vec{P}^{(2)}(\vec{r}, t)$ i.e., we will consider $\underline{\chi}^{(n)} = 0$ for $n > 2$. We consider that the input electric field has a frequency spectrum composed of discrete angular frequencies $\omega_1, \omega_2, \dots, \omega_n$. Now the Fourier transform of Eq. 4.11 for second order nonlinearity can be expressed by Eq. 4.12 [72].

$$\tilde{P}_i^{(2)}(\omega_n + \omega_m) = \epsilon_0 \sum_{jk} \sum_{(nm)} \tilde{\chi}_{ijk}^{(2)}(\omega_n + \omega_m; \omega_n, \omega_m) \tilde{E}_j(\omega_n) \tilde{E}_k(\omega_m) \quad (4.12)$$

For the sake of readability, we drop the spatial dependence in the quantities of Eq. 4.12 where $\tilde{\chi}^{(2)}$ is the second order nonlinear electric susceptibility. The (nm) notation means that, the summation over the n and m indices includes only those sets of frequencies ω_n, ω_m which result in a frequency $\omega_n + \omega_m$. The indices i, j, k = 1, 2, 3 indicate the three orthogonal coordinates of the nonlinear optical medium. When we think about mixing an electrical signal with optical signal, we can consider the electrical signal, ω_n (usually in the GHz range) varies very slowly compared to the optical signal i.e., the frequency of the electrical signal, ω_m (usually in the GHz range) is very low compared to the frequency of the optical signal, $\omega_n = \omega_c$ (~193 THz for optical communication). Then Eq. 4.12 can be re-written as,

$$\tilde{P}_i^{(2)}(\omega_c + \omega_m \approx \omega_c) = \epsilon_0 \sum_{jk} \tilde{\chi}_{ijk}^{(2)}(\omega_c + \omega_m; \omega_c, \omega_m) \tilde{E}_j(\omega_c) \tilde{E}_k(\omega_m) \quad (4.13)$$

The factor of 2 in Eq. 4.13 comes as a result of the permutation symmetry of the input electric fields which causes a permutation symmetry of the susceptibility tensor i.e., $\tilde{\chi}_{ijk}^{(2)}(\omega_c + \omega_m; \omega_c, \omega_m) = \tilde{\chi}_{ikj}^{(2)}(\omega_c + \omega_m; \omega_m, \omega_c)$. Applying the Einstein notation for sums and using the Eq. 4.13 the electric field displacement, $\tilde{D}_i(\omega_c)$ spectrum can be written as,

$$\tilde{D}_i(\omega_c) = \tilde{P}_L + \tilde{P}_{NL} = \epsilon_0 \left(\delta_{ij} + \tilde{\chi}_{ijk}^{(1)}(\omega_c) \right) + 2\epsilon_0 \tilde{\chi}_{ijk}^{(2)} \tilde{E}_j(\omega_c) \tilde{E}_k(\omega_m) \quad (4.14)$$

Here, δ_{ij} is known as the Kronecker Delta which is expressed by Eq. 4.15.

$$\delta_{ij} = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases} \quad (4.15)$$

The low frequency (ω_m) electric field dependent electric permittivity of the second order nonlinear optical medium can be defined as,

$$\epsilon_{ij} = \epsilon_0 \left(\delta_{ij} + \tilde{\chi}_{ijk}^{(1)} + 2\epsilon_0 \tilde{\chi}_{ijk}^{(2)} (\omega_m) \tilde{E}_k (\omega_m) \right) = \epsilon_0 (\epsilon_{r,ij} + \Delta\epsilon_{r,ij}) \quad (4.16)$$

The change of relative permittivity of the nonlinear optical medium upon applying a low frequency electric field, $\Delta\epsilon_{r,ij}$ is,

$$\Delta\epsilon_{r,ij} = 2\tilde{\chi}_{ijk}^{(2)} \tilde{E}_k (\omega_m) \quad (4.17)$$

A special case for the Eq. 4.16 is found when both the low frequency electric field and the electric field of the optical field are polarized along the axis 3. In such case, $\tilde{\chi}_{ijk}^{(2)}$ can be assumed to be the dominant component of the electric permittivity tensor. Thus, the change of permittivity is given by,

$$\Delta\epsilon_r = 2\tilde{\chi}_{333}^{(2)} \tilde{E}_3 (\omega_m) \quad (4.18)$$

If the refractive index of the medium before applying the low frequency electric field is n and the small change of the refractive index of the optical medium is Δn then,

$$\epsilon_r + \Delta\epsilon_r = (n + \Delta n)^2 \approx n^2 + 2n\Delta n \quad (4.19)$$

Comparing Eq. 4.18 and 4.19 the expression we get,

$$\Delta n = \frac{\Delta\epsilon_r}{2n} = \frac{\tilde{\chi}_{333}^{(2)}}{n} \tilde{E}_3 (\omega_m) \quad (4.20)$$

Eq. 4.20 is known as the famous Pockels equation which describes the change of refractive index in an optical medium depends linearly on the applied low frequency electric field. This is why the Pockels effect is also known as the linear EO effect.

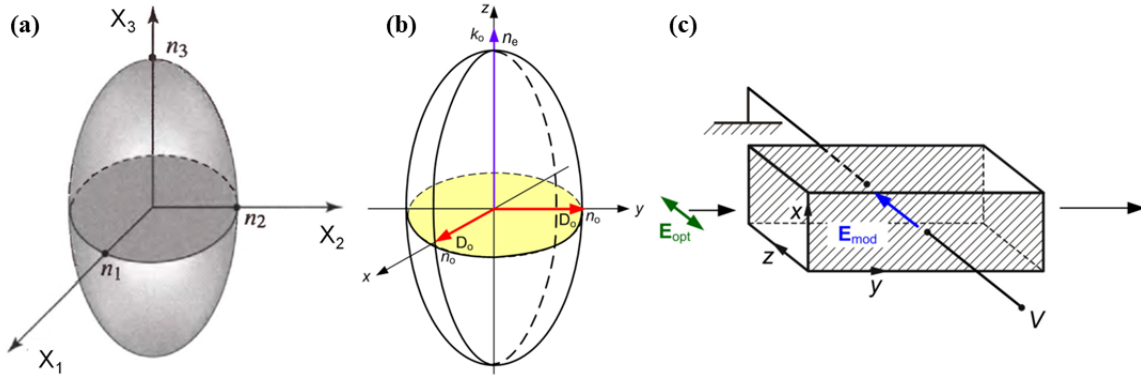


Figure 4.1: (a) The index ellipsoid with refractive index of n_1 , n_2 and n_3 along the principal axes X_1 , X_2 and X_3 respectively. (b) The index ellipsoid of a uniaxial crystal for light propagation along one of the principal axes. (c) Light propagation through a transverse EO modulator. Figures were reproduced from Ref. [71].

For an anisotropic crystalline material like LN the Pockels effect can be described as the deformation of the optical index ellipsoid of the crystal. Fig. 4.1 shows the schematic diagram of an index ellipsoid or the optical indicatrix of an anisotropic medium where X_1 , X_2 and X_3 are the principal axes. The

refractive indices along the three principal axes are n_1 , n_2 and n_3 respectively. Mathematically the index ellipsoid can be defined as,

$$\left(\frac{1}{n^2}\right)_1 X_1^2 + \left(\frac{1}{n^2}\right)_2 X_2^2 + \left(\frac{1}{n^2}\right)_3 X_3^2 + 2\left(\frac{1}{n^2}\right)_4 X_2 X_3 + 2\left(\frac{1}{n^2}\right)_5 X_1 X_3 + 2\left(\frac{1}{n^2}\right)_6 X_1 X_2 = 1 \quad (4.21)$$

In such case, the EO coefficient, r_{ijk} is preferred over the second order susceptibility tensor, $\tilde{\chi}_{ijk}^{(2)}$ from Eq. 4.17 while dealing with the EO modulation of the optical field. The change of the refractive index in terms of EO coefficient matrix is defined by,

$$\Delta\left(\frac{1}{n^2}\right)_i = \sum r_{ijk} E_k \quad (4.22)$$

When the frequency of the optical wave, ω_c is far away from the resonance of the optical medium then we can consider the Kleinman symmetry or permutation symmetry of the input fields. Thus, the number of independent components of $\tilde{\chi}_{ijk}^{(2)}$ is reduced to 18 from 27. The tensors $\tilde{\chi}_{ijk}^{(2)}$ are expressed by contracted notations $\tilde{\chi}_{ih}^{(2)}$ and r_{ih} respectively where,

$$h = \begin{cases} 1 & \text{for } jk=11 \\ 2 & \text{for } jk=22 \\ 3 & \text{for } jk=33 \\ 4 & \text{for } jk=23 \text{ or } 32 \\ 5 & \text{for } jk=13 \text{ or } 31 \\ 6 & \text{for } jk=12 \text{ or } 21 \end{cases} \quad (4.23)$$

The $h=jk$ contracted notations were also use in Eq. 4.21. Then Eq. 4.22 can be written in a matrix form.

$$\begin{bmatrix} \Delta\left(\frac{1}{n^2}\right)_1 \\ \Delta\left(\frac{1}{n^2}\right)_2 \\ \Delta\left(\frac{1}{n^2}\right)_3 \\ \Delta\left(\frac{1}{n^2}\right)_4 \\ \Delta\left(\frac{1}{n^2}\right)_5 \\ \Delta\left(\frac{1}{n^2}\right)_6 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \\ r_{41} & r_{42} & r_{43} \\ r_{51} & r_{52} & r_{53} \\ r_{61} & r_{62} & r_{63} \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ E_3 \end{bmatrix} \quad (4.24)$$

LN has the 3m symmetry which makes $r_{12} = -r_{22} = -r_{61}$, $r_{13} = r_{23}$, $r_{42} = r_{51}$ and $r_{11} = r_{21} = r_{31} = r_{32} = r_{41} = r_{43} = r_{52} = r_{53} = r_{62} = r_{63} = 0$. Thus, the Eq. 4.24 can be modified as,

$$\begin{bmatrix} \Delta\left(\frac{1}{n^2}\right)_1 \\ \Delta\left(\frac{1}{n^2}\right)_2 \\ \Delta\left(\frac{1}{n^2}\right)_3 \\ \Delta\left(\frac{1}{n^2}\right)_4 \\ \Delta\left(\frac{1}{n^2}\right)_5 \\ \Delta\left(\frac{1}{n^2}\right)_6 \end{bmatrix} = \begin{bmatrix} 0 & r_{12} & r_{13} \\ 0 & -r_{12} & r_{13} \\ 0 & 0 & r_{33} \\ 0 & r_{51} & 0 \\ r_{51} & 0 & 0 \\ r_{12} & 0 & 0 \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ E_3 \end{bmatrix} \quad (4.25)$$

In LN modulators the direction of light propagation is normally considered along a principal axis. Hence, we can consider X_1 , X_2 and X_3 as x, y and z-axis respectively. Moreover, LN is a uniaxial crystal i.e., the refractive indices along two principal axes are the same which is different from the refractive index of the third principal axes. The identical refractive indices along two principal axes are known as the ordinary principal axes, $n_o = n_x = n_y = n_1 = n_2$ whereas the refractive index along the third principal axis is known as the extraordinary refractive index, $n_e = n_z = n_3$ (shown in Fig. 4.1b). For LN EO modulators the low frequency electric field is applied perpendicular to the direction of the light propagation direction. Generally, the preferred direction of optical wave propagation is along the y-axis as depicted in Fig. 4.1c. Therefore, we also call these modulators as the transverse EO modulators. The low frequency electrical field is applied along the z-axis (Z-cut LN). Altogether after applying an external low frequency electric field along the z-axis, $\vec{E}_{\text{mod}} = E_z \vec{e}_z$ the modified index ellipsoid becomes,

$$\left(\frac{1}{n_o^2} + r_{13}E_z\right)x^2 + \left(\frac{1}{n_o^2} + r_{13}E_z\right)y^2 + \left(\frac{1}{n_e^2} + r_{33}E_z\right)z^2 = 1 \quad (4.26)$$

For a z-polarized plane wave, the field induced refractive index change can be found from the following expression,

$$\Delta n = -\frac{1}{2}n_e^3 r_{33} E_z \quad (4.27)$$

If the plane wave propagates through LN for a distance of L along z-direction then the corresponding accumulated phase shift can be given by,

$$\Delta\phi = -\frac{2\pi}{\lambda_c} \Delta n L = \frac{n_e^3 r_{33} \pi E_z L}{\lambda_c} \quad (4.28)$$

Here, $\lambda_c = 2\pi c/\omega_c$ and c are the wavelength and speed of the optical wave free in free space. If the modulating low frequency electric field E_z depends on the externally applied voltage V_m and the separation between the electrodes is d, then we can consider $E_z = V_m/d$. This leads to a half-wave voltage can be expressed by,

$$V_\pi = \frac{d}{L} \frac{\lambda_c}{n_e^3 r_{33}} \quad (4.29)$$

We can see from Eq. 4.29 that if r_{33} increases the V_π of the device decreases. LN has a large EO coefficient of $r_{33} = 31$ pm/V and transparent over a wide range of wavelength (340-4600 nm) which makes it an ideal material for EO modulation.

4.1.2 LNOI Platform-Based EO Modulators

In previous section, we considered plane wave propagation through LN. But in commercial EO modulators light propagates through a waveguide. Therefore, instead of considering the change of refractive index, Δn we consider the change of effective refractive index, Δn_{eff} while designing the modulator. Conventional LN modulators are based on low index contrast ($\Delta n \sim 0.02$) LN waveguides fabricated by titanium in-diffusion [31] which have large optical modes leading to power hungry, bulky modulators with high operating voltages. In recent years, LNOI platform based EO modulators have shown excellent performance which are based on high contrast rib waveguides fabricated by controlled dry etching of LN [43, 44, 45]. In integrated waveguides like this the

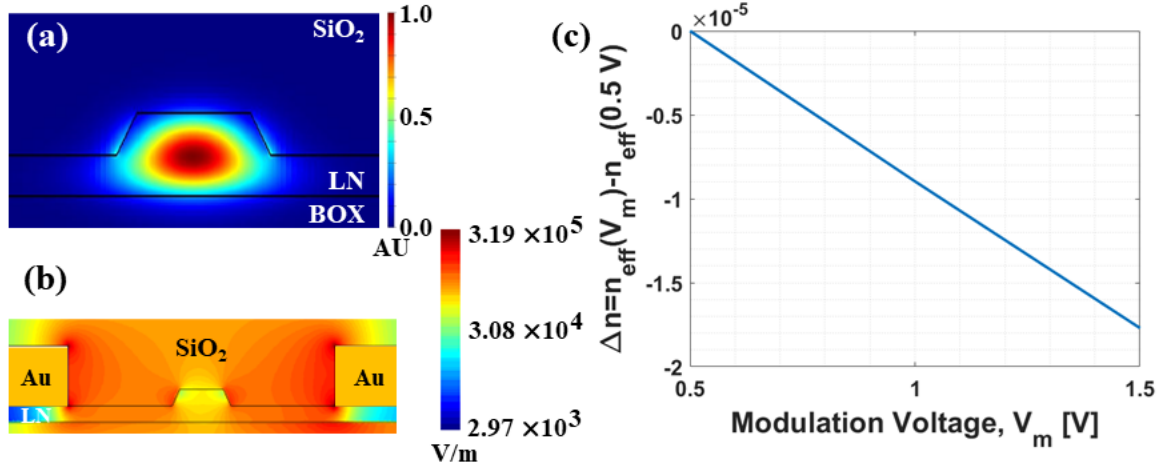


Figure 4.2: (a) The fundamental optical TE mode confined in the core region of the state-of-the-art etched TFLN based rib waveguide. (b) Electrical field distribution around the waveguide at one arm of a push-pull MZI modulator for $V_m = 0.5$ V. (c) Change of refractive index, $\Delta n_{\text{eff}}(V_m)$ with respect to V_m . Dimensions from Ref. [43] were used in Lumerical Mode and Device simulations.

electric field distribution is also not quite uniform around the waveguide core region. Therefore, straightforward use of Eq. 4.29 will not give correct estimate of V_π of the device. Commercial simulation tools like Lumerical Mode and Device are used to estimate the $V_\pi L$ of the device. Fig. 4.2a shows the fundamental optical TE mode of a state-of-the-art TFLN based etched LN rib waveguide simulated in Lumerical Mode. For such devices X-cut TFLN (x-axis is perpendicular to the TFLN surface) and MZI architecture is preferred where modulation voltage, V_m is applied at the center conductor of a CPW or GSG transmission line while grounding the other conductors. Thus, by using push-pull operation the modulation is performed. Fig. 4.2b shows the electrical field distribution around the waveguide region of one arm of the MZI for $V_m = 0.5$ V. This electric field distribution is each mesh is used in Eq. 4.25 to calculate the modified index ellipsoid of LN followed by modified effective refractive index $n_{\text{eff}}(V_m)$ in the respective meshes. For different V_m the change of refractive index, $\Delta n_{\text{eff}}(V_m)$ can be calculated which can be plotted in Fig. 4.2c. The following equation set can be used to calculate the $V_\pi L$ of a push-pull MZI modulator.

$$V_\pi = V_{m,\text{max}} - V_{m,\text{min}} \quad (4.30)$$

$$\Delta n_{\text{eff}} = n_{\text{eff}}(V_{m,\text{max}}) - n_{\text{eff}}(V_{m,\text{min}}) \quad (4.31)$$

$$L_\pi = \frac{\lambda}{4\Delta n_{\text{eff}}} \quad (4.32)$$

$$V_\pi L = V_\pi L_\pi = \frac{\lambda (V_{m,\text{max}} - V_{m,\text{min}})}{4(n_{\text{eff}}(V_{m,\text{max}}) - n_{\text{eff}}(V_{m,\text{min}}))} \quad (4.33)$$

Here, $V_{m,\text{min}}$ and $V_{m,\text{max}}$ are the minimum and maximum modulation voltage applied during the voltage simulation sweep of the modulator structure. The main disadvantage of these devices is that the electrical field confinement within the core region is very small because of lower dielectric constant of silicon dioxide ($\epsilon_{\text{SiO}_2} = 3.9$) cladding compared to LN ($\epsilon_{\text{LN}} = 28$). In this thesis, this problem has been addressed and lithium niobate-barium titanate (LN-BTO) hybrid modulator has

been proposed as a potential solution.

4.2 Lithium Niobate-Barium Titanate Hybrid Modulator

In this section, we will first discuss about the effect of dielectric constant of the cladding material of an etched LN rib waveguide (ϵ_{cl} on electrical field confinement in the waveguide core region i.e., the V_π of the device [73]. We optimized the modulator dimensions considering BTO as the cladding material and estimated the bandwidth of the modulator [74].

4.2.1 Effect of Cladding Material on TFLN Based Modulator Performance

We consider the cross-sectional view of a TFLN based modulator is shown in Fig. 4.3a. The LNOI wafer has a stack of x-cut TFLN/BOX/silicon where the thicknesses of x-cut TFLN, BOX and silicon are 600 nm, 4.7 μm and 500 μm respectively. The most important part of this modulator is the LN rib waveguide which has a core thickness of 300 nm having a sidewall angle of 65° whereas the total thickness of the LN thin film is 600 nm leaving a slab thickness of 300 nm. A width of 800 nm of the top surface of the LN waveguide core ensures the single mode operation. For the transmission line, 1.1 μm thick gold (Au) conductors were placed on the LN slab on both sides of the LN waveguide core with a gap of $\sim 5 \mu\text{m}$ between two metal lines (corresponding to a gap of 1.96 μm between the edge of the waveguide and the metal). The whole structure is covered with a cladding of 800 nm. All these dimensions are adapted from [43] where silicon dioxide was used as a cladding. Lumerical Mode and Device simulations show that, for a silicon dioxide cladding the $V_\pi L$ is $\sim 2.19 \text{ V.cm}$ for an MZI configuration with a calculated metal absorption loss of $\sim 0.3 \text{ dB/cm}$. The simulated value of $V_\pi L$ is in a good agreement with the $V_\pi L$ ($\sim 2.2 \text{ V.cm}$) reported in [43]. We repeated these simulations for air and BTO cladding keeping all the dimensions same. Simulation results show that, the $V_\pi L$ values are $\sim 2.3 \text{ V.cm}$ & $\sim 1.75 \text{ V.cm}$ for air & BTO claddings respectively. TFLN modulator with BTO cladding can achieve such low $V_\pi L$ because of the significantly higher static electric field drop across the LN waveguide core. We see this significant improvement of $V_\pi L$ because BTO (~ 40) [23] has a higher dielectric constant than air, silicon dioxide and even LN (~ 28). The enhancement of the static electric field drop across the LN core can be seen from Fig. 4.3b to 4.3d and the corresponding normalized optical field intensity distributions can be seen from Fig. 4.3e to 4.3g. From Fig. 4.3h it is seen that, the n_{eff} of the LN rib waveguide drops faster with the applied voltage while using BTO cladding than the air and silicon dioxide claddings.

High confinement of both the optical mode and the RF electric field within the core region of the LN rib waveguide are necessary for the best device performance. Hence, an ideal cladding material should have a very low refractive index at 1550 nm wavelength and a very high dielectric constant in the RF domain. Therefore, although BTO cladding improves the $V_\pi L$ of the device shown in Fig. 4.3a compared to silicon dioxide cladding, it increases the optical loss due to metal absorption since BTO has a slightly higher refractive index than silicon dioxide [12]. Hence, before diving into the comparison of different cladding materials we established a colormap showing $V_\pi L$ for different refractive indices and dielectric constants considering a modulator structure similar to

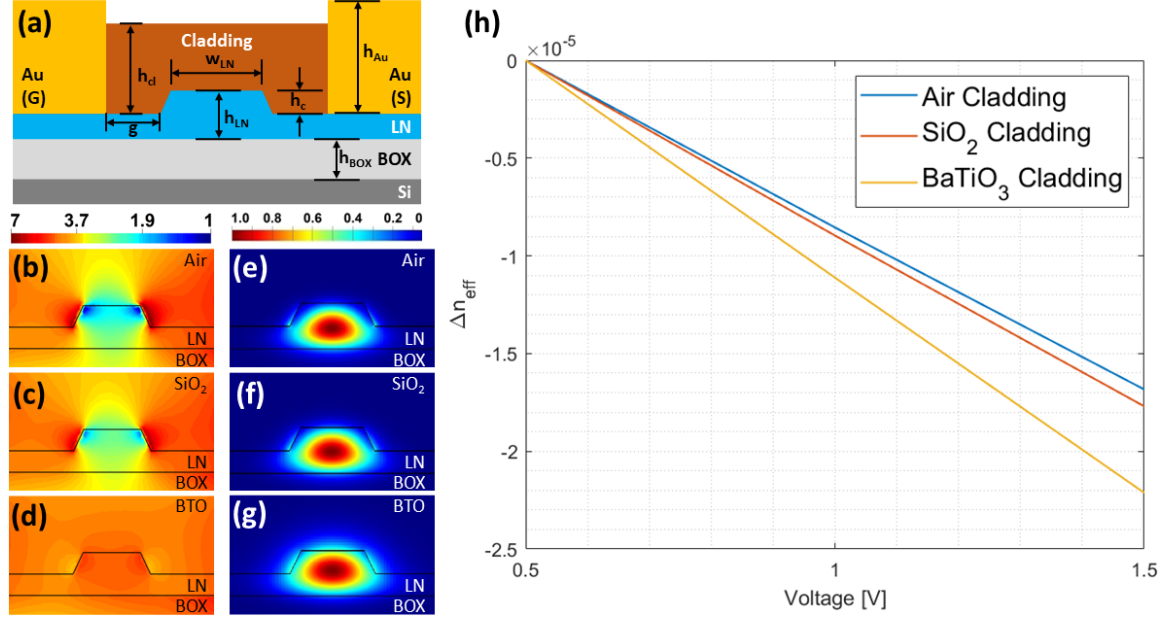


Figure 4.3: (a) Cross-sectional view of TFLN based modulator (dimensions are adapted from [1]). Electrostatic field distributions of TFLN modulators using (b) air, (c) silicon dioxide & (d) BTO claddings (the scale is normalized by 2.21×10^4 V/m) and corresponding electric field intensities for the optical modes (e), (f) & (g) respectively). (h) Change of optical effective indices for the TFLN rib waveguides for different claddings.

Fig. 4.3a. All the dimensions such as the thickness of the LN layer (c), the LN etch depth (h_c), the metal layer thickness (h_{Au}), the cladding thickness (h_{cl}) and the BOX layer thickness (h_{BOX}) are considered same as Ref. [43] i.e., h_{LN} , h_c , h_{Au} , h_{cl} and h_{BOX} are considered 600 nm, 300 nm, 1.1 μ m, 800 nm and 4.7 μ m respectively. A refractive index sweep of the cladding material (n_{cl}) was done from 1.4 to 2.1 with a step size of 0.05. For each n_{cl} the LN waveguide core width (w_{LN}) is maximized ($w_{LN,max}$) in such a way that the waveguide supports only the fundamental TE mode. All the values of n_{cl} are considered for 1550 nm wavelength. The blue curve of Fig. 4.4a shows the $w_{LN,max}$ with the change n_{cl} at 1550 nm wavelength. Waveguides with core widths less than $w_{LN,max}$ support only the fundamental TE mode, but for smaller core widths the evanescent tail in the cladding region will expand further away from the core region which will limit the gap between the metal layer and the waveguide core (g). For each set of n_{cl} and $w_{LN,max}$ the gap, g is minimized ($g_{0.3dB/cm}$) in such a way that the optical loss due to the metal absorption remains just below 0.3 dB/cm. The red curve of Fig. 4.4a shows the change of $g_{0.3dB/cm}$ with the change n_{cl} at 1550 nm wavelength. For each set of n_{cl} , $w_{LN,max}$ and $g_{0.3dB/cm}$ a sweep of DC dielectric constant of the cladding material (ϵ_{cl}) has been done from 3 to 45 with a step size of 3 in order to calculate the $V_{\pi}L$. Thus, a 15×15 matrix of $V_{\pi}L$ was calculated for all combinations of n_{cl} and ϵ_{cl} . Next a cubic interpolation of the 15×15 $V_{\pi}L$ is carried out to convert this into a 1401×1401 $V_{\pi}L$ matrix which is depicted in a colormap in Fig. 4.4b. Fig. 4.4b also includes the contour lines of different $V_{\pi}L$ and shows where different cladding materials such as silicon dioxide (SiO_2), magnesium oxide (MgO), aluminium oxide (Al_2O_3), barium titanate ($BaTiO_3$ or BTO). BTO (with $n_{cl} = 1.85$ [12] and $\epsilon_{cl} = 40$ [23]) sits on the left side of the 1.9 V.cm $V_{\pi}L$ contour line whereas SiO_2 (with $n_{cl} = 1.4431$ [75] and $\epsilon_{cl} = 3.9$ [62]), MgO (with $n_{cl} = 1.7146$ [76] and $\epsilon_{cl} = 9.9$ [77]) and Al_2O_3 (with $n_{cl} = 1.6216$

[75] and $\epsilon_{cl} = 7.8$ [78]) on the right side of 1.9 V.cm $V_{\pi}L$ contour line. Therefore, it is clear that BTO is a better cladding material than other aforementioned materials in order to achieve a lower $V_{\pi}L$.

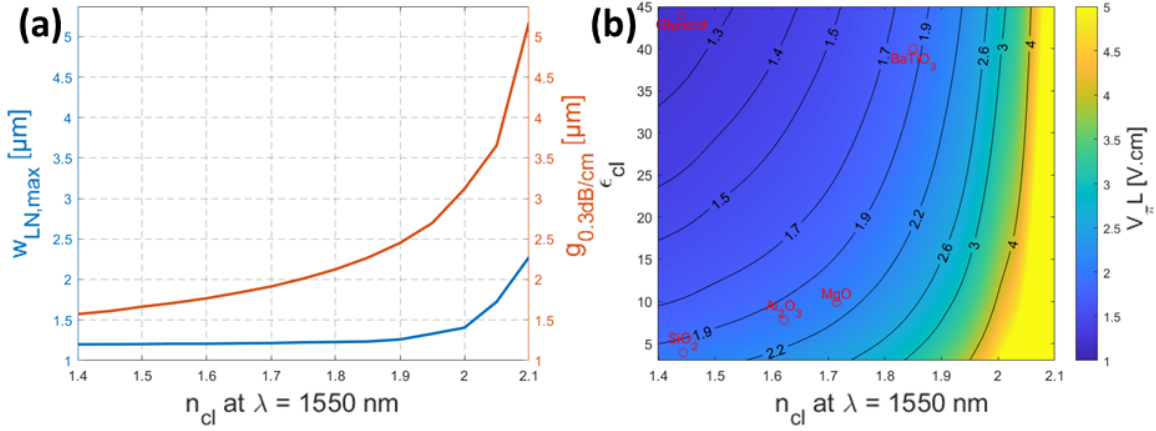


Figure 4.4: (a) Change of $w_{LN,max}$ and $g_{0.3dB/cm}$ with respect to the change of cladding material's refractive index, n_{cl} at 1550 nm wavelength. (b) Colormap of 1401×1401 $V_{\pi}L$ matrix calculated by cubic interpolation of the previously calculated 15×15 $V_{\pi}L$ matrix with a contour of different $V_{\pi}L$ values. The prospect of different cladding materials such as SiO₂, MgO, Al₂O₃ and BaTiO₃ are benchmarked on this $V_{\pi}L$ colormap.

4.2.2 Optimization of LN-BTO Hybrid Modulator

4.2.2.1 Waveguide Structure Optimization

According to Fig. 4.4b, BTO shows better potentiality as a cladding material. Hence, we moved forward with the optimization of LN-BTO Hybrid Modulator. The use of BTO as a cladding material greatly enhances the static electric confinement within the LN waveguide core region. However, the optical confinement within the LN core region is compromised due to the lower refractive index contrast between LN (2.21) and BTO (1.85) [12] at 1550 nm which eventually increases the chance of higher optical metal absorption loss. Hence, the waveguide structure i.e., the LN etch depth, h_c , the gap between the LN core and the metal electrode, g , the metal electrode thickness, h_{Au} etc. need to be optimized to realize an efficient LN-BTO hybrid modulator. In our first proposal, we showed a cross-sectional view of the LN-BTO hybrid modulator like Fig. 4.5a [73]. In this structure we considered an additional buffer BTO layer with a thickness, h_{buf} between the LN slab and Au electrodes since it was observed previously that such a buffer dielectric layer is able to reduce the optical loss due to the metal absorption to some extent [79].

We start the optimization process by considering the same LNOI wafer stack of x-cut LN/BOX/silicon where the LN, BOX and silicon layers have a thickness of 600 nm, 4.7 μm and 500 μm respectively. Other device thicknesses such as, the electrode layer thickness, h_{Au} , and the BTO buffer layer thickness, h_{buf} are considered to be 100 nm and 200 nm respectively. The sidewall angle of the LN waveguide core is 65°. Initially we consider the etch depth of LN waveguide core, h_c to be 300 nm. For this core height a core width, $w_{LN} = 1.2$ μm will ensure the dominant single TE mode operation. The gap between the Au electrode and the LN waveguide core edge, $g \sim 1.8$ μm (metal to metal

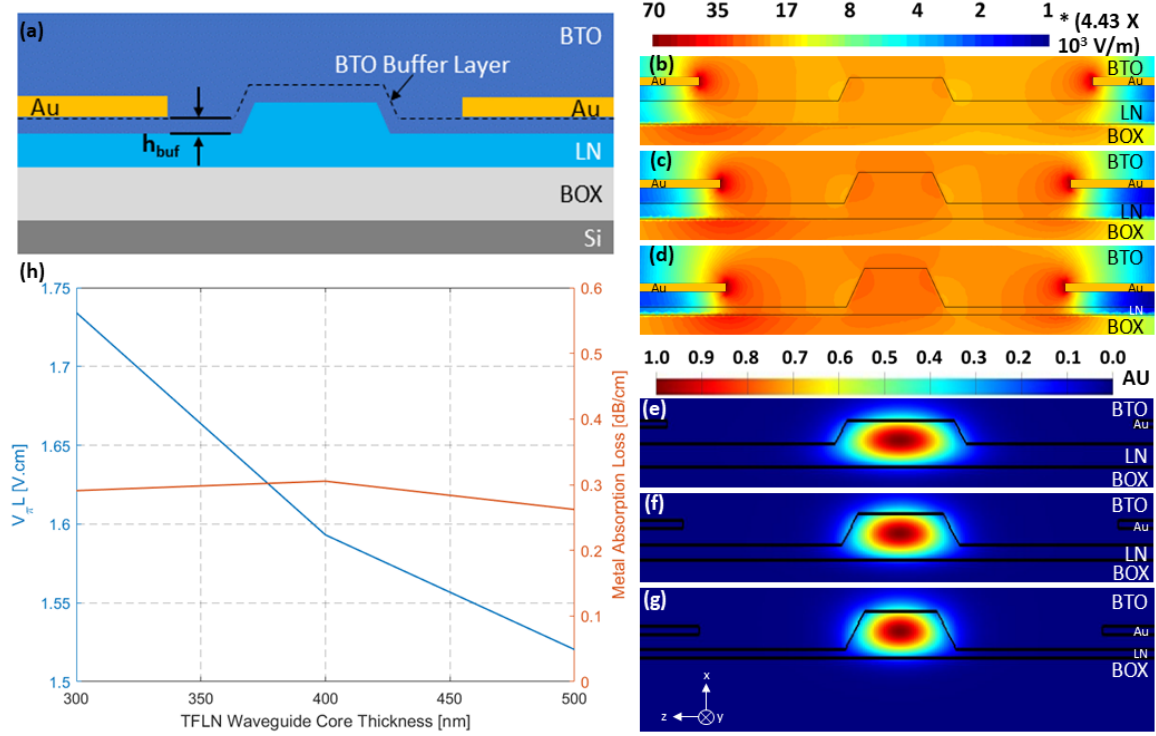


Figure 4.5: (a) Cross-sectional view of LN-BTO Hybrid Modulator with $h_{buf} = 200$ nm. Lumerical Device simulations for the static electric field distribution ($|E_{z,elec}|$) considering (b) $h_c = 300$ nm, (c) $h_c = 400$ nm & (d) $h_c = 500$ nm respectively. The scale of electric field is normalized to 4.43×10^3 V/m. Lumerical Mode simulations for the electric field intensities ($|E_{z,opt}|^2$) for the optical modes considering (e) $h_c = 300$ nm, (f) $h_c = 400$ nm & (g) $h_c = 500$ nm respectively. (h) Change of $V_{\pi}L$ and metal absorption loss of the optimized structure with the change of waveguide core thickness.

gap of $\sim 5 \mu\text{m}$) was found to give a metal absorption just below ~ 0.3 dB/cm. The simulated value of the $V_{\pi}L$ of this modulator is ~ 1.73 V.cm which shows $\sim 21\%$ improvement than Ref. [43]. We further extended the simulations by considering LN core thicknesses of 400 nm and 500 nm. For single mode operation, 950 nm & 800 nm of LN core top surface widths were chosen and $\sim 4.6 \mu\text{m}$ & $\sim 4.3 \mu\text{m}$ metal to metal gaps were chosen so that metal absorption loss remains below 0.3 dB/cm for the respective LN core thicknesses. Fig. 4.5b-4.5d show the static electric field distributions for LN core thicknesses of 300 nm, 400 nm & 500 nm respectively when 0.5 V is applied between the two electrodes placed on two sides of the LN waveguide core. Fig. 4.5e-4.5g show the TE optical mode field distribution at 1550 nm wavelength for the respective LN core thicknesses. For 400 nm and 500 nm LN core thicknesses the $V_{\pi}L$ was calculated to be ~ 1.59 V.cm and ~ 1.52 V.cm respectively maintaining the optical losses below 0.3 dB/cm margin that is depicted in Fig. 4.5h. Therefore, it is possible to reduce the $V_{\pi}L$ of the device by etching LN more whereas the optical scattering loss due to the sidewall roughness of the LN waveguide is expected to be low because the refractive index contrast is comparatively higher when BTO cladding is used.

As a next step, we simulate the optical loss due to metal absorption (solid lines) as well as the $V_{\pi}L$ (broken lines) of an LN-BTO hybrid modulator for different BTO buffer thicknesses which is shown in Fig. 4.6a. It was noticed that, for 300 nm LN core thickness the optical loss (solid blue line)

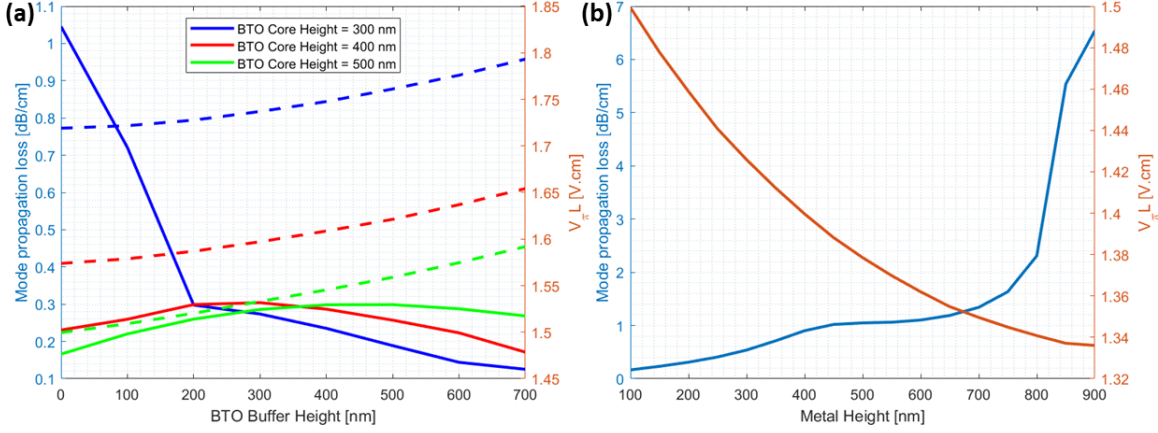


Figure 4.6: (a) The change of optical absorption loss and $V_{\pi}L$ with the change of BTO buffer layer thickness, h_{buf} for different LN core thicknesses, h_c . (b) The change of optical absorption loss and $V_{\pi}L$ with the change of metal electrode thickness, h_{Au} considering $h_c = 500$ nm and $h_{buf} = 0$ nm. All the results were obtained from the Lumerical Mode and Device simulations.

due to metal absorption sharply decreases up to 200 nm BTO buffer layer thickness. If the thickness of the BTO buffer layer is further increased, the optical loss decreases very slowly. This happens because after 200 nm of BTO buffer thickness the top surface of the metal layer goes above the top surface of the LN core and the overlap of the optical mode with the metal layer reduces. Thus, the inclusion of a buffer cladding layer between the metal electrodes and the LN slab layer reduces the optical loss as claimed in Ref. [79]. For higher LN core thicknesses this is not entirely true. As for example, for 400 nm & 500 nm LN core thicknesses the optical losses (solid red & green lines respectively) increase up to 300 nm & 500 nm of BTO buffer layer thicknesses respectively and start to reduce after these thicknesses respectively. The behavior of the optical absorption loss can be explained by the fact that, for higher LN etch depths the confinement of the optical mode in the LN slab region reduces and the evanescent wave is exposed more in the BTO cladding region. Therefore, increasing the BTO buffer layer thickness exposes the metal layer more towards the evanescent wave of the optical mode in the BTO cladding region. For BTO buffer layer thicknesses greater than 300 nm and LN waveguides with thicknesses of 400 nm the metal electrode losses become negligible. For all LN core thicknesses, the $V_{\pi}L$ (broken blue, red & green lines) increase monotonously with the increase of BTO buffer layer thickness. Hence, in the subsequent simulation we fix the LN core thickness at 500 nm. We also avoid having any BTO buffer layer since this not only increases the optical loss for higher LN etch depths but also increases the fabrication complexity.

Next, we simulate the LN-BTO hybrid modulator to get the optical losses and $V_{\pi}L$ for different metal electrode thicknesses. Fig. 4.6b shows that the optical absorption loss increases slowly when the metal electrode thickness is increased from 100 nm to 400 nm. The optical loss remains almost the same (~ 1 dB/cm) until the metal thickness reaches up to 600 nm. After 600 nm of metal thickness the optical loss increases sharply. The $V_{\pi}L$ reduces almost linearly with the increase of the metal electrode thickness. Considering both the optical loss and the $V_{\pi}L$ curves an optimum metal thickness of 600 nm is chosen so that the optical loss remains near ~ 1 dB/cm and a $V_{\pi}L$ of ~ 1.36 V.cm can be achieved. To the best of our knowledge, this is the lowest $V_{\pi}L$ value for all LNOI

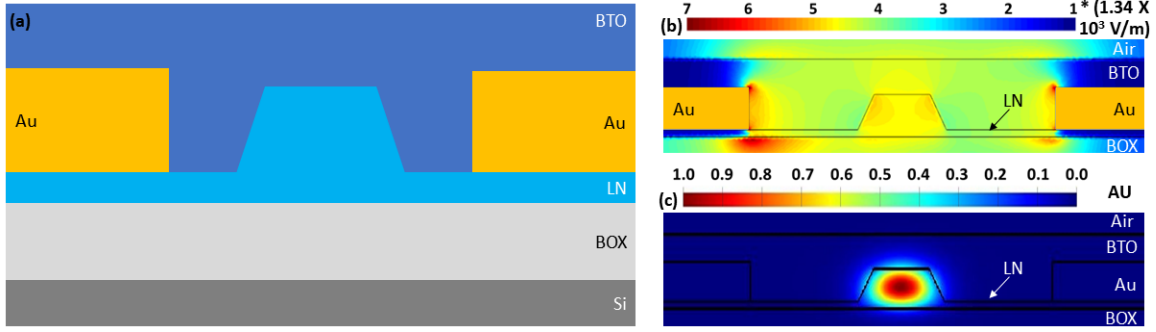


Figure 4.7: (a) Cross-sectional view of optimized LN-BTO Hybrid Modulator. (b) Lumerical Device simulation for the static electric field distribution of the optimized waveguide region of an LN-BTO hybrid modulator. considering $h_c = 500$ nm, $h_{buf} = 0$ nm, and $h_{Au} = 600$ nm. The scale of electric field is normalized to 1.34×10^3 V/m. (c) Lumerical Device simulation for the electric field intensities for the optical mode of the optimized waveguide region of an LN-BTO hybrid modulator considering $h_c = 500$ nm, $h_{buf} = 0$ nm, and $h_{Au} = 600$ nm.

based modulators. The cross-sectional view of the optimized LN-BTO hybrid modulator looks like Fig. 4.7a. Fig. 4.7b shows the static electric field distribution of the optimized device when 0.5 V is applied between the two electrodes placed on two sides of the LN waveguide core and Fig. 4.7c shows the optical mode field distribution of the optimized modulator at 1550 nm wavelength.

4.2.2.2 Transmission Lines Design

Once the waveguide structure is optimized for the best possible trade-off between the optical loss and $V_\pi L$, we can turn our attention towards the transmission line i.e., the CPW design. The RF electrical field is applied using a GSG transmission line. A CPW with infinite ground width has several advantages such as, low dispersion, low radiation loss, suppression of parasitic RF modes etc. A CPW having two to three times the ground width compared to the central conductor width acts very similar to a CPW with infinite width ground electrodes [80]. The RF electric field is well confined between the center and ground electrodes, especially with the LN waveguide core, which can be seen from the RF electric mode field distribution at 10 GHz shown in Fig 4.8a. To account for the conformal deposition of BTO we have added BTO regions above the Au electrodes. This

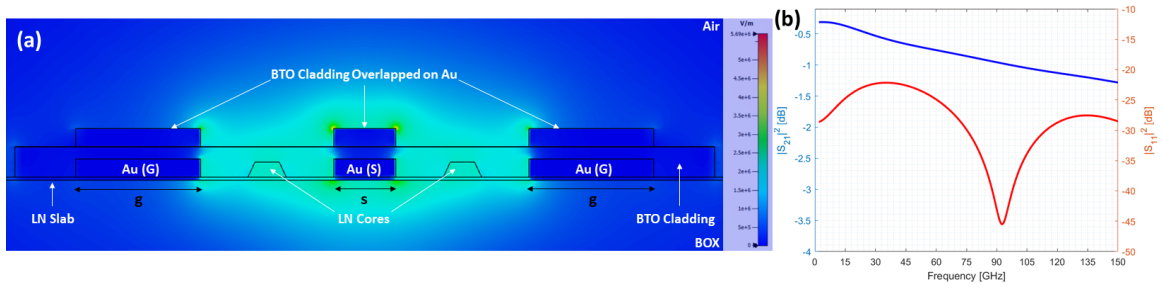


Figure 4.8: (a) The change of optical absorption loss and $V_\pi L$ with the change of BTO buffer layer thickness, h_{buf} for different LN core thicknesses, h_c . (b) The change of optical absorption loss and $V_\pi L$ with the change of metal electrode thickness, h_{Au} considering $h_c = 500$ nm and $h_{buf} = 0$ nm. All the results were obtained from the Lumerical Mode and Device simulations.

CPW structure is broadband and optimized for $50\ \Omega$ having a center electrode width, $s = 2.05\ \mu\text{m}$ and ground electrode width, $g = 4.11\ \mu\text{m}$. Fig. 4.8b shows magnitude of the S-parameters, $|S_{11}|^2$ and $|S_{21}|^2$ in dB which quantify the reflected and the transmitted power respectively through the designed CPW for a modulator length, L of $500\ \mu\text{m}$. The S-parameters were generated by using the Time Domain solver of the Finite-Difference Time-Domain (FDTD) simulation tool, CST Microwave Studio. Frequency independent dielectric constants of LN and BTO are considered as 28 and 40 respectively which are same as the values considered while doing the DC electric field distribution simulations using the Lumerical Device module. The eigen mode solution for the fundamental RF mode provides the broadband characteristic impedance that is shown in Fig. 4.9a. The characteristics impedance of the designed CPW is very well matched to $50\ \Omega$ over a very broad range of frequency which reduces the reflection. The CST generated S-parameter can be used to extract the complex propagation constant, γ . The imaginary part of the of the complex $^*S_{21}$ is the propagation constant, β that can be used to get the microwave effective index, n_{mw} using the expression, $\beta = \frac{2\pi f_{\text{RF}}}{c} n_{\text{mw}}$. The microwave effective indices over a broad frequency range shown in Fig. 4.9b are not well matched with the optical group refractive index ($n_{\text{opt}} = 2.18$) at $1550\ \text{nm}$

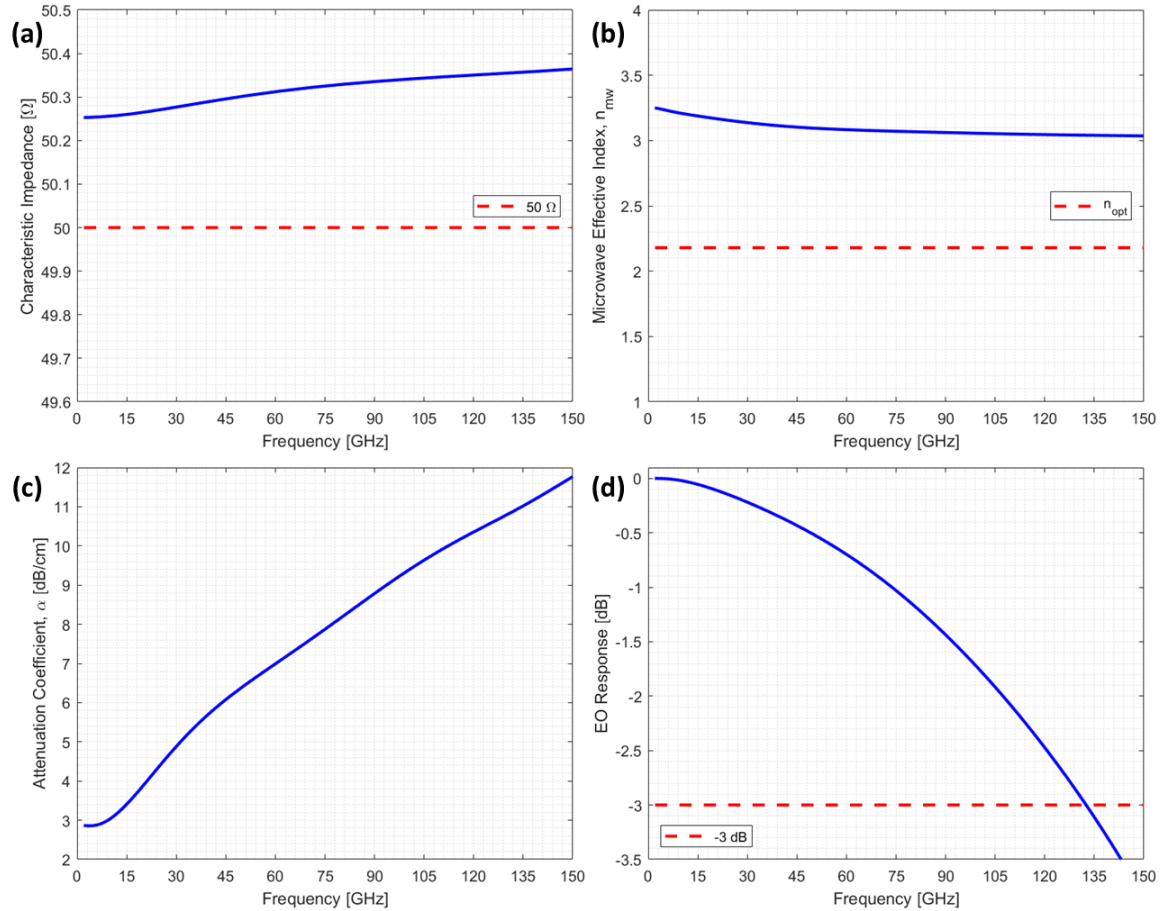


Figure 4.9: (a) The change of optical absorption loss and $V_{\pi}L$ with the change of BTO buffer layer thickness, h_{buf} for different LN core thicknesses, h_c . (b) The change of optical absorption loss and $V_{\pi}L$ with the change of metal electrode thickness, h_{Au} considering $h_c = 500\ \text{nm}$ and $h_{\text{buf}} = 0\ \text{nm}$. All the results were obtained from the Lumerical Mode and Device simulations.

wavelength. The real part of the complex S_{21} gives us the attenuation constant, α of the microwave signal, which is shown in Fig. 4.9c. The index matching between the microwave mode and the optical mode as well as the reduction of the microwave loss can be performed by designing a segmented CPW which is beyond the scope of this thesis [81].

4.2.2.3 Bandwidth Estimation

The simulated values of attenuation coefficient, α and the microwave refractive indices, n_{mw} over a broad frequency range have been used to calculate the electro-optic response of the designed LN-BTO hybrid modulator. Theoretically the bandwidth of the electro-optic response can be calculated using Eq. 2.31 and 2.31. Since Fig. 4.9a shows that the CPW transmission line is well-matched to 50 Ω , the impedance mismatch factors ρ_1 and ρ_2 can be ignored. Thus, the modulation index for bandwidth estimation of LN-BTO hybrid modulator can be simplified as the following expression [82],

$$m(f_{\text{RF}}) = 20 \log \left[e^{-\frac{\alpha L}{2}} \left\{ \frac{\sinh^2\left(\frac{\alpha L}{2}\right) + \sin^2\left(\frac{bL}{2}\right)}{\left(\frac{\alpha L}{2}\right)^2 + \left(\frac{bL}{2}\right)^2} \right\}^{\frac{1}{2}} \right] \quad (4.34)$$

Here, b is defined by the expression, $b = \frac{2\pi f_{\text{RF}}}{c} (n_{\text{mw}} - n_{\text{opt}})$ that quantifies the effect of velocity mismatch between the optical and RF mode on the electro-optic bandwidth of the modulator. The electro-optic response of the modulator is shown in Fig. 4.9d. The electro-optics response of the modulator shows that the 3 dB cut-off frequency or the 3 dB bandwidth of the designed modulator is expected to be around ~132 GHz.

4.2.2.4 Comparison of Etched TFLN Modulators

The performances of the state-of-the-art etched TFLN modulators with uniform CPW electrodes are summarized in Table 4.1. Most of these modulators have used low dielectric constant materials like air [33], SiO₂ [43, 82, 84] or ultra-violet curable adhesive (UV15) [83]. For the first time, we proposed a TFLN modulator which exploited a high dielectric constant material like BTO [73] as a cladding material which showed significant reduction of $V_{\pi}L$ with low optical loss. Later, we have further reduced the $V_{\pi}L$ to 1.36 V.cm for and optical loss less than 1 dB/cm by optimizing multiple physical parameters [74]. Very soon after presenting our first proposal [73], a composite cladding consist of a SiO₂ and a high dielectric constant material like (MgO, Al₂O₃ or glycerol) has been used in etched TFLN modulators [85]. Although this is a similar concept to our work, the device structure is completely different. However, BTO thin films are more suitable than the liquid

Table 4.1: Etched TFLN Modulator Comparison

Ref.	Cladding Material	Etch Depth	$V_{\pi}L$	Optical Loss	EO Bandwidth
[83]	UV15	1.3 μm	9.4 V.cm	1 dB/cm	40 GHz
[43]	SiO ₂	300 nm	2.2 V.cm	0.2 dB/cm	100 GHz
[44]	Air	180 nm	2.2 V.cm	0.98 dB/cm	>70 GHz
[82]	SiO ₂	300 nm	1.75 V.cm	0.7 dB/cm	>40 GHz
[84]	SiO ₂	300 nm	1.32 V.cm	2.7 dB/cm	>3 GHz
[73]	BaTiO ₃	500 nm	1.52 V.cm	0.3 dB/cm	-
[85]	SiO ₂ +Glycerol	200 nm	1.41 V.cm	1.25 dB/cm	>40 GHz
[74]	BaTiO ₃	500 nm	1.36 V.cm	1 dB/cm	132 GHz

Glycerol for practical applications. Besides, our optimized structure estimates lower $V_{\pi}L$ (~ 1.36 V.cm), lower optical propagation loss (~ 1 dB/cm) and higher EO bandwidth (~ 132 GHz) compared to the modulator reported in Ref. [85].

4.3 Conclusion

In this chapter, we report the design of an optimized LN-BTO hybrid modulator where the silicon dioxide cladding of a conventional thin film LN based modulator with etched LN rib waveguide is replaced with a cladding material with very high dielectric constant. Using a LN rib waveguide realized by etching LN by 500 nm on a 600 nm thin film LN layer in the LN-BTO hybrid modulator, a $V_{\pi}L$ of ~ 1.36 V.cm with an optical loss as low as ~ 1 dB/cm. To the best of our knowledge this is the lowest $V_{\pi}L$ values for a thin film LN based modulator. A CPW of 600 nm thick Au layer has been optimized for 50Ω to reduce the reflection of the RF signal at the point of insertion. Despite velocity mismatch between the optical and RF modes the modulator is expected to have a 3 dB bandwidth beyond 100 GHz (theoretical f_{3dB} of ~ 132 GHz) because of the device length as short as a 500 μm with low RF loss. A longer device can be realized by tuning the BTO cladding thickness and by designing a segmented CPW to mitigate the velocity mismatch between RF and optical modes.

Chapter 5

Summary and Outlook

5.1 Summary

Optical modulators are key devices for future data-hungry optical communication systems and microwave photonic circuits. High-speed modulators with low driving voltages are necessary to realize energy and cost-effective optical communication links. Although platform offers integrated optical modulators with low driving voltages, often these devices are limited their bandwidth. On the other hand, the TFLN-based LNOI platform has emerged as a possible solution to minimize the bandwidth limitations. These modulators have lower driving voltages compared to the bulk LN platform but have higher driving voltages compared to the silicon devices. This thesis addressed all these limitations and proposed possible solutions for both platforms. The key accomplishments of this work are summarized as follows:

First demonstration of CC-Si modulator with silicon dioxide cladding with flat EO response up to 68 GHz: Our observation of plasma dispersion effects in foundry fabricated silicon strip waveguides was documented for the first time. For a single drive CC-Si MZI structure we were able to induce sidebands of up to 50 MHz around a carrier signal near telecom wavelengths by introducing sinusoidal modulation signals at one arm of the MZI modulator. By using a self-heterodyne measurement setup, we were able to determine the $V_{\pi}L$ of ~ 9.4 V.cm at a modulation frequency of 20 MHz. We have also carried out the EO measurement of this modulator and observed almost flat EO frequency response up to 68 GHz. The modulator demonstrated an f_{3dB} of ~ 9 GHz, and with additional improvements, the bandwidth may exceed 40 GHz. We have also demonstrated a CC-Si MZI modulator in SPP configuration with a flat frequency response up to 68 GHz and a $V_{\pi}L$ of ~ 87.8 V.cm for an RF modulation sinusoidal signal of 15 GHz.

First demonstration of CC-Si modulator with barium titanate cladding with flat EO response up to 67 GHz: We fabricated the CC-Si modulator with barium titanate cladding in SPP configuration for the first time which includes a foundry fabricated PN diode waveguide fabrication and BEOL post-fabrication process of metal contact layer and barium titanate cladding layer. We carried out the EO measurement of this device and noticed that EO frequency response up of this modulator is very flat up to 67 GHz. We also estimated a $V_{\pi}L$ of ~ 2 V.m for an RF modulation

sinusoidal signal of 15 GHz.

Demonstration of silicon MRR-based optical phase shifter with ultra-low intensity variation for higher order modulation formats: We demonstrated an optical phase shifter based on the SOI platform by combining the through and drop ports of a silicon add-drop MRR modulator. Our proposed phase shifter, which combines the through and drop ports, exhibits ~ 4.29 dB less maximum intensity variation than the maximum intensity variation observed at the through port for $\sim 2\pi$ phase shift, according to the measured intensity spectra for various bias voltages. By integrating such an optical phase shifter into one arm of an integrated MZI structure, we were also able to extract the phase spectrum. By driving the silicon MRR modulator with a forward bias voltage, we were also able to achieve a full 2π phase shift of a CW carrier optical signal with a maximum intensity variation of ~ 4.38 dB. This gave us a V_π of 250 mV for a CW optical carrier used in telecommunication. In conclusion, we generated an 8-PSK constellation map by utilizing the extracted phase and intensity at various forward bias voltages.

First design of a LN-BTO hybrid modulator with lowest estimated half-wave-voltage-length product: We proposed a novel EO modulator based on the LNOI platform. The modulator utilizes a cladding material with a very high dielectric constant, BTO around an etched LN waveguide. An MZI-based EO modulator has a $V_\pi L$ as low as ~ 1.36 V.cm, while maintaining optical loss due to metal absorption below ~ 1 dB/cm. The microwave simulations carried out in CST Studio Suite generated data that allowed for an estimation of an f_{3dB} of ~ 132 GHz.

5.2 Outlook and Future Work

The silicon strip waveguide-based CC-Si modulator that we have demonstrated opens the door of enhancing the bandwidth of silicon modulators by eliminating the RC time constant. The CC-Si modulators we presented in this thesis is first-generation proof-of-concept devices which leave vast room for further improvement of the device metrics. The CC-Si modulator with silicon dioxide cladding has very high $V_\pi L$ because of the low dielectric constant of silicon dioxide cladding. Additionally, for the safety of signal generator no DC bias to the CC-Si modulators which may cause the increase of $V_\pi L$ in CC-Si MZI modulator in SPP configuration. When no DC bias is applied higher chirping may occur because of positive Δn_{eff} in both MZI arms which result in lower differential phase shift compared to an MZI with negative DC bias. For the single drive CC-Si modulator with silicon dioxide cladding, we measured the electrical S-parameters of the transmission line of the modulator and estimated a 3-dB bandwidth of ~ 9.3 GHz. Later, we carried out the EO measurement of CC-Si MZI modulator in both the single drive and SPP configurations and observed a very flat frequency response up to 68 GHz. Although the demonstrated CC-Si modulator with silicon dioxide cladding modulator is ideally free of RC time delays, the bandwidth limitation comes from other factors like velocity mismatch between the optical mode & the RF mode and higher microwave propagation loss of the GS transmission line. We showed that the modulator can achieve more than 40 GHz of 3-dB bandwidth if we can reduce the microwave loss of the modulator transmission line and match the velocities of the microwave and optical modes. The velocity mismatch can be minimized by optimizing the cross-sectional geometry of the modulator or by designing a segmented

TWE [86]. The high microwave loss comes from the highly doped silicon slabs which can be reduced by replacing the highly doped silicon slabs with a metal layer. However, the Vial layer cannot be placed directly on the BOX layer. Hence, low loss transmission lines for these types of modulators composed of only metal layers are still out of the scope of a standard silicon photonic foundry. The bandwidth also depends on the DC bias. Since for the safety of the signal generator we did not apply any DC bias, for half-cycle of the RF modulation sinusoidal signal the PN diode will be in forward bias which in the end may limit the bandwidth of the device.

The electrical field confinement within the waveguide region can potentially be improved by introducing a high dielectric constant material, barium titanate. We demonstrated a CC-Si MZI modulator in SPP configuration which creates the possibility of enhancing the bandwidth of silicon modulators as well as improving the electrical field confinement. The EO measurement of the CC-Si MZI modulator is carried out for RF sinusoidal signals up to 67 GHz which leads to a very flat frequency response that once again proves the validity of capacitive coupling for bandwidth enhancement. For a 2 mm device a $V_\pi L$ of ~ 2 V.m is calculated for an RF signal with 15 GHz frequency with different voltage amplitudes. Although the demonstrated CC-Si device shows the potential of high bandwidth, the $V_\pi L$ of this device is still very high. But with the help of EO simulations we qualitatively show that the device performance can be improved to a great extent by the proper alignment of metal electrodes close to the waveguide and by the deposition of thicker metal electrodes as well as thicker barium titanate cladding. Additionally, chirping arises because of not applying a DC bias at the DC contact which eventually increases the $V_\pi L$. Also, the $V_\pi L$ is calculated at a fairly high speed (15 GHz) since the OSA does not have the resolution to measure the sideband at lower frequencies. In such high frequencies, a lot of other things like impedance mismatch, microwave propagation loss, velocity mismatch etc. come to play which not only increase the $V_\pi L$ in higher frequencies but also reduce the device bandwidth. Moreover, by not applying a DC bias we are operating the CC-Si phase shifters in forward bias for half period of a sinusoidal signal. Thus, the device bandwidth is further compromised. All these limitations can be minimized significantly by optimizing the fabrication process flow and by properly biasing the CC-Si modulator. The concept of capacitive coupling of RF electric field using high-k material to the waveguide region is already found effective in the SOH platform [12, 22, 23, 24] and the LNOI platform [73, 74]. This device shows the potential capacitive coupling using high dielectric constant material like barium titanate on silicon modulator for future high-speed devices as well. However, the standard silicon photonic foundries do not offer the deposition of high-k material cladding which is a very easy step to integrate and compatible. Nevertheless, we believe that the challenges faced by our proposed CC-Si modulator will eventually be solved with the progress of fabrication techniques that will allow great improvement in the device's performances.

The silicon MRR-based optical phase shifter has a low V_π since we operated the MRR modulator in forward bias regime where carrier injection phenomenon is exploited to modulate the light. Hence, the speed of this device will be limited to <1 GHz [11]. On the other hand, for high-speed operation the modulator has to be operated in reverse bias regime in order to change the PN diode depletion region width to modulate light [11]. But, in that case the V_π of the device increases and achieving a full $\sim 2\pi$ phase shift becomes very difficult. A ring resonator with very narrow resonance

may be able to resolve this issue. Also, we expect to have a very low optical loss modulation which is unavoidable because of the amplitude-phase coupling of the silicon modulators. By using lower doping concentrations in the waveguide region this problem can be minimized which is beyond the scope of foundry fabricated devices since the doping concentration levels are fixed and confidential for the foundries. Another possible solution to reduce the optical loss modulation and increase the speed could be to use capacitive coupling to modulate the light instead of ohmic contacts [12, 59]. However, with more careful design and optimization this device can achieve higher order modulation format with high speed, low driving voltage and low optical insertion loss.

Compared to the conventional LNOI based modulators, the main challenges faced by the proposed LN-BTO hybrid modulator lie in two major things the quality of BTO film for desired optical & RF properties during deposition and the patterning of thicker BTO film. Firstly, high-quality stoichiometric crystalline BTO thin film can be deposited by molecular beam epitaxy (MBE) [87, 88] or metalorganic chemical vapour deposition (MOCVD) [89] method which can provide dielectric constant as high as 2200. But such films generally have higher refractive index ($n_{\text{BTO}} = 2.286$) as well in the telecommunication wavelengths which can jeopardize the optical confinement in the LN waveguide core region [88]. Moreover, the above-mentioned deposition methods may require very high deposition temperature (up to 900°C) which is not compatible with pre-existing metal layers. On the other hand, amorphous BTO thin films can be deposited by RF magnetron sputtering method which has lower dielectric constant ($\epsilon_{\text{BTO}} \sim 40$) [23] compared to the crystalline BTO thin films deposited by MBE or MOCVD method. However, sputtered BTO films also have a comparatively lower refractive index ($n_{\text{BTO}} = 1.85$) at 1550 nm wavelength which ensures optical mode confinement in the LN waveguide core region [12]. Besides, the required deposition temperature for amorphous sputtered BTO thin film is much lower compared to the MBE or MOCVD method (typically below 200°C) which is compatible with pre-existing metal layers [12, 22]. Therefore, RF magnetron sputtering deposition method should be preferred over that other deposition methods. Secondly, etching of BTO thin film is extremely difficult since pure physical etching techniques are normally used for etching BTO [88]. This may cause the damage of underlying layers. Hence, standard e-beam lithography followed by lift-off of sputtered BTO thin film is preferred for patterning the BTO thin film. Although the deposition of thicker BTO film using an RF magnetron sputter system is quite challenging, it can still be achieved at the cost of longer deposition time. Also, the liftoff process of thicker BTO thin films is quite challenging, but using thick bilayer of e-beam resists the lift-off process can be optimized for thicker BTO films. Nevertheless, with the progress in fabrication techniques we believe that the challenges faced by our proposed modulator structure will eventually be solved that will allow great improvement in the device performances.

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